

Sliper QML C++ — Kaynak Kod

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CMakeLists.txt

```
1 cmake_minimum_required(VERSION 3.16)

2 project(sliper VERSION 1.0)

3

4 set(CMAKE_CXX_STANDARD 17)

5 set(CMAKE_PREFIX_PATH "C:/Qt/6.7.3/mingw_64")

6

7 find_package(Qt6 REQUIRED COMPONENTS Core Quick Charts Widgets Sql PrintSupport Network)

8

9 qt_standard_project_setup()

10 qt_policy(SET QTP0001 NEW)

11

12 qt_add_executable(sliper

13     main.cpp

14     src/Backend.h

15     src/Backend.cpp

16     src/SensorManager.h

17     src/SensorManager.cpp

18     src/Calculator.h

19     src/Calculator.cpp

20     src/Database.h

21     src/Database.cpp

22     src/ReportManager.h

23     src/ReportManager.cpp

24     src/WifiManager.h

25     src/WifiManager.cpp

26 )

27

28 qt_add_qml_module(sliper

29     URI sliper

30     VERSION 1.0
```

```
31
32     QML_FILES
33         qml/Main.qml
34         qml/pages/LoginPage.qml
35         qml/pages/DashboardPage.qml
36         qml/pages/OlcumPage.qml
37         qml/pages/SonucularPage.qml
38         qml/pages/GecmisPage.qml
39         qml/pages/KalibrasyonPage.qml
40         qml/components/SuTerazisi.qml
41         qml/components/TarihSecici.qml
42
43     RESOURCES
44         qml/assets/logo.png
45         qml/assets/egim_sensor.png
46         qml/assets/loadcell.png
47         qml/assets/mesafe_sensor.png
48 )
49
50 target_link_libraries(slipe PRIVATE
51     Qt6::Core
52     Qt6::Quick
53     Qt6::Charts
54     Qt6::Widgets
55     Qt6::Sql
56     Qt6::PrintSupport
57     Qt6::Network
58 )
```

main.cpp

```
1 #include <QmlContext>

2 #include <QApplication>

3 #include <QmlApplicationEngine>

4 #include <QDebug>

5 #include <QtGlobal>

6 #include "src/Backend.h"

7 #include "src/SensorManager.h"

8 #include "src/Calculator.h"

9 #include "src/Database.h"

10 #include "src/ReportManager.h"

11 #include "src/WifiManager.h"

12

13 int main(int argc, char *argv[])

14 {

15     qputenv("QT_QUICK_CONTROLS_STYLE", "Basic");

16

17     QApplication app(argc, argv);

18     QApplication::setApplicationName("SLIPER");

19

20     Backend backend;

21     SensorManager sensorManager;

22     Calculator calculator;

23     Database database;

24     ReportManager reportManager;

25     WifiManager wifiManager(&sensorManager, &database);

26

27     QmlApplicationEngine engine;

28

29     engine.rootContext()->setContextProperty("backend", &backend);

30     engine.rootContext()->setContextProperty("sensorManager", &sensorManager);
```

```
31     engine.rootContext()->setContextProperty("calculator", &calculator);
32     engine.rootContext()->setContextProperty("database", &database);
33     engine.rootContext()->setContextProperty("reportManager", &reportManager);
34     engine.rootContext()->setContextProperty("wifiManager", &wifiManager);
35
36     QObject::connect(
37         &engine,
38         &QQmlApplicationEngine::objectCreationFailed,
39         &app,
40         []() {
41             qWarning() << "QML nesnesi olusturulamadi!";
42             QCoreApplication::exit(-1);
43         },
44         Qt::QueuedConnection);
45
46     engine.loadFromModule("sliper", "Main");
47
48     return app.exec();
49 }
```

qml/Main.qml

```
1 import QtQuick 6.7

2 import QtQuick.Controls 6.7

3 import QtQuick.Window 6.7

4 import QtQuick.Layouts 6.7

5 import "pages"

6

7 ApplicationWindow {

8     id: pencere

9     visibility: Window.Maximized

10    title: "SLIPER Analiz"

11

12    background: Rectangle {

13        color: "#060d17"

14    }

15

16    StackLayout {

17        id: anaStack

18        anchors.fill: parent

19        currentIndex: 0

20

21        LoginPage {

22            onGirisBasarili: anaStack.currentIndex = 1

23        }

24

25        DashboardPage {

26        }

27    }

28 }
```

qml/components/SuTerazisi.qml

```
1 import QtQuick 6.7
2
3 Item {
4     id: root
5     property real egimX: 0
6     property real egimY: 0
7     property real maxAci: 15
8
9     width: 200
10    height: 200
11
12    readonly property real yaricap: (width / 2) - 20
13    readonly property bool duz: Math.abs(egimX) < 0.5 && Math.abs(egimY) < 0.5
14
15    readonly property real kabarcikOffsetX: Math.max(-1, Math.min(1, egimY / maxAci)) * yaricap
16    readonly property real kabarcikOffsetY: Math.max(-1, Math.min(1, egimX / maxAci)) * yaricap
17
18    Rectangle {
19        anchors.fill: parent
20        radius: width / 2
21        color: "#0f1420"
22        border.color: "#1e2a3f"
23        border.width: 2
24    }
25
26    Rectangle {
27        anchors.centerIn: parent
28        width: root.yaricap * 1.2
29        height: width
30        radius: width / 2
```

```
31     color: "transparent"
32     border.color: "#1e2a3f"
33     border.width: 1
34 }
35
36 Rectangle {
37     anchors.centerIn: parent
38     width: parent.width - 20
39     height: 1
40     color: "#1e2a3f"
41 }
42
43 Rectangle {
44     anchors.centerIn: parent
45     width: 1
46     height: parent.height - 20
47     color: "#1e2a3f"
48 }
49
50 Rectangle {
51     id: kabarcik
52     width: 28
53     height: 28
54     radius: 14
55     color: root.duz ? "#16a34a" : "#dc2626"
56     border.color: root.duz ? "#4ade80" : "#f87171"
57     border.width: 2
58
59     x: (root.width / 2) - (width / 2) + root.kabarcikOffsetX
60     y: (root.height / 2) - (height / 2) + root.kabarcikOffsetY
```



```
61
62     Behavior on x { NumberAnimation { duration: 150; easing.type: Easing.OutQuad } }
63     Behavior on y { NumberAnimation { duration: 150; easing.type: Easing.OutQuad } }
64     Behavior on color { ColorAnimation { duration: 200 } }
65     Behavior on border.color { ColorAnimation { duration: 200 } }
66 }
67
68 Text {
69     anchors.top: parent.bottom
70     anchors.horizontalCenter: parent.horizontalCenter
71     anchors.topMargin: 8
72     text: root.duz ? "Düz" : "Ayarlanıyor..."
73     color: root.duz ? "#4ade80" : "#6b7280"
74     font.family: "Segoe UI"
75     font.pixelSize: 12
76     font.bold: root.duz
77 }
78 }
```

qml/components/TarihSecici.qml

```
1 import QtQuick 6.7
2 import QtQuick.Controls 6.7
3
4 Item {
5     id: root
6     width: 130
7     height: 36
8
9     // Kontrollü bileşen: seçili tarih dışarıdan (ör. gecmisSayfasi.baslangicTarihi)
10    // bağlanır, kullanıcı bir gün seçtiğinde sadece tarihSecildi sinyali yayılır –
11    // root.secilenTarih burada asla doğrudan atanmaz ki dışarıdaki binding kopmasın.
12    property string secilenTarih: ""
13    property string placeholderText: "gg.aa.yyyy"
14    property date gorunenAy: new Date()
15
16    readonly property int hucreBoyutu: 28
17    readonly property int hucreAraligi: 4
18    readonly property int gridGenisligi: 7 * hucreBoyutu + 6 * hucreAraligi
19
20    signal tarihSecildi(string tarihStr)
21
22    function ikiHane(n) {
23        return n < 10 ? "0" + n : "" + n
24    }
25
26    function tarihStr(d) {
27        return ikiHane(d.getDate()) + "." + ikiHane(d.getMonth() + 1) + "." + d.getFullYear()
28    }
29
30    function gunSayisi(yil, ay) {
```

```

31         return new Date(yil, ay + 1, 0).getDate()
32     }
33
34     // Takvim ızgarasını (önceki/sonraki aydan taşan günler dahil, Pazartesi başlangıçlı,
35     // her zaman 6 hafta = 42 hücre) saf QtQuick ile hesaplar; Qt.labs.calendar modülüne
36     // bağımlı değildir çünkü bu Qt kurulumunda mevcut değil.
37     function ayGunleriniHesapla(ay) {
38         var yil = ay.getFullYear()
39         var ayIndeks = ay.getMonth()
40         var ilkGun = new Date(yil, ayIndeks, 1)
41         var haftaBaslangicKaymasi = (ilkGun.getDay() + 6) % 7 // Pazartesi = 0
42         var baslangic = new Date(yil, ayIndeks, 1 - haftaBaslangicKaymasi)
43
44         var gunler = []
45         for (var i = 0; i < 42; i++) {
46             gunler.push(new Date(baslangic.getFullYear(), baslangic.getMonth(), baslangic.getDate() + i))
47         }
48         return gunler
49     }
50
51     property var gunHucreleri: ayGunleriniHesapla(gorunenAy)
52
53     readonly property var ayAdlari: ["Ocak", "Şubat", "Mart", "Nisan", "Mayıs", "Haziran",
54                                     "Temmuz", "Ağustos", "Eylül", "Ekim", "Kasım", "Aralık"]
55     readonly property var haftaGunleri: ["Pt", "Sa", "Ça", "Pe", "Cu", "Ct", "Pz"]
56
57     Rectangle {
58         id: kutu
59         anchors.fill: parent
60         radius: 8

```

```

61     color: "#0f1420"
62     border.color: (alan.containsMouse || takvimPopup.visible) ? "#3b82f6" : "#1e2a3f"
63     border.width: 1
64     Behavior on border.color { ColorAnimation { duration: 120 } }
65
66     Text {
67         anchors.left: parent.left
68         anchors.right: simgeMetni.left
69         anchors.leftMargin: 10
70         anchors.verticalCenter: parent.verticalCenter
71         elide: Text.ElideRight
72         text: root.secilenTarih.length > 0 ? root.secilenTarih : root.placeholderText
73         color: root.secilenTarih.length > 0 ? "#dce8f5" : "#4b5563"
74         font.family: "Segoe UI"
75         font.pixelSize: 13
76     }
77
78     Text {
79         id: simgeMetni
80         anchors.right: parent.right
81         anchors.rightMargin: 10
82         anchors.verticalCenter: parent.verticalCenter
83         text: "📅"
84         font.pixelSize: 12
85         color: "#6b7280"
86     }
87
88     MouseArea {
89         id: alan
90         anchors.fill: parent
91         hoverEnabled: true

```

```
92         cursorShape: Qt.PointingHandCursor
93     onClicked: {
94         if (root.secilenTarih.length === 10) {
95             var parcalar = root.secilenTarih.split(".")
96             root.gorunenAy = new Date(parseInt(parcalar[2], 10), parseInt(parcalar[1], 10) - 1, 1)
97         } else {
98             root.gorunenAy = new Date()
99         }
100         takvimPopup.open()
101     }
102 }
103 }
104
105 Popup {
106     id: takvimPopup
107     y: kutu.height + 6
108     width: root.gridGenisligi + 28
109     padding: 14
110     modal: true
111     focus: true
112     closePolicy: Popup.CloseOnEscape | Popup.CloseOnPressOutside
113
114     background: Rectangle {
115         color: "#0f1420"
116         radius: 12
117         border.color: "#1e2a3f"
118         border.width: 1
119     }
120
121     Overlay.modal: Rectangle {
```



```

152             onClicked: root.gorunenAy = new Date(root.gorunenAy.getFullYear(),
root.gorunenAy.getMonth() - 1, 1)

153         }

154     }

155

156     Text {
157         width: parent.width - 56
158         anchors.verticalCenter: parent.verticalCenter
159         horizontalAlignment: Text.AlignHCenter
160         text: root.ayAdlari[root.gorunenAy.getMonth()] + " " + root.gorunenAy.getFullYear()
161         color: "#dce8f5"
162         font.family: "Segoe UI"
163         font.pixelSize: 13
164         font.bold: true
165     }

166

167     Rectangle {
168         width: 28
169         height: 28
170         radius: 6
171         anchors.verticalCenter: parent.verticalCenter
172         color: sonrakiAlan.containsMouse ? "#1e2a3f" : "transparent"
173         Behavior on color { ColorAnimation { duration: 120 } }
174
175     Text {
176         anchors.centerIn: parent
177         text: ">"
178         color: "#9ca3af"
179         font.pixelSize: 16
180     }
181

```

```

182         MouseArea {
183             id: sonrakiAlan
184             anchors.fill: parent
185             hoverEnabled: true
186             cursorShape: Qt.PointingHandCursor
187             onClicked: root.gorunenAy = new Date(root.gorunenAy.getFullYear(),
root.gorunenAy.getMonth() + 1, 1)
188         }
189     }
190 }
191
192 Row {
193     width: root.gridGenisligi
194     spacing: root.hucreAraligi
195
196     Repeater {
197         model: root.haftaGunleri
198
199         delegate: Text {
200             width: root.hucreBoyutu
201             horizontalAlignment: Text.AlignHCenter
202             text: modelData
203             color: "#6b7280"
204             font.family: "Segoe UI"
205             font.pixelSize: 10
206             font.bold: true
207         }
208     }
209 }
210
211 Grid {

```



```

212         width: root.gridGenisligi
213
214         columns: 7
215
216         rowSpacing: root.hucreAraligi
217         columnSpacing: root.hucreAraligi
218
219
220     Repeater {
221
222         model: root.gunHucreleri
223
224
225         delegate: Rectangle {
226
227             id: gunHucresi
228
229             width: root.hucreBoyutu
230
231             height: root.hucreBoyutu
232
233             radius: root.hucreBoyutu / 2
234
235             property bool buAyMi: modelData.getMonth() === root.gorunenAy.getMonth()
236
237             property bool seciliMi: root.secilenTarih === root.tarihStr(modelData)
238
239             property bool bugunMu: root.tarihStr(modelData) === root.tarihStr(new Date())
240
241             color: seciliMi ? "#3b82f6" : (gunAlani.containsMouse ? "#1e2a3f" : "transparent")
242
243             border.color: (bugunMu && !seciliMi) ? "#3b82f6" : "transparent"
244
245             border.width: 1
246
247
248             Text {
249
250                 anchors.centerIn: parent
251
252                 text: modelData.getDate()
253
254                 color: gunHucresi.buAyMi ? (gunHucresi.seciliMi ? "ffffff" : "#dce8f5") :
255 "#374151"
256
257                 font.family: "Segoe UI"
258
259                 font.pixelSize: 12
260
261                 font.bold: gunHucresi.seciliMi
262
263             }
264
265
266             MouseArea {

```

```

242             id: gunAlani
243             anchors.fill: parent
244             hoverEnabled: true
245             cursorShape: Qt.PointingHandCursor
246             onClicked: {
247                 root.tarihSecildi(root.tarihStr(modelData))
248                 takvimPopup.close()
249             }
250         }
251     }
252 }
253
254
255 Rectangle {
256     width: root.gridGenisligi
257     height: 1
258     color: "#1e2a3f"
259 }
260
261 Rectangle {
262     width: root.gridGenisligi
263     height: 30
264     radius: 6
265     color: temizleAlani.containsMouse ? "#1e2a3f" : "transparent"
266     Behavior on color { ColorAnimation { duration: 120 } }
267
268     Text {
269         anchors.centerIn: parent
270         text: "Temizle"
271         color: "#9ca3af"
272         font.family: "Segoe UI"

```

```
273         font.pixelSize: 12
274     }
275
276     MouseArea {
277         id: temizleAlani
278         anchors.fill: parent
279         hoverEnabled: true
280         cursorShape: Qt.PointingHandCursor
281         onClicked: {
282             root.tarihSecildi("")
283             takvimPopup.close()
284         }
285     }
286 }
287 }
288 }
289 }
```

qml/pages/DashboardPage.qml

```
1 import QtQuick 6.7
2 import QtQuick.Controls 6.7
3 import QtQuick.Layouts 6.7
4
5 Rectangle {
6     color: "#0a0e17"
7
8     readonly property var navOgeleri: [
9         { etiket: "Ölçüm", ikon: "📊" },
10        { etiket: "Sonuçlar", ikon: "📈" },
11        { etiket: "Geçmiş", ikon: "🕒" },
12        { etiket: "Kalibrasyon", ikon: "⚙️" }
13    ]
14
15    Row {
16        anchors.fill: parent
17        spacing: 0
18
19        Rectangle {
20            id: sidebar
21            width: 230
22            height: parent.height
23
24            gradient: Gradient {
25                GradientStop { position: 0.0; color: "#121b30" }
26                GradientStop { position: 1.0; color: "#080b12" }
27            }
28
29            Image {
30                anchors.top: parent.top
```

```
31         anchors.horizontalCenter: parent.horizontalCenter
32         anchors.topMargin: 24
33         source: "../assets/logo.png"
34         width: 144
35         height: 144
36         fillMode: Image.PreserveAspectFit
37     }
38
39     Rectangle {
40         id: baglantiDurumu
41         anchors.top: parent.top
42         anchors.left: parent.left
43         anchors.right: parent.right
44         anchors.leftMargin: 24
45         anchors.rightMargin: 24
46         anchors.topMargin: 178
47         height: 36
48         radius: 6
49         color: baglantiAlani.hoverli ? "#101a2c" : "transparent"
50         Behavior on color { ColorAnimation { duration: 120 } }
51
52     Row {
53         anchors.left: parent.left
54         anchors.leftMargin: 8
55         anchors.verticalCenter: parent.verticalCenter
56         spacing: 8
57
58         Rectangle {
59             width: 8
60             height: 8
```

```

61         radius: 4

62         anchors.verticalCenter: parent.verticalCenter

63         color: wifiManager.baglandi ? "#16a34a" : (wifiManager.baglaniyor ? "#f59e0b" :
"#dc2626")

64     }

65

66     Text {

67         text: wifiManager.baglandi ? "SLIPER-ESP32 Bağlı" : (wifiManager.baglaniyor ?
"Bağlanıyor..." : "Bağlı Değil")

68         color: "#9ca3af"

69         font.family: "Segoe UI"

70         font.pixelSize: 12

71         anchors.verticalCenter: parent.verticalCenter

72     }

73 }

74

75 MouseArea {

76     id: baglantiAlani

77     anchors.fill: parent

78     hoverEnabled: true

79     cursorShape: Qt.PointingHandCursor

80     enabled: !wifiManager.baglandi && !wifiManager.baglaniyor

81     onClicked: wifiManager.baglan()

82 }

83 }

84

85 Rectangle {

86     anchors.top: parent.top

87     anchors.left: parent.left

88     anchors.right: parent.right

89     anchors.leftMargin: 24

90     anchors.rightMargin: 24

```

```
91         anchors.topMargin: 224
92         height: 1
93         color: "#1f2c46"
94     }
95
96     Column {
97         anchors.top: parent.top
98         anchors.topMargin: 242
99         width: parent.width
100        spacing: 2
101
102        Repeater {
103            model: navOgeleri
104
105            Rectangle {
106                id: navOgesi
107                width: parent.width
108                height: 46
109                property bool aktif: icerikStack.currentIndex === index
110                property bool hoverli: false
111                color: aktif ? "#182644" : (hoverli ? "#101a2c" : "transparent")
112                Behavior on color { ColorAnimation { duration: 120 } }
113
114                Rectangle {
115                    width: 3
116                    height: parent.height
117                    color: "#3b82f6"
118                    visible: navOgesi.aktif
119                }
120
```

```

121         Row {
122             anchors.left: parent.left
123             anchors.leftMargin: 24
124             anchors.verticalCenter: parent.verticalCenter
125             spacing: 12
126
127             Text {
128                 text: modelData.ikon
129                 font.pixelSize: 15
130                 anchors.verticalCenter: parent.verticalCenter
131             }
132
133             Text {
134                 text: modelData.etiket
135                 color: navOgesi.aktif ? "#6ea8ff" : (navOgesi.hoverli ? "#dce8f5" :
"#9ca3af")
136                 font.family: "Segoe UI"
137                 font.pixelSize: 14
138                 font.bold: navOgesi.aktif
139                 anchors.verticalCenter: parent.verticalCenter
140                 Behavior on color { ColorAnimation { duration: 120 } }
141             }
142         }
143
144         MouseArea {
145             anchors.fill: parent
146             hoverEnabled: true
147             cursorShape: Qt.PointingHandCursor
148             onEntered: navOgesi.hoverli = true
149             onExited: navOgesi.hoverli = false
150             onClicked: icerikStack.currentIndex = index

```



```

151         }
152     }
153 }
154 }
155 }
156
157 Rectangle {
158     id: sidebarAyraci
159     width: 3
160     height: parent.height
161     color: "#05070b"
162
163     Rectangle {
164         anchors.right: parent.right
165         anchors.top: parent.top
166         anchors.bottom: parent.bottom
167         width: 1
168         color: "#403b82f6"
169     }
170 }
171
172 StackLayout {
173     id: icerikStack
174     width: parent.width - sidebar.width - sidebarAyraci.width
175     height: parent.height
176     currentIndex: 0
177
178     OlcumPage {
179         onOlcumTamamlandi: function(id) {
180             sonuclarSayfasi.olcumId = id
181             icerikStack.currentIndex = 1

```

```
182         }
183     }
184
185     SonuclarPage {
186         id: sonuclarSayfasi
187     }
188
189     GecmisPage {
190         onTestSecildi: function(id) {
191             sonuclarSayfasi.olcumId = id
192             icerikStack.currentIndex = 1
193         }
194     }
195
196     KalibrasyonPage {}
197 }
198 }
199 }
```

qml/pages/GecmisPage.qml

```
1 import QtQuick 6.7
2 import QtQuick.Controls 6.7
3 import "../components"
4
5 Rectangle {
6     id: gecmisSayfasi
7     color: "#0a0e17"
8
9     signal testSecildi(int id)
10
11     property var olcumListesi: []
12     property string baslangicTarihi: ""
13     property string bitisTarihi: ""
14
15     function listeyiYenile() {
16         olcumListesi = database.tumOlcumleriGetir()
17     }
18
19     onVisibleChanged: {
20         if (visible) {
21             listeyiYenile()
22         }
23     }
24
25     Component.onCompleted: listeyiYenile()
26
27     function tarihiAyristir(tarihStr) {
28         if (!tarihStr) return null
29         var tarihKismi = tarihStr.split(" ")[0].split(".")
30         if (tarihKismi.length !== 3) return null
```

```
31     var gun = parseInt(tarihKismi[0], 10)
32     var ay = parseInt(tarihKismi[1], 10)
33     var yil = parseInt(tarihKismi[2], 10)
34     if (isNaN(gun) || isNaN(ay) || isNaN(yil)) return null
35     return new Date(yil, ay - 1, gun)
36 }
37
38 function gecerliOzelTarih(metin) {
39     if (!metin || metin.length !== 10) return null
40     return tarihiAyristir(metin)
41 }
42
43 function filtrelenmisListeyiHesapla() {
44     var sonuc = []
45     var aramaMetni = aramaKutusu.text.toLowerCase()
46     var ozelBaslangic = gecerliOzelTarih(gecmisSayfasi.baslangicTarihi)
47     var ozelBitis = gecerliOzelTarih(gecmisSayfasi.bitisTarihi)
48     var simdi = new Date()
49
50     for (var i = 0; i < olcumListesi.length; i++) {
51         var kayit = olcumListesi[i]
52         var tarihObj = tarihiAyristir(kayit.tarih)
53
54         if (aramaMetni.length > 0) {
55             varmusteriEslesir = kayit.musteri.toLowerCase().indexOf(aramaMetni) !== -1
56             var receteEslesir = kayit.recete.toLowerCase().indexOf(aramaMetni) !== -1
57             if (!musteriEslesir && !receteEslesir) continue
58         }
59
60         if (ozelBaslangic || ozelBitis) {
```

```

61         if (ozelBaslangic && tarihObj && tarihObj < ozelBaslangic) continue
62         if (ozelBitis && tarihObj) {
63             var bitisSonu = new Date(ozelBitis.getFullYear(), ozelBitis.getMonth(),
ozelBitis.getDate(), 23, 59, 59)
64             if (tarihObj > bitisSonu) continue
65         }
66     } else if (tarihObj) {
67         if (filtreKutusu.currentIndex === 1) {
68             var esik7 = new Date(simdi.getTime() - 7 * 24 * 60 * 60 * 1000)
69             if (tarihObj < esik7) continue
70         } else if (filtreKutusu.currentIndex === 2) {
71             var esik30 = new Date(simdi.getTime() - 30 * 24 * 60 * 60 * 1000)
72             if (tarihObj < esik30) continue
73         }
74     }
75
76     sonuc.push(kayit)
77 }
78
79 return sonuc
80 }
81
82 property var filtrelenmisListe: filtrelenmisListeyiHesapla()
83
84 Column {
85     anchors.fill: parent
86     anchors.margins: 20
87     spacing: 16
88
89     Row {
90         width: parent.width

```

```

91         spacing: 12
92
93     TextField {
94         id: aramaKutusu
95         width: parent.width - filtreKutusu.width - parent.spacing
96         height: 40
97         placeholderText: "Müşteri veya reçete ara..."
98         placeholderTextColor: "#4b5563"
99         color: "#ffffff"
100        font.pixelSize: 13
101        leftPadding: 34
102        verticalAlignment: TextInput.AlignVCenter
103        background: Rectangle {
104            color: "#0f1420"
105            radius: 8
106            border.color: aramaKutusu.activeFocus ? "#3b82f6" : "#1e2a3f"
107            border.width: 1
108        }
109
110        Text {
111            anchors.left: parent.left
112            anchors.leftMargin: 12
113            anchors.verticalCenter: parent.verticalCenter
114            text: "🔍"
115            color: "#4b5563"
116            font.pixelSize: 13
117        }
118    }
119
120    ComboBox {
121        id: filtreKutusu

```

```
122         width: 160
123         height: 40
124         model: ["Tüm Testler", "Son 7 Gün", "Son 30 Gün"]
125
126         background: Rectangle {
127             color: "#0f1420"
128             radius: 8
129             border.color: filtreKutusu.activeFocus ? "#3b82f6" : "#1e2a3f"
130             border.width: 1
131         }
132
133         contentItem: Text {
134             text: filtreKutusu.displayText
135             color: "#9ca3af"
136             font.pixelSize: 13
137             leftPadding: 10
138             verticalAlignment: Text.AlignVCenter
139         }
140     }
141 }
142
143 Row {
144     width: parent.width
145     spacing: 10
146
147     Text {
148         text: "Özel Tarih:"
149         color: "#9ca3af"
150         font.family: "Segoe UI"
151         font.pixelSize: 12
```

```
152         anchors.verticalCenter: parent.verticalCenter
153     }
154
155     TarihSecici {
156         id: baslangicSecici
157         anchors.verticalCenter: parent.verticalCenter
158         secilenTarih: gecmisSayfasi.baslangicTarihi
159         onTarihSecildi: function(t) { gecmisSayfasi.baslangicTarihi = t }
160     }
161
162     Text {
163         text: "-"
164         color: "#4b5563"
165         font.pixelSize: 13
166         anchors.verticalCenter: parent.verticalCenter
167     }
168
169     TarihSecici {
170         id: bitisSecici
171         anchors.verticalCenter: parent.verticalCenter
172         secilenTarih: gecmisSayfasi.bitisTarihi
173         onTarihSecildi: function(t) { gecmisSayfasi.bitisTarihi = t }
174     }
175
176     Rectangle {
177         width: temizleMetni.implicitWidth + 20
178         height: 36
179         radius: 8
180         color: temizleMouse.containsMouse ? "#162033" : "#0f1420"
181         border.color: "#1e2a3f"
182         border.width: 1
```



```
183         visible: gecmisSayfasi.baslangicTarihi.length > 0 || gecmisSayfasi.bitisTarihi.length > 0
184         anchors.verticalCenter: parent.verticalCenter
185
186         Text {
187             id: temizleMetni
188             anchors.centerIn: parent
189             text: "Temizle"
190             color: "#9ca3af"
191             font.family: "Segoe UI"
192             font.pixelSize: 12
193         }
194
195         MouseArea {
196             id: temizleMouse
197             anchors.fill: parent
198             hoverEnabled: true
199             cursorShape: Qt.PointingHandCursor
200             onClicked: {
201                 gecmisSayfasi.baslangicTarihi = ""
202                 gecmisSayfasi.bitisTarihi = ""
203             }
204         }
205     }
206 }
207
208 ListView {
209     width: parent.width
210     height: parent.height - 96
211     clip: true
212     spacing: 10
```

```
213
214     model: gecmisSayfasi.filtrelenmisListe
215
216     delegate: Rectangle {
217         id: gecmisKarti
218         width: parent.width
219         height: 76
220         radius: 10
221         color: "#0f1420"
222         border.color: kartAlani.containsMouse ? "#3b82f6" : "#1e2a3f"
223         border.width: 1
224         Behavior on border.color { ColorAnimation { duration: 120 } }
225
226         MouseArea {
227             id: kartAlani
228             anchors.fill: parent
229             hoverEnabled: true
230             acceptedButtons: Qt.NoButton
231         }
232
233         Rectangle {
234             id: rozet
235             width: 44
236             height: 44
237             radius: 10
238             anchors.left: parent.left
239             anchors.verticalCenter: parent.verticalCenter
240             anchors.leftMargin: 16
241             color: "#132335"
242
243             Text {
```

```
244         anchors.centerIn: parent
245         text: modelData.musteri.charAt(0).toUpperCase()
246         color: "#3b82f6"
247         font.family: "Segoe UI"
248         font.pixelSize: 18
249         font.bold: true
250     }
251 }
252
253 Column {
254     anchors.left: rozet.right
255     anchors.verticalCenter: parent.verticalCenter
256     anchors.leftMargin: 14
257     spacing: 4
258
259     Row {
260         spacing: 8
261
262         Text {
263             text: modelData.tarih
264             color: "#6b7280"
265             font.family: "Segoe UI"
266             font.pixelSize: 11
267         }
268
269         Rectangle {
270             width: strokeMetni.implicitWidth + 12
271             height: 16
272             radius: 8
273             color: "#2a2010"
```

```
274         anchors.verticalCenter: parent.verticalCenter
275
276         Text {
277             id: strokeMetni
278             anchors.centerIn: parent
279             text: modelData.strokeSayisi + " stroke"
280             color: "#e8a020"
281             font.family: "Segoe UI"
282             font.pixelSize: 10
283             font.bold: true
284         }
285     }
286 }
287
288 Text {
289     text: "Müşteri: " + modelData.musteri + " • Reçete: " + modelData.recete
290     color: "#dce8f5"
291     font.family: "Segoe UI"
292     font.pixelSize: 13
293     font.bold: true
294 }
295 }
296
297 Row {
298     anchors.right: parent.right
299     anchors.verticalCenter: parent.verticalCenter
300     anchors.rightMargin: 16
301     spacing: 8
302
303     Rectangle {
304         id: silButonu
```

```

305         width: 34
306         height: 34
307         radius: 6
308         color: silMouse.containsMouse ? "#3f1d24" : "#1a1420"
309         border.color: silMouse.containsMouse ? "#ef4444" : "#2a2030"
310         border.width: 1
311         Behavior on color { ColorAnimation { duration: 120 } }
312
313         Text {
314             anchors.centerIn: parent
315             text: "🗑"
316             font.pixelSize: 14
317         }
318
319         MouseArea {
320             id: silMouse
321             anchors.fill: parent
322             hoverEnabled: true
323             cursorShape: Qt.PointingHandCursor
324             onClicked: {
325                 silOnayPopup.hedefId = modelData.id
326                 silOnayPopup.hedefMusteri = modelData.musteri
327                 silOnayPopup.open()
328             }
329         }
330     }
331
332     Button {
333         anchors.verticalCenter: parent.verticalCenter
334         height: 34

```

```
335         text: "Görüntüle"
336         font.pixelSize: 12
337
338         onClicked: testSecildi(modelData.id)
339
340         background: Rectangle {
341             radius: 6
342             color: parent.hovered ? "#4f8cf7" : "#3b82f6"
343             Behavior on color { ColorAnimation { duration: 120 } }
344         }
345
346         contentItem: Text {
347             text: parent.text
348             color: "#ffffff"
349             font: parent.font
350             horizontalAlignment: Text.AlignHCenter
351             verticalAlignment: Text.AlignVCenter
352             leftPadding: 8
353             rightPadding: 8
354         }
355     }
356 }
357 }
358 }
359 }
360
361 Popup {
362     id: silOnayPopup
363     anchors.centerIn: parent
364     width: 380
365     padding: 24
```

```
366     modal: true
367     focus: true
368     closePolicy: Popup.CloseOnEscape
369
370     property int hedefId: -1
371     property string hedefMusteri: ""
372
373     onOpened: {
374         sifreGirisAlani.text = ""
375         hataMetni.visible = false
376         sifreGirisAlani.forceActiveFocus()
377     }
378
379     background: Rectangle {
380         color: "#0f1420"
381         radius: 14
382         border.color: "#1e2a3f"
383         border.width: 1
384     }
385
386     contentItem: Column {
387         width: silOnayPopup.availableWidth
388         spacing: 16
389
390         Text {
391             text: "Kaydı Sil"
392             color: "#ffffff"
393             font.family: "Segoe UI"
394             font.pixelSize: 17
395             font.bold: true
```

```
396         }
397
398         Text {
399             width: parent.width
400             wrapMode: Text.WordWrap
401             text: "" + silOnayPopup.hedefMusteri + "" ölçümünü silmek istediğinize emin misiniz? Bu
işlem geri alınamaz."
402             color: "#9ca3af"
403             font.family: "Segoe UI"
404             font.pixelSize: 13
405         }
406
407         Column {
408             width: parent.width
409             spacing: 6
410
411             Text {
412                 text: "$İFRE"
413                 color: "#6b7280"
414                 font.family: "Segoe UI"
415                 font.pixelSize: 10
416                 font.letterSpacing: 1
417             }
418
419             TextField {
420                 id: sifreGirisAlani
421                 width: parent.width
422                 height: 40
423                 echoMode: TextInput.Password
424                 placeholderText: "$ifrenizi girin"
425                 placeholderTextColor: "#4b5563"
```



```
426         color: "#ffffff"
427
428         font.pixelSize: 13
429
430         leftPadding: 12
431
432         verticalAlignment: TextInput.AlignVCenter
433
434         background: Rectangle {
435             color: "#0a0e17"
436
437             radius: 6
438
439             border.color: sifreGirisAlani.activeFocus ? "#3b82f6" : "#1e2a3f"
440
441             border.width: 1
442         }
443
444         Keys.onReturnPressed: silOnaylaButonu.clicked()
445     }
446
447
448
449     Text {
450         id: hataMetni
451
452         text: "Şifre hatalı, tekrar deneyin."
453
454         color: "#ef4444"
455
456         font.family: "Segoe UI"
457
458         font.pixelSize: 11
459
460         visible: false
461     }
462 }
463
464
465
466 Row {
467
468     width: parent.width
469
470     spacing: 10
471
472     layoutDirection: Qt.RightToLeft
473
474
475     Button {
476
477         id: silOnaylaButonu
478
479         width: (parent.width - 10) / 2
```

```
457         height: 40
458         text: "Sil"
459         font.pixelSize: 13
460         font.bold: true
461
462         onClicked: {
463             if (backend.sifreDogrula(sifreGirisAlani.text)) {
464                 database.olcumSil(silOnayPopup.hedefId)
465                 gecmisSayfasi.listeyiYenile()
466                 silOnayPopup.close()
467             } else {
468                 hataMetni.visible = true
469                 sifreGirisAlani.text = ""
470                 sifreGirisAlani.forceActiveFocus()
471             }
472         }
473
474         background: Rectangle {
475             radius: 8
476             color: parent.pressed ? "#b91c1c" : (parent.hovered ? "#ef4444" : "#dc2626")
477             Behavior on color { ColorAnimation { duration: 120 } }
478         }
479
480         contentItem: Text {
481             text: parent.text
482             color: "#ffffff"
483             font: parent.font
484             horizontalAlignment: Text.AlignHCenter
485             verticalAlignment: Text.AlignVCenter
486         }
```

```

487         }
488
489     Button {
490         width: (parent.width - 10) / 2
491         height: 40
492         text: "İptal"
493         font.pixelSize: 13
494
495         onClicked: silOnayPopup.close()
496
497         background: Rectangle {
498             radius: 8
499             color: parent.hovered ? "#162033" : "#0f1420"
500             border.color: "#1e2a3f"
501             border.width: 1
502             Behavior on color { ColorAnimation { duration: 120 } }
503         }
504
505         contentItem: Text {
506             text: parent.text
507             color: "#9ca3af"
508             font: parent.font
509             horizontalAlignment: Text.AlignHCenter
510             verticalAlignment: Text.AlignVCenter
511         }
512     }
513 }
514 }
515 }
516 }

```

qml/pages/KalibrasyonPage.qml

```
1 import QtQuick 6.7
2 import QtQuick.Controls 6.7
3 import QtQuick.Layouts 6.7
4 import "../components"
5
6 Rectangle {
7     color: "#0a0e17"
8
9     property int seciliSensor: -1
10    property int loadCellAdimi: 0
11
12    property bool egimKayitliVar: false
13    property string egimSonTarih: ""
14    property bool egimBildirimGoster: false
15    property bool egimBildirimBasarili: true
16    property string egimBildirimMesaj: ""
17
18    property bool loadCellKayitliVar: false
19    property string loadCellSonTarih: ""
20    property int loadCellNoktaSayisi: 0
21    property var loadCellNoktalari: []
22    onLoadCellNoktalariChanged: loadCellAdimi = loadCellNoktalari.length
23    property int loadCellToplamNokta: 10
24    property bool loadCellSayiSecildi: false
25    property int loadCellDuzenlemeIndex: -1
26
27    property bool loadCellAraEkranGoster: false
28    property bool loadCellGoruntuleModu: false
29    property bool loadCellTamDuzenlemeModu: false
30    property bool loadCellYenidenOnayGoster: false
```

```
31     property bool loadCellTamamlandiOnayGoster: false
32
33     property bool loadCellBildirimGoster: false
34     property bool loadCellBildirimBasarili: true
35     property string loadCellBildirimMesaj: ""
36
37     property string mesafeSonTarih: ""
38
39     property bool mesafeOlcumKayitliVar: false
40     property int mesafeOlcumNoktaSayisi: 0
41     property var mesafeNoktalari: []
42     property int mesafeToplamNokta: 10
43     property bool mesafeSayiSecildi: false
44     property int mesafeDuzenlemeIndex: -1
45
46     property bool mesafeAraEkranGoster: false
47     property bool mesafeGoruntuleModu: false
48     property bool mesafeTamDuzenlemeModu: false
49     property bool mesafeYenidenOnayGoster: false
50     property bool mesafeTamamlandiOnayGoster: false
51
52     property bool mesafeBildirimGoster: false
53     property bool mesafeBildirimBasarili: true
54     property string mesafeBildirimMesaj: ""
55
56     function loadCellSiralıKopya(dizi) {
57         var kopya = dizi.slice()
58         kopya.sort(function(a, b) { return a.hamDeger - b.hamDeger })
59         return kopya
60     }
```

```

61
62 function loadCellKaydetVeBildir() {
63     var sirali = loadCellSiraliKopya(loadCellNoktalari)
64     var basarili = database.loadCellNoktalariKaydet(sirali)
65     loadCellBildirimBasarili = basarili
66     loadCellBildirimMesaj = basarili ? "✓ Kalibrasyon kaydedildi" : "✗ Kalibrasyon kaydedilemedi, tekrar
deneyin"
67     loadCellBildirimGoster = true
68     loadCellBildirimTimer.restart()
69 }
70
71 Timer {
72     id: loadCellBildirimTimer
73     interval: loadCellBildirimBasarili ? 1800 : 2500
74     repeat: false
75     onTriggered: {
76         loadCellBildirimGoster = false
77         if (loadCellBildirimBasarili) {
78             seciliSensor = -1
79         }
80     }
81 }
82
83 function mesafeSiraliKopya(dizi) {
84     var kopya = dizi.slice()
85     kopya.sort(function(a, b) { return a.hamDeger - b.hamDeger })
86     return kopya
87 }
88
89 function mesafeKaydetVeBildir() {
90     var sirali = mesafeSiraliKopya(mesafeNoktalari)

```

```

91     var basarili = database.mesafeNoktalarKaydet(sirali)
92     if (basarili) {
93         wifiManager.kalibrasyonYenidenYukle()
94     }
95     mesafeBildirimBasarili = basarili
96     mesafeBildirimMesaj = basarili ? "✓ Kalibrasyon kaydedildi" : "✗ Kalibrasyon kaydedilemedi, tekrar
deneyin"
97     mesafeBildirimGoster = true
98     mesafeBildirimTimer.restart()
99 }
100
101 Timer {
102     id: mesafeBildirimTimer
103     interval: mesafeBildirimBasarili ? 1800 : 2500
104     repeat: false
105     onTriggered: {
106         mesafeBildirimGoster = false
107         if (mesafeBildirimBasarili) {
108             seciliSensor = -1
109         }
110     }
111 }
112
113 function egimSifirlaVeBildir() {
114     var biasX = sensorManager.hamAccelX
115     var biasY = sensorManager.hamAccelY
116     var basarili = database.egimKalibrasyonuKaydet(biasX, biasY, 0.0, 1.0, 1.0, 1.0)
117     if (basarili) {
118         wifiManager.kalibrasyonYenidenYukle()
119     }
120     egimBildirimBasarili = basarili

```

```
121         egimBildirimMesaj = basarili ? "✓ Kalibrasyon kaydedildi" : "✗ Kalibrasyon kaydedilemedi, tekrar
deneyin"

122         egimBildirimGoster = true

123         egimBildirimTimer.restart()

124     }

125

126     Timer {

127         id: egimBildirimTimer

128         interval: egimBildirimBasarili ? 1800 : 2500

129         repeat: false

130         onTriggered: {

131             egimBildirimGoster = false

132             if (egimBildirimBasarili) {

133                 seciliSensor = -1

134             }

135         }

136     }

137

138     // Sensör kartlarının "Kalibre Edildi" rozeti ve tarihi kalıcı depodan (SQLite)

139     // okunur; sadece o an kalibrasyon yapılıncaya değil, sayfa her açıldığında yeniden

140     // yüklenmeli ki uygulama yeniden başlatıldığında da rozet doğru görünsün.

141     function kalibrasyonDurumunuYukle() {

142         var egimKal = database.egimKalibrasyonuGetir()

143         egimKayitliVar = egimKal.mevcut

144         egimSonTarih = egimKal.mevcut ? egimKal.tarih : ""

145

146         var lcNoktalar = database.loadCellNoktalarıGetir()

147         loadCellKayitliVar = lcNoktalar.length > 0

148         loadCellNoktaSayisi = lcNoktalar.length

149         loadCellSonTarih = loadCellKayitliVar ? database.kalibrasyonTarihiGetir("loadcell") : ""

150     }
```



```
151         var mNoktalar = database.mesafeNoktalarGetir()
152         mesafeOlcumKayitliVar = mNoktalar.length > 0
153         mesafeOlcumNoktaSayisi = mNoktalar.length
154         mesafeSonTarih = mesafeOlcumKayitliVar ? database.kalibrasyonTarihiGetir("mesafe_olcum") : ""
155     }
156
157     Component.onCompleted: kalibrasyonDurumunuYukle()
158
159     onVisibleChanged: {
160         if (visible) {
161             kalibrasyonDurumunuYukle()
162         }
163     }
164
165     onSeciliSensorChanged: {
166         if (seciliSensor === 0) {
167             var egimKal = database.egimKalibrasyonuGetir()
168             egimKayitliVar = egimKal.mevcut
169             egimSonTarih = egimKal.mevcut ? egimKal.tarih : ""
170             egimBildirimGoster = false
171         } else if (seciliSensor === 1) {
172             var noktalar = database.loadCellNoktalarGetir()
173             loadCellKayitliVar = noktalar.length > 0
174             loadCellNoktaSayisi = noktalar.length
175             loadCellSonTarih = loadCellKayitliVar ? database.kalibrasyonTarihiGetir("loadcell") : ""
176             loadCellNoktalar = []
177             loadCellSayiSecildi = false
178             loadCellDuzenlemeIndex = -1
179             loadCellTamDuzenlemeModu = false
180             loadCellGoruntuleModu = false
181             loadCellYenidenOnayGoster = false
```

```

182         loadCellTamamlandiOnayGoster = false
183         loadCellBildirimGoster = false
184         loadCellAraEkranGoster = loadCellKayitliVar
185     } else if (seciliSensor === 2) {
186         var mNoktalar = database.mesafeNoktalarGetir()
187         mesafeOlcumKayitliVar = mNoktalar.length > 0
188         mesafeOlcumNoktaSayisi = mNoktalar.length
189         mesafeSonTarih = mesafeOlcumKayitliVar ? database.kalibrasyonTarihiGetir("mesafe_olcum") : ""
190         mesafeNoktalar = []
191         mesafeSayiSecildi = false
192         mesafeDuzenlemeIndex = -1
193         mesafeTamDuzenlemeModu = false
194         mesafeGoruntuleModu = false
195         mesafeYenidenOnayGoster = false
196         mesafeTamamlandiOnayGoster = false
197         mesafeBildirimGoster = false
198         mesafeAraEkranGoster = mesafeOlcumKayitliVar
199     }
200 }
201
202 StackLayout {
203     anchors.fill: parent
204     currentIndex: seciliSensor === -1 ? 0 : 1
205
206     Item {
207         Text {
208             anchors.top: parent.top
209             anchors.left: parent.left
210             anchors.margins: 20
211             text: "KALİBRASYON – Sensör Seçin"

```

```
212         color: "#6b7280"
213         font.family: "Segoe UI"
214         font.pixelSize: 12
215         font.bold: true
216         font.letterSpacing: 1
217     }
218
219     Row {
220         anchors.centerIn: parent
221         spacing: 24
222
223         Rectangle {
224             id: egimKarti
225             width: 350
226             height: 450
227             radius: 12
228             color: "#0f1420"
229             border.color: egimKarti.hovered ? "#14b8a6" : "#1e2a3f"
230             border.width: 1
231             property bool hovered: false
232             scale: hovered ? 1.03 : 1.0
233             Behavior on border.color { ColorAnimation { duration: 120 } }
234             Behavior on scale { NumberAnimation { duration: 120; easing.type: Easing.OutQuad } }
235
236             Column {
237                 anchors.centerIn: parent
238                 spacing: 14
239
240                 Item {
241                     anchors.horizontalCenter: parent.horizontalCenter
242                     width: 210
```

```
243             height: 150
244
245             Image {
246                 anchors.centerIn: parent
247                 width: parent.width
248                 height: parent.height
249                 source: "../assets/egim_sensor.png"
250                 fillMode: Image.PreserveAspectFit
251             }
252         }
253
254         Text { anchors.horizontalCenter: parent.horizontalCenter; text: "Eğim Sensörü";
color: "#dce8f5"; font.family: "Segoe UI"; font.pixelSize: 24; font.bold: true }
255
256         Text { anchors.horizontalCenter: parent.horizontalCenter; text: "Yatay hizalama
kalibrasyonu"; color: "#6b7280"; font.family: "Segoe UI"; font.pixelSize: 14 }
257
258         Rectangle {
259             anchors.horizontalCenter: parent.horizontalCenter
260             visible: egimKayitliVar
261             width: kaliMetni1.implicitWidth + 16
262             height: 20
263             radius: 10
264             color: "#123321"
265
266             Text {
267                 id: kaliMetni1
268                 anchors.centerIn: parent
269                 text: "✓ Kalibre Edildi: " + egimSonTarih
270                 color: "#4ade80"
271                 font.pixelSize: 10
272                 font.bold: true
```

```

272         }
273     }
274 }
275
276 MouseArea {
277     anchors.fill: parent
278     hoverEnabled: true
279     cursorShape: Qt.PointingHandCursor
280     onEntered: egimKarti.hovered = true
281     onExited: egimKarti.hovered = false
282     onClicked: seciliSensor = 0
283 }
284 }
285
286 Rectangle {
287     id: loadCellKarti
288     width: 350
289     height: 450
290     radius: 12
291     color: "#0f1420"
292     border.color: loadCellKarti.hovered ? "#3b82f6" : "#1e2a3f"
293     border.width: 1
294     property bool hovered: false
295     scale: hovered ? 1.03 : 1.0
296     Behavior on border.color { ColorAnimation { duration: 120 } }
297     Behavior on scale { NumberAnimation { duration: 120; easing.type: Easing.OutQuad } }
298
299     Column {
300         anchors.centerIn: parent
301         spacing: 14
302

```

```

303         Item {
304             anchors.horizontalCenter: parent.horizontalCenter
305             width: 210
306             height: 150
307
308             Image {
309                 anchors.centerIn: parent
310                 width: parent.width
311                 height: parent.height
312                 source: "../assets/loadcell.png"
313                 fillMode: Image.PreserveAspectFit
314             }
315         }
316
317         Text { anchors.horizontalCenter: parent.horizontalCenter; text: "Load Cell"; color:
"#dce8f5"; font.family: "Segoe UI"; font.pixelSize: 24; font.bold: true }
318         Text { anchors.horizontalCenter: parent.horizontalCenter; text: "Basınç sensörü
kalibrasyonu"; color: "#6b7280"; font.family: "Segoe UI"; font.pixelSize: 14 }
319
320         Rectangle {
321             anchors.horizontalCenter: parent.horizontalCenter
322             visible: loadCellKayitliVar
323             width: kaliMetni2.implicitWidth + 16
324             height: 20
325             radius: 10
326             color: "#123321"
327
328             Text {
329                 id: kaliMetni2
330                 anchors.centerIn: parent
331                 text: "✓ Kalibre Edildi: " + loadCellSonTarih

```

```
332             color: "#4ade80"
333             font.pixelSize: 10
334             font.bold: true
335         }
336     }
337 }
338
339 MouseArea {
340     anchors.fill: parent
341     hoverEnabled: true
342     cursorShape: Qt.PointingHandCursor
343     onEntered: loadCellKarti.hovered = true
344     onExited: loadCellKarti.hovered = false
345     onClicked: seciliSensor = 1
346 }
347 }
348
349 Rectangle {
350     id: mesafeKarti
351     width: 350
352     height: 450
353     radius: 12
354     color: "#0f1420"
355     border.color: mesafeKarti.hovered ? "#9333ea" : "#1e2a3f"
356     border.width: 1
357     property bool hovered: false
358     scale: hovered ? 1.03 : 1.0
359     Behavior on border.color { ColorAnimation { duration: 120 } }
360     Behavior on scale { NumberAnimation { duration: 120; easing.type: Easing.OutQuad } }
361
362     Column {
```

```
363         anchors.centerIn: parent
364
365         spacing: 14
366
367         Item {
368             anchors.horizontalCenter: parent.horizontalCenter
369             width: 210
370             height: 150
371
372             Image {
373                 anchors.centerIn: parent
374                 width: parent.width
375                 height: parent.height
376                 source: "../assets/mesafe_sensor.png"
377                 fillMode: Image.PreserveAspectFit
378             }
379
380             Text { anchors.horizontalCenter: parent.horizontalCenter; text: "Mesafe Sensörü";
color: "#dce8f5"; font.family: "Segoe UI"; font.pixelSize: 24; font.bold: true }
381
382             Text { anchors.horizontalCenter: parent.horizontalCenter; text: "Konum ölçümü
kalibrasyonu"; color: "#6b7280"; font.family: "Segoe UI"; font.pixelSize: 14 }
383
384             Rectangle {
385                 anchors.horizontalCenter: parent.horizontalCenter
386                 visible: mesafeOlcumKayitliVar
387                 width: kaliMetni3.implicitWidth + 16
388                 height: 20
389                 radius: 10
390                 color: "#123321"
391
392                 Text {
```



```
392         id: kaliMetni3
393         anchors.centerIn: parent
394         text: "✓ Kalibre Edildi: " + mesafeSonTarih
395         color: "#4ade80"
396         font.pixelSize: 10
397         font.bold: true
398     }
399 }
400 }
401
402 MouseArea {
403     anchors.fill: parent
404     hoverEnabled: true
405     cursorShape: Qt.PointingHandCursor
406     onEntered: mesafeKarti.hovered = true
407     onExited: mesafeKarti.hovered = false
408     onClicked: seciliSensor = 2
409 }
410 }
411 }
412 }
413
414 Item {
415     Row {
416         anchors.top: parent.top
417         anchors.left: parent.left
418         anchors.margins: 20
419         spacing: 12
420
421         Button {
422             text: "← Geri"
```

```
423         font.pixelSize: 13
424
425         background: Rectangle {
426             radius: 6
427             color: parent.hovered ? "#1e2a3f" : "#0f1420"
428             border.color: "#1e2a3f"
429             border.width: 1
430             Behavior on color { ColorAnimation { duration: 120 } }
431         }
432
433         contentItem: Text {
434             text: parent.text
435             color: "#dce8f5"
436             font: parent.font
437             horizontalAlignment: Text.AlignHCenter
438             verticalAlignment: Text.AlignVCenter
439             leftPadding: 10
440             rightPadding: 10
441         }
442
443         onClicked: seciliSensor = -1
444     }
445
446     Rectangle {
447         width: 8
448         height: 8
449         radius: 4
450         anchors.verticalCenter: parent.verticalCenter
451         color: {
452             if (seciliSensor === 0) return "#14b8a6"
```

```
453         if (seciliSensor === 1) return "#3b82f6"
454         if (seciliSensor === 2) return "#9333ea"
455         return "#6b7280"
456     }
457 }
458
459 Text {
460     anchors.verticalCenter: parent.verticalCenter
461     text: {
462         if (seciliSensor === 0) return "Eğim Sensörü Kalibrasyonu"
463         if (seciliSensor === 1) return "Load Cell Kalibrasyonu"
464         if (seciliSensor === 2) return "Mesafe Sensörü Kalibrasyonu"
465         return ""
466     }
467     color: "#dce8f5"
468     font.family: "Segoe UI"
469     font.pixelSize: 16
470     font.bold: true
471 }
472 }
473
474 Item {
475     anchors.top: parent.top
476     anchors.left: parent.left
477     anchors.right: parent.right
478     anchors.bottom: parent.bottom
479     anchors.topMargin: 70
480     visible: seciliSensor === 0
481
482     Column {
483         anchors.centerIn: parent
```

```
484         spacing: 20
485
486
487     Rectangle {
488         anchors.horizontalCenter: parent.horizontalCenter
489         visible: egimKayitliVar
490         width: egimSonKaliMetni.implicitWidth + 20
491         height: 26
492         radius: 13
493         color: "#0f1420"
494         border.color: "#1e2a3f"
495         border.width: 1
496
497         Text {
498             id: egimSonKaliMetni
499             anchors.centerIn: parent
500             text: "Son kalibrasyon: " + egimSonTarih
501             color: "#6b7280"
502             font.pixelSize: 11
503         }
504     }
505
506     Text {
507         width: parent.width
508         horizontalAlignment: Text.AlignHCenter
509         wrapMode: Text.WordWrap
510         text: "Cihazı düz ve sabit bir yüzeye yerleştirin"
511         color: "#dce8f5"
512         font.family: "Segoe UI"
513         font.pixelSize: 14
```

```

514         }
515
516         SuTerazisi {
517             anchors.horizontalCenter: parent.horizontalCenter
518             egimX: sensorManager.egimX
519             egimY: sensorManager.egimY
520         }
521
522         Rectangle {
523             anchors.horizontalCenter: parent.horizontalCenter
524             width: 220
525             height: 72
526             radius: 12
527             color: "#0f2e2a"
528             border.color: "#14b8a6"
529             border.width: 1
530
531             Text {
532                 anchors.centerIn: parent
533                 text: "X: " + sensorManager.egimX.toFixed(1) + "°   Y: " +
sensorManager.egimY.toFixed(1) + "°"
534                 color: "#2dd4bf"
535                 font.family: "Segoe UI"
536                 font.pixelSize: 22
537                 font.bold: true
538             }
539         }
540
541         Text {
542             width: parent.width
543             horizontalAlignment: Text.AlignHCenter

```

```
544         wrapMode: Text.WordWrap
545
546         visible: !wifiManager.baglandi
547
548         text: "Cihaz bağlı değil – kalibrasyon kaydedilemez"
549
550         color: "#f87171"
551
552         font.family: "Segoe UI"
553
554         font.pixelSize: 11
555     }
556
557     Button {
558
559         width: parent.width
560
561         height: 44
562
563         text: "Sıfırla (Zero)"
564
565         font.pixelSize: 14
566
567         font.bold: true
568
569         enabled: wifiManager.baglandi
570
571
572         onClicked: egimSifirlaVeBildir()
573
574
575         background: Rectangle {
576
577             radius: 8
578
579             color: parent.enabled ? (parent.hovered ? "#2dd4bf" : "#14b8a6") : "#1e2a3f"
580
581             Behavior on color { ColorAnimation { duration: 120 } }
582         }
583
584         contentItem: Text {
585
586             text: parent.text
587
588             color: "#ffffff"
589
590             font: parent.font
591
592             horizontalAlignment: Text.AlignHCenter
593
594             verticalAlignment: Text.AlignVCenter
595         }
596     }
```

```

575         }
576     }
577
578     Rectangle {
579         id: egimBildirimKutusu
580         anchors.top: parent.top
581         anchors.horizontalCenter: parent.horizontalCenter
582         anchors.topMargin: 12
583         z: 100
584         width: egimBildirimMetni.implicitWidth + 24
585         height: 32
586         radius: 16
587         color: egimBildirimBasarili ? "#123321" : "#3a1414"
588         opacity: egimBildirimGoster ? 1 : 0
589         visible: opacity > 0
590         Behavior on opacity { NumberAnimation { duration: 200 } }
591
592         Text {
593             id: egimBildirimMetni
594             anchors.centerIn: parent
595             text: egimBildirimMesaj
596             color: egimBildirimBasarili ? "#4ade80" : "#f87171"
597             font.family: "Segoe UI"
598             font.pixelSize: 12
599             font.bold: true
600         }
601     }
602 }
603
604 Item {

```

```
605         anchors.top: parent.top
606         anchors.left: parent.left
607         anchors.right: parent.right
608         anchors.bottom: parent.bottom
609         anchors.topMargin: 70
610         visible: seciliSensor === 1
611
612     Column {
613         anchors.centerIn: parent
614         spacing: 24
615         width: 420
616         visible: loadCellAraEkranGoster
617
618         Rectangle {
619             anchors.horizontalCenter: parent.horizontalCenter
620             width: araEkranBadge.implicitWidth + 20
621             height: 26
622             radius: 13
623             color: "#0f1420"
624             border.color: "#1e2a3f"
625             border.width: 1
626
627             Text {
628                 id: araEkranBadge
629                 anchors.centerIn: parent
630                 text: "Kayıtlı: " + loadCellNoktaSayisi + " nokta"
631                 color: "#6b7280"
632                 font.pixelSize: 11
633             }
634         }
635     }
```



```
636         Text {
637             width: parent.width
638             horizontalAlignment: Text.AlignHCenter
639             wrapMode: Text.WordWrap
640             text: "Bu sensör için kayıtlı bir kalibrasyon bulundu"
641             color: "#dce8f5"
642             font.family: "Segoe UI"
643             font.pixelSize: 16
644             font.bold: true
645         }
646
647     Row {
648         anchors.horizontalCenter: parent.horizontalCenter
649         spacing: 20
650
651         Rectangle {
652             id: goruntuleKarti
653             width: 190
654             height: 180
655             radius: 14
656             color: "#0f1420"
657             border.color: goruntuleKarti.hovered ? "#3b82f6" : "#1e2a3f"
658             border.width: 1
659             property bool hovered: false
660             Behavior on border.color { ColorAnimation { duration: 120 } }
661
662             Column {
663                 anchors.centerIn: parent
664                 spacing: 12
665                 width: parent.width - 28
```

```

666
667 Canvas {
668     anchors.horizontalCenter: parent.horizontalCenter
669     width: 32
670     height: 32
671     onPaint: {
672         var ctx = getContext("2d")
673         ctx.clearRect(0, 0, width, height)
674         ctx.lineWidth = 1.8
675         ctx.lineCap = "round"
676         ctx.lineJoin = "round"
677         ctx.strokeStyle = "#3b82f6"
678
679         ctx.beginPath()
680         ctx.moveTo(3, 16)
681         ctx.bezierCurveTo(7.5, 8.5, 12.5, 6.8, 16, 6.8)
682         ctx.bezierCurveTo(19.5, 6.8, 24.5, 8.5, 29, 16)
683         ctx.bezierCurveTo(24.5, 23.5, 19.5, 25.2, 16, 25.2)
684         ctx.bezierCurveTo(12.5, 25.2, 7.5, 23.5, 3, 16)
685         ctx.stroke()
686
687         ctx.beginPath()
688         ctx.arc(16, 16, 4.4, 0, Math.PI * 2, false)
689         ctx.stroke()
690     }
691 }
692
693 Text {
694     width: parent.width
695     horizontalAlignment: Text.AlignHCenter
696     wrapMode: Text.WordWrap

```

```
697         text: "Mevcut Kalibrasyonu Görüntüle"
698         color: "#dce8f5"
699         font.family: "Segoe UI"
700         font.pixelSize: 13
701         font.bold: true
702     }
703
704     Text {
705         width: parent.width
706         horizontalAlignment: Text.AlignHCenter
707         wrapMode: Text.WordWrap
708         text: "Kayıtlı noktaları incele"
709         color: "#6b7280"
710         font.family: "Segoe UI"
711         font.pixelSize: 11
712     }
713 }
714
715 MouseArea {
716     anchors.fill: parent
717     hoverEnabled: true
718     cursorShape: Qt.PointingHandCursor
719     onEntered: goruntuleKarti.hovered = true
720     onExited: goruntuleKarti.hovered = false
721     onClicked: {
722         loadCellNoktalari = database.loadCellNoktalariGetir()
723         loadCellAraEkranGoster = false
724         loadCellGoruntuleModu = true
725         loadCellDuzenlemeIndex = -1
726     }
```

```

727         }
728     }
729
730     Rectangle {
731         id: yenidenKarti
732         width: 190
733         height: 180
734         radius: 14
735         color: "#0f1420"
736         border.color: yenidenKarti.hovered ? "#9ca3af" : "#1e2a3f"
737         border.width: 1
738         property bool hovered: false
739         Behavior on border.color { ColorAnimation { duration: 120 } }
740
741         Column {
742             anchors.centerIn: parent
743             spacing: 12
744             width: parent.width - 28
745
746             Canvas {
747                 anchors.horizontalCenter: parent.horizontalCenter
748                 width: 32
749                 height: 32
750                 onPaint: {
751                     var ctx = getContext("2d")
752                     ctx.clearRect(0, 0, width, height)
753                     ctx.strokeStyle = "#9ca3af"
754                     ctx.fillStyle = "#9ca3af"
755                     ctx.lineWidth = 3.4
756                     ctx.lineCap = "butt"
757

```

```
758         function yay0ku(startDeg, endDeg) {
759             var s = startDeg * Math.PI / 180
760             var e = endDeg * Math.PI / 180
761             var cx = 16, cy = 16, r = 10.5
762
763             ctx.beginPath()
764             ctx.arc(cx, cy, r, s, e, false)
765             ctx.stroke()
766
767             var ex = cx + r * Math.cos(e)
768             var ey = cy + r * Math.sin(e)
769             var tx = -Math.sin(e)
770             var ty = Math.cos(e)
771             var px = -ty
772             var py = tx
773             var geri = 6.2
774             var yaricap = 3.6
775             var bx = ex - tx * geri
776             var by = ey - ty * geri
777
778             ctx.beginPath()
779             ctx.moveTo(ex, ey)
780             ctx.lineTo(bx + px * yaricap, by + py * yaricap)
781             ctx.lineTo(bx - px * yaricap, by - py * yaricap)
782             ctx.closePath()
783             ctx.fill()
784         }
785
786         yay0ku(195, 345)
787         yay0ku(15, 165)
```

```
788         }
789     }
790
791     Text {
792         width: parent.width
793         horizontalAlignment: Text.AlignHCenter
794         wrapMode: Text.WordWrap
795         text: "Yeniden Kalibre Et"
796         color: "#dce8f5"
797         font.family: "Segoe UI"
798         font.pixelSize: 13
799         font.bold: true
800     }
801
802     Text {
803         width: parent.width
804         horizontalAlignment: Text.AlignHCenter
805         wrapMode: Text.WordWrap
806         text: "Sıfırdan yeni ölçüm al"
807         color: "#6b7280"
808         font.family: "Segoe UI"
809         font.pixelSize: 11
810     }
811 }
812
813 MouseArea {
814     anchors.fill: parent
815     hoverEnabled: true
816     cursorShape: Qt.PointingHandCursor
817     onEntered: yenidenKarti.hovered = true
818     onExited: yenidenKarti.hovered = false
```

```
819             onClicked: loadCellYenidenOnayGoster = true
820         }
821     }
822 }
823 }
824
825 Column {
826     anchors.centerIn: parent
827     spacing: 16
828     width: 340
829     visible: loadCellGoruntuleModu
830
831     Text {
832         text: "KAYITLI KALİBRASYON NOKTALARI"
833         color: "#6b7280"
834         font.family: "Segoe UI"
835         font.pixelSize: 11
836         font.bold: true
837         font.letterSpacing: 1
838     }
839
840     Rectangle {
841         width: parent.width
842         height: 320
843         radius: 14
844         color: "#0f1420"
845         border.color: "#1e2a3f"
846         border.width: 1
847
848         ListView {
```

```
849         anchors.fill: parent
850
851         anchors.margins: 16
852
853         clip: true
854
855         spacing: 8
856
857
858
859
860
861
862
863
864
865
866
867
868
869
870
871
872
873
874
875
876
877
878
879         delegate: Rectangle {
880
881             width: ListView.view.width
882
883             height: 48
884
885             radius: 8
886
887             color: "#0a0e17"
888
889             border.color: "#1e2a3f"
890
891             border.width: 1
892
893
894
895             Row {
896
897                 anchors.fill: parent
898
899                 anchors.leftMargin: 12
900
901                 anchors.rightMargin: 12
902
903                 spacing: 8
904
905
906
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996
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998
999
```



```
880         anchors.centerIn: parent
881         text: index + 1
882         color: "#3b82f6"
883         font.pixelSize: 10
884         font.bold: true
885     }
886 }
887
888 Column {
889     anchors.verticalCenter: parent.verticalCenter
890     spacing: 1
891
892     Text {
893         text: modelData.hedefKg.toFixed(2) + " kg"
894         color: "#dce8f5"
895         font.family: "Segoe UI"
896         font.pixelSize: 13
897         font.bold: true
898     }
899
900     Text {
901         text: "ADC: " + modelData.hamDeger.toFixed(0)
902         color: "#6b7280"
903         font.family: "Segoe UI"
904         font.pixelSize: 10
905     }
906 }
907 }
908 }
909 }
```

```

910     }
911
912     Button {
913         width: parent.width
914         height: 46
915         text: "Düzenle"
916         font.pixelSize: 14
917         font.bold: true
918
919         onClicked: {
920             loadCellToplamNokta = loadCellNoktalari.length
921             loadCellGoruntuleModu = false
922             loadCellTamDuzenlemeModu = true
923             loadCellSayiSecildi = true
924             loadCellDuzenlemeIndex = -1
925         }
926
927         background: Rectangle {
928             radius: 8
929             color: parent.hovered ? "#4f8cf7" : "#3b82f6"
930             Behavior on color { ColorAnimation { duration: 120 } }
931         }
932
933         contentItem: Text {
934             text: parent.text
935             color: "#ffffff"
936             font: parent.font
937             horizontalAlignment: Text.AlignHCenter
938             verticalAlignment: Text.AlignVCenter
939         }
940     }

```

```
941     }
942
943     Rectangle {
944         anchors.fill: parent
945         color: "#000000b3"
946         visible: loadCellYenidenOnayGoster
947         z: 200
948
949         MouseArea { anchors.fill: parent }
950
951         Rectangle {
952             anchors.centerIn: parent
953             width: 360
954             height: 200
955             radius: 14
956             color: "#0f1420"
957             border.color: "#1e2a3f"
958             border.width: 1
959
960             Column {
961                 anchors.centerIn: parent
962                 spacing: 18
963                 width: parent.width - 40
964
965                 Text {
966                     width: parent.width
967                     horizontalAlignment: Text.AlignHCenter
968                     wrapMode: Text.WordWrap
969                     text: "Mevcut kalibrasyon silinecek, emin misiniz?"
970                     color: "#dce8f5"
```

```
971         font.family: "Segoe UI"
972         font.pixelSize: 14
973         font.bold: true
974     }
975
976     Row {
977         anchors.horizontalCenter: parent.horizontalCenter
978         spacing: 12
979
980         Button {
981             width: 150
982             height: 42
983             text: "Vazgeç"
984
985             onClicked: loadCellYenidenOnayGoster = false
986
987             background: Rectangle {
988                 radius: 8
989                 color: parent.hovered ? "#1e2a3f" : "transparent"
990                 border.color: "#1e2a3f"
991                 border.width: 1
992                 Behavior on color { ColorAnimation { duration: 120 } }
993             }
994
995             contentItem: Text {
996                 text: parent.text
997                 color: "#9ca3af"
998                 font: parent.font
999                 horizontalAlignment: Text.AlignHCenter
1000                 verticalAlignment: Text.AlignVCenter
1001             }
```

```

1002     }
1003
1004     Button {
1005         width: 150
1006         height: 42
1007         text: "Evet, Yeniden Başla"
1008         font.bold: true
1009
1010         onClicked: {
1011             loadCellYenidenOnayGoster = false
1012             loadCellAraEkranGoster = false
1013             loadCellGoruntuleModu = false
1014             loadCellTamDuzenlemeModu = false
1015             loadCellNoktalari = []
1016             loadCellSayiSecildi = false
1017             loadCellDuzenlemeIndex = -1
1018         }
1019
1020         background: Rectangle {
1021             radius: 8
1022             color: parent.hovered ? "#b91c1c" : "#991b1b"
1023             Behavior on color { ColorAnimation { duration: 120 } }
1024         }
1025
1026         contentItem: Text {
1027             text: parent.text
1028             color: "#ffffff"
1029             font: parent.font
1030             horizontalAlignment: Text.AlignHCenter
1031             verticalAlignment: Text.AlignVCenter

```

```

1032         }
1033     }
1034 }
1035 }
1036 }
1037 }
1038
1039 Rectangle {
1040     anchors.fill: parent
1041     color: "#000000b3"
1042     visible: loadCellTamamlandiOnayGoster
1043     z: 200
1044
1045     MouseArea { anchors.fill: parent }
1046
1047     Rectangle {
1048         anchors.centerIn: parent
1049         width: 380
1050         height: 220
1051         radius: 14
1052         color: "#0f1420"
1053         border.color: "#16a34a"
1054         border.width: 1
1055
1056         Column {
1057             anchors.centerIn: parent
1058             spacing: 18
1059             width: parent.width - 40
1060
1061             Column {
1062                 width: parent.width

```

```
1063         spacing: 6
1064
1065         Text {
1066             width: parent.width
1067             horizontalAlignment: Text.AlignHCenter
1068             text: "Kalibrasyon Tamamlandı"
1069             color: "#4ade80"
1070             font.family: "Segoe UI"
1071             font.pixelSize: 17
1072             font.bold: true
1073         }
1074
1075         Text {
1076             width: parent.width
1077             horizontalAlignment: Text.AlignHCenter
1078             wrapMode: Text.WordWrap
1079             text: "Toplam " + loadCellToplamNokta + " nokta girildi. Bu kalibrasyonu
kaydetmek istiyor musunuz?"
1080             color: "#9ca3af"
1081             font.family: "Segoe UI"
1082             font.pixelSize: 13
1083         }
1084     }
1085
1086     Row {
1087         anchors.horizontalCenter: parent.horizontalCenter
1088         spacing: 12
1089
1090         Button {
1091             width: 150
1092             height: 42
```

```

1093         text: "Noktaları Kontrol Et"
1094
1095         onClicked: loadCellTamamlandiOnayGoster = false
1096
1097         background: Rectangle {
1098             radius: 8
1099             color: parent.hovered ? "#1e2a3f" : "transparent"
1100             border.color: "#1e2a3f"
1101             border.width: 1
1102             Behavior on color { ColorAnimation { duration: 120 } }
1103         }
1104
1105         contentItem: Text {
1106             text: parent.text
1107             color: "#9ca3af"
1108             font: parent.font
1109             horizontalAlignment: Text.AlignHCenter
1110             verticalAlignment: Text.AlignVCenter
1111         }
1112     }
1113
1114     Button {
1115         width: 150
1116         height: 42
1117         text: "Kaydet ve Bitir"
1118         font.bold: true
1119
1120         onClicked: {
1121             loadCellTamamlandiOnayGoster = false
1122             loadCellKaydetVeBildir()

```



```

1123         }
1124
1125         background: Rectangle {
1126             radius: 8
1127             color: parent.hovered ? "#22c55e" : "#16a34a"
1128             Behavior on color { ColorAnimation { duration: 120 } }
1129         }
1130
1131         contentItem: Text {
1132             text: parent.text
1133             color: "#ffffff"
1134             font: parent.font
1135             horizontalAlignment: Text.AlignHCenter
1136             verticalAlignment: Text.AlignVCenter
1137         }
1138     }
1139 }
1140 }
1141 }
1142 }
1143
1144 Rectangle {
1145     id: loadCellBildirirKutusu
1146     anchors.top: parent.top
1147     anchors.horizontalCenter: parent.horizontalCenter
1148     anchors.topMargin: 12
1149     z: 100
1150     width: loadCellBildirirMetni.implicitWidth + 24
1151     height: 32
1152     radius: 16
1153     color: loadCellBildirirBasarili ? "#123321" : "#3a1414"

```

```
1154         opacity: loadCellBildirimGoster ? 1 : 0
1155
1156         visible: opacity > 0
1157
1158         Behavior on opacity { NumberAnimation { duration: 200 } }
1159
1160         Text {
1161             id: loadCellBildirimMetni
1162             anchors.centerIn: parent
1163             text: loadCellBildirimMesaj
1164             color: loadCellBildirimBasarili ? "#4ade80" : "#f87171"
1165             font.family: "Segoe UI"
1166             font.pixelSize: 12
1167             font.bold: true
1168         }
1169     }
1170
1171     Column {
1172         anchors.centerIn: parent
1173         spacing: 24
1174         width: 360
1175         visible: !loadCellSayiSecildi && !loadCellAraEkranGoster && !loadCellGoruntuleModu
1176
1177         Text {
1178             width: parent.width
1179             horizontalAlignment: Text.AlignHCenter
1180             wrapMode: Text.WordWrap
1181             text: "Kaç noktalı kalibrasyon yapmak istersiniz?"
1182             color: "#dce8f5"
1183             font.family: "Segoe UI"
1184             font.pixelSize: 16
1185             font.bold: true
```

```
1184     }
1185
1186     Text {
1187         width: parent.width
1188         horizontalAlignment: Text.AlignHCenter
1189         wrapMode: Text.WordWrap
1190         text: "Daha fazla nokta, daha hassas kalibrasyon sağlar"
1191         color: "#6b7280"
1192         font.family: "Segoe UI"
1193         font.pixelSize: 12
1194     }
1195
1196     Grid {
1197         anchors.horizontalCenter: parent.horizontalCenter
1198         columns: 4
1199         spacing: 12
1200
1201         Repeater {
1202             model: [4, 6, 8, 10]
1203
1204             Rectangle {
1205                 id: secKarti
1206                 width: 72
1207                 height: 72
1208                 radius: 12
1209                 color: secKarti.hovered ? "#182644" : "#0f1420"
1210                 border.color: "#3b82f6"
1211                 border.width: 1
1212                 property bool hovered: false
1213                 Behavior on color { ColorAnimation { duration: 120 } }
1214
```

```
1215         Text {
1216             anchors.centerIn: parent
1217             text: modelData
1218             color: "#dce8f5"
1219             font.family: "Segoe UI"
1220             font.pixelSize: 26
1221             font.bold: true
1222         }
1223
1224         MouseArea {
1225             anchors.fill: parent
1226             hoverEnabled: true
1227             cursorShape: Qt.PointingHandCursor
1228             onEntered: secKarti.hovered = true
1229             onExited: secKarti.hovered = false
1230             onClicked: {
1231                 loadCellToplamNokta = modelData
1232                 loadCellNoktalari = []
1233                 loadCellDuzenlemeIndex = -1
1234                 loadCellSayiSecildi = true
1235             }
1236         }
1237     }
1238 }
1239 }
1240 }
1241
1242 Row {
1243     anchors.centerIn: parent
1244     spacing: 24
```

```
1245         visible: loadCellSayiSecildi
1246
1247     Rectangle {
1248         width: 340
1249         height: 500
1250         radius: 14
1251         color: "#0f1420"
1252         border.color: "#1e2a3f"
1253         border.width: 1
1254
1255     Column {
1256         anchors.centerIn: parent
1257         spacing: 16
1258         width: parent.width - 40
1259
1260         Rectangle {
1261             anchors.horizontalCenter: parent.horizontalCenter
1262             visible: loadCellKayitliVar && !loadCellTamDuzenlemeModu
1263             width: sonKaliMetni2.implicitWidth + 20
1264             height: 24
1265             radius: 12
1266             color: "#0a0e17"
1267             border.color: "#1e2a3f"
1268             border.width: 1
1269
1270             Text {
1271                 id: sonKaliMetni2
1272                 anchors.centerIn: parent
1273                 text: "Kayıtlı: " + loadCellNoktaSayisi + " nokta"
1274                 color: "#6b7280"
1275                 font.pixelSize: 11
```

```

1276         }
1277     }
1278
1279     Text {
1280         anchors.horizontalCenter: parent.horizontalCenter
1281         text: loadCellTamDuzenlemeModu
1282             ? "Kalibrasyonu Düzenle"
1283             : "Nokta " + (loadCellNoktalari.length + 1) + " / " +
loadCellToplamNokta
1284         color: "#3b82f6"
1285         font.family: "Segoe UI"
1286         font.pixelSize: 13
1287         font.bold: true
1288         font.letterSpacing: 1
1289     }
1290
1291     Rectangle {
1292         width: parent.width
1293         height: 6
1294         radius: 3
1295         color: "#1e2a3f"
1296
1297         Rectangle {
1298             width: parent.width * (loadCellNoktalari.length / loadCellToplamNokta)
1299             height: parent.height
1300             radius: 3
1301             color: "#3b82f6"
1302             Behavior on width { NumberAnimation { duration: 150 } }
1303         }
1304     }
1305

```

```
1306         Text {
1307             width: parent.width
1308             horizontalAlignment: Text.AlignHCenter
1309             wrapMode: Text.WordWrap
1310             text: loadCellDuzenlemeIndex >= 0
1311                 ? "Nokta " + (loadCellDuzenlemeIndex + 1) + " düzenleniyor"
1312                 : (loadCellTamDuzenlemeModu
1313                     ? "Düzenlemek istediğiniz noktaya sağdaki listeden tıklayın"
1314                     : "Hedef ağırlığı gir, sonra o ağırlığı load cell'e koy")
1315             color: loadCellDuzenlemeIndex >= 0 ? "#3b82f6" : "#9ca3af"
1316             font.family: "Segoe UI"
1317             font.bold: loadCellDuzenlemeIndex >= 0
1318             font.pixelSize: 13
1319         }
1320
1321         TextField {
1322             id: hedefKgKutusu
1323             anchors.horizontalCenter: parent.horizontalCenter
1324             width: 180
1325             height: 44
1326             placeholderText: "kg"
1327             placeholderTextColor: "#4b5563"
1328             color: "ffffff"
1329             font.pixelSize: 16
1330             horizontalAlignment: TextInput.AlignHCenter
1331             validator: DoubleValidator { bottom: 0; top: 500; decimals: 2 }
1332             background: Rectangle {
1333                 color: "#0a0e17"
1334                 radius: 10
1335                 border.color: hedefKgKutusu.activeFocus ? "#3b82f6" : "#1e2a3f"
```

```

1336         border.width: 1
1337     }
1338 }
1339
1340 Rectangle {
1341     anchors.horizontalCenter: parent.horizontalCenter
1342     width: parent.width
1343     height: 64
1344     radius: 12
1345     color: "#132335"
1346     border.color: wifiManager.baglandi ? "#3b82f6" : "#dc2626"
1347     border.width: 1
1348     Behavior on border.color { ColorAnimation { duration: 120 } }
1349
1350     Text {
1351         anchors.centerIn: parent
1352         text: "Ham değ r: " + sensorManager.hamAgirlik.toFixed(0)
1353         color: wifiManager.baglandi ? "#3b82f6" : "#f87171"
1354         font.family: "Segoe UI"
1355         font.pixelSize: 20
1356         font.bold: true
1357     }
1358 }
1359
1360 Text {
1361     width: parent.width
1362     horizontalAlignment: Text.AlignHCenter
1363     wrapMode: Text.WordWrap
1364     visible: !wifiManager.baglandi
1365     text: "Cihaz baėlı deėil – ham değ r 0 olarak kaydedilecek"
1366     color: "#f87171"

```



```

1367         font.family: "Segoe UI"
1368         font.pixelSize: 11
1369     }
1370
1371     Button {
1372         id: noktaKaydetButonu
1373         width: parent.width
1374         height: 46
1375         property bool tumNoktalarGirildi: !loadCellTamDuzenlemeModu &&
loadCellDuzenlemeIndex < 0 && loadCellNoktalari.length >= loadCellToplamNokta
1376         text: loadCellDuzenlemeIndex >= 0
1377             ? "Noktayı Güncelle"
1378             : (tumNoktalarGirildi
1379                 ? "Kalibrasyonu Tamamla"
1380                 : (loadCellNoktalari.length >= loadCellToplamNokta - 1 ? "Son
Noktayı Kaydet" : "Bu Noktayı Kaydet"))
1381         font.pixelSize: 14
1382         font.bold: true
1383         enabled: tumNoktalarGirildi || (hedefKgKutusu.text.length > 0 &&
(loadCellDuzenlemeIndex >= 0 || loadCellNoktalari.length < loadCellToplamNokta))
1384
1385         onClicked: {
1386             if (tumNoktalarGirildi) {
1387                 loadCellTamamlandiOnayGoster = true
1388                 return
1389             }
1390
1391             var yeniNokta = { hedefKg: parseFloat(hedefKgKutusu.text), hamDeger:
sensorManager.hamAgirlik }
1392
1393             if (loadCellDuzenlemeIndex >= 0) {
1394                 // Var olan bir noktayı yerinde güncelliyoruz, diğer noktalar
etkilenmiyor

```

```

1395         var guncelDizi = loadCellNoktalari.slice()
1396         guncelDizi[loadCellDuzenlemeIndex] = yeniNokta
1397         loadCellNoktalari = guncelDizi
1398         loadCellDuzenlemeIndex = -1
1399         hedefKgKutusu.text = ""
1400         return
1401     }
1402
1403     var yeniDizi = loadCellNoktalari.slice()
1404     yeniDizi.push(yeniNokta)
1405     loadCellNoktalari = yeniDizi
1406     hedefKgKutusu.text = ""
1407
1408     if (loadCellNoktalari.length >= loadCellToplamNokta) {
1409         loadCellTamamlandiOnayGoster = true
1410     }
1411 }
1412
1413 background: Rectangle {
1414     radius: 10
1415     color: parent.enabled ? (parent.hovered ? "#4f8cf7" : "#3b82f6") :
"#1e2a3f"
1416     Behavior on color { ColorAnimation { duration: 120 } }
1417 }
1418
1419 contentItem: Text {
1420     text: parent.text
1421     color: "ffffff"
1422     font: parent.font
1423     horizontalAlignment: Text.AlignHCenter
1424     verticalAlignment: Text.AlignVCenter

```

```
1425         }
1426     }
1427
1428     Button {
1429         width: parent.width
1430         height: 46
1431         visible: loadCellTamDuzenlemeModu
1432         text: "Değişiklikleri Kaydet"
1433         font.pixelSize: 14
1434         font.bold: true
1435
1436         onClicked: loadCellKaydetVeBildir()
1437
1438         background: Rectangle {
1439             radius: 10
1440             color: parent.hovered ? "#22c55e" : "#16a34a"
1441             Behavior on color { ColorAnimation { duration: 120 } }
1442         }
1443
1444         contentItem: Text {
1445             text: parent.text
1446             color: "#ffffff"
1447             font: parent.font
1448             horizontalAlignment: Text.AlignHCenter
1449             verticalAlignment: Text.AlignVCenter
1450         }
1451     }
1452
1453     Button {
1454         width: parent.width
```

```

1455         height: 38
1456         visible: loadCellDuzenlemeIndex >= 0
1457         text: "Vazgeç"
1458         font.pixelSize: 13
1459
1460         onClicked: {
1461             loadCellDuzenlemeIndex = -1
1462             hedefKgKutusu.text = ""
1463         }
1464
1465         background: Rectangle {
1466             radius: 8
1467             color: parent.hovered ? "#1e2a3f" : "transparent"
1468             border.color: "#1e2a3f"
1469             border.width: 1
1470             Behavior on color { ColorAnimation { duration: 120 } }
1471         }
1472
1473         contentItem: Text {
1474             text: parent.text
1475             color: "#9ca3af"
1476             font: parent.font
1477             horizontalAlignment: Text.AlignHCenter
1478             verticalAlignment: Text.AlignVCenter
1479         }
1480     }
1481 }
1482 }
1483
1484 Rectangle {
1485     width: 300

```

```
1486         height: 500
1487         radius: 14
1488         color: "#0f1420"
1489         border.color: "#1e2a3f"
1490         border.width: 1
1491
1492     Column {
1493         anchors.top: parent.top
1494         anchors.left: parent.left
1495         anchors.right: parent.right
1496         anchors.margins: 20
1497         spacing: 4
1498
1499         Text {
1500             text: "KAYDEDİLEN NOKTALAR"
1501             color: "#6b7280"
1502             font.family: "Segoe UI"
1503             font.pixelSize: 11
1504             font.bold: true
1505             font.letterSpacing: 1
1506         }
1507
1508         Text {
1509             text: "Düzeltilmek için bir noktaya tıkla"
1510             color: "#4b5563"
1511             font.family: "Segoe UI"
1512             font.pixelSize: 10
1513         }
1514     }
1515
```

```

1516         ListView {
1517             anchors.top: parent.top
1518             anchors.left: parent.left
1519             anchors.right: parent.right
1520             anchors.bottom: parent.bottom
1521             anchors.margins: 16
1522             anchors.topMargin: 60
1523             clip: true
1524             spacing: 8
1525
1526             model: loadCellNoktalari
1527
1528             delegate: Rectangle {
1529                 width: ListView.view.width
1530                 height: 48
1531                 radius: 8
1532                 color: satirAlani.containsMouse ? "#182644" : "#0a0e17"
1533                 border.color: index === loadCellDuzenlemeIndex ? "#3b82f6" : "#1e2a3f"
1534                 border.width: index === loadCellDuzenlemeIndex ? 2 : 1
1535                 Behavior on color { ColorAnimation { duration: 120 } }
1536
1537                 Row {
1538                     anchors.fill: parent
1539                     anchors.leftMargin: 12
1540                     anchors.rightMargin: 12
1541                     spacing: 8
1542
1543                     Rectangle {
1544                         width: 22
1545                         height: 22
1546                         radius: 11

```

```
1547         anchors.verticalCenter: parent.verticalCenter
1548         color: "#182644"
1549         border.color: "#3b82f6"
1550         border.width: 1
1551
1552     Text {
1553         anchors.centerIn: parent
1554         text: index + 1
1555         color: "#3b82f6"
1556         font.pixelSize: 10
1557         font.bold: true
1558     }
1559 }
1560
1561 Column {
1562     anchors.verticalCenter: parent.verticalCenter
1563     spacing: 1
1564
1565     Text {
1566         text: modelData.hedefKg.toFixed(2) + " kg"
1567         color: "#dce8f5"
1568         font.family: "Segoe UI"
1569         font.pixelSize: 13
1570         font.bold: true
1571     }
1572
1573     Text {
1574         text: "ADC: " + modelData.hamDeger.toFixed(0)
1575         color: "#6b7280"
1576         font.family: "Segoe UI"
```

```
1577         font.pixelSize: 10
1578     }
1579 }
1580 }
1581
1582 Text {
1583     anchors.right: parent.right
1584     anchors.verticalCenter: parent.verticalCenter
1585     anchors.rightMargin: 12
1586     text: "↖"
1587     color: "#6b7280"
1588     font.pixelSize: 13
1589     visible: satirAlani.containsMouse
1590 }
1591
1592 MouseArea {
1593     id: satirAlani
1594     anchors.fill: parent
1595     hoverEnabled: true
1596     cursorShape: Qt.PointingHandCursor
1597     onClicked: {
1598         hedefKgKutusu.text = modelData.hedefKg.toString()
1599         loadCellDuzenlemeIndex = index
1600     }
1601 }
1602 }
1603 }
1604 }
1605 }
1606 }
1607
```



```

1608         Item {
1609             anchors.top: parent.top
1610             anchors.left: parent.left
1611             anchors.right: parent.right
1612             anchors.bottom: parent.bottom
1613             anchors.topMargin: 70
1614             visible: seciliSensor === 2
1615
1616             // =====
1617             // OLCUM KALIBRASYONU – cok nokta (load cell birebir kopyasi)
1618             // =====
1619             Item {
1620                 anchors.fill: parent
1621
1622                 Column {
1623                     anchors.centerIn: parent
1624                     spacing: 24
1625                     width: 420
1626                     visible: mesafeAraEkranGoster
1627
1628                     Rectangle {
1629                         anchors.horizontalCenter: parent.horizontalCenter
1630                         width: mesafeAraEkranBadge.implicitWidth + 20
1631                         height: 26
1632                         radius: 13
1633                         color: "#0f1420"
1634                         border.color: "#1e2a3f"
1635                         border.width: 1
1636
1637                         Text {

```

```

1638         id: mesafeAraEkranBadge
1639         anchors.centerIn: parent
1640         text: "Kayıtlı: " + mesafeOlcumNoktaSayisi + " nokta"
1641         color: "#6b7280"
1642         font.pixelSize: 11
1643     }
1644 }
1645
1646 Text {
1647     width: parent.width
1648     horizontalAlignment: Text.AlignHCenter
1649     wrapMode: Text.WordWrap
1650     text: "Bu sensör için kayıtlı bir kalibrasyon bulundu"
1651     color: "#dce8f5"
1652     font.family: "Segoe UI"
1653     font.pixelSize: 16
1654     font.bold: true
1655 }
1656
1657 Row {
1658     anchors.horizontalCenter: parent.horizontalCenter
1659     spacing: 20
1660
1661     Rectangle {
1662         id: mesafeGoruntuleKarti
1663         width: 190
1664         height: 180
1665         radius: 14
1666         color: "#0f1420"
1667         border.color: mesafeGoruntuleKarti.hovered ? "#9333ea" : "#1e2a3f"
1668         border.width: 1

```

```
1669         property bool hovered: false
1670
1671         Behavior on border.color { ColorAnimation { duration: 120 } }
1672
1673         Column {
1674             anchors.centerIn: parent
1675             spacing: 12
1676             width: parent.width - 28
1677
1678             Canvas {
1679                 anchors.horizontalCenter: parent.horizontalCenter
1680                 width: 32
1681                 height: 32
1682                 onPaint: {
1683                     var ctx = getContext("2d")
1684                     ctx.clearRect(0, 0, width, height)
1685                     ctx.lineWidth = 1.8
1686                     ctx.lineCap = "round"
1687                     ctx.lineJoin = "round"
1688                     ctx.strokeStyle = "#9333ea"
1689
1690                     ctx.beginPath()
1691                     ctx.moveTo(3, 16)
1692                     ctx.bezierCurveTo(7.5, 8.5, 12.5, 6.8, 16, 6.8)
1693                     ctx.bezierCurveTo(19.5, 6.8, 24.5, 8.5, 29, 16)
1694                     ctx.bezierCurveTo(24.5, 23.5, 19.5, 25.2, 16, 25.2)
1695                     ctx.bezierCurveTo(12.5, 25.2, 7.5, 23.5, 3, 16)
1696                     ctx.stroke()
1697
1698                     ctx.beginPath()
1699                     ctx.arc(16, 16, 4.4, 0, Math.PI * 2, false)
```

```
1699             ctx.stroke()
1700         }
1701     }
1702
1703     Text {
1704         width: parent.width
1705         horizontalAlignment: Text.AlignHCenter
1706         wrapMode: Text.WordWrap
1707         text: "Mevcut Kalibrasyonu Görüntüle"
1708         color: "#dce8f5"
1709         font.family: "Segoe UI"
1710         font.pixelSize: 13
1711         font.bold: true
1712     }
1713
1714     Text {
1715         width: parent.width
1716         horizontalAlignment: Text.AlignHCenter
1717         wrapMode: Text.WordWrap
1718         text: "Kayıtlı noktaları incele"
1719         color: "#6b7280"
1720         font.family: "Segoe UI"
1721         font.pixelSize: 11
1722     }
1723 }
1724
1725 MouseArea {
1726     anchors.fill: parent
1727     hoverEnabled: true
1728     cursorShape: Qt.PointingHandCursor
1729     onEntered: mesafeGoruntuleKarti.hovered = true
```

```

1730         onExited: mesafeGoruntuleKarti.hovered = false
1731     onClicked: {
1732         mesafeNoktalari = database.mesafeNoktalariGetir()
1733         mesafeAraEkranGoster = false
1734         mesafeGoruntuleModu = true
1735         mesafeDuzenlemeIndex = -1
1736     }
1737 }
1738 }
1739
1740 Rectangle {
1741     id: mesafeYenidenKarti
1742     width: 190
1743     height: 180
1744     radius: 14
1745     color: "#0f1420"
1746     border.color: mesafeYenidenKarti.hovered ? "#9ca3af" : "#1e2a3f"
1747     border.width: 1
1748     property bool hovered: false
1749     Behavior on border.color { ColorAnimation { duration: 120 } }
1750
1751     Column {
1752         anchors.centerIn: parent
1753         spacing: 12
1754         width: parent.width - 28
1755
1756         Canvas {
1757             anchors.horizontalCenter: parent.horizontalCenter
1758             width: 32
1759             height: 32

```

```
1760         onPaint: {
1761             var ctx = getContext("2d")
1762             ctx.clearRect(0, 0, width, height)
1763             ctx.strokeStyle = "#9ca3af"
1764             ctx.fillStyle = "#9ca3af"
1765             ctx.lineWidth = 3.4
1766             ctx.lineCap = "butt"
1767
1768             function yayOku(startDeg, endDeg) {
1769                 var s = startDeg * Math.PI / 180
1770                 var e = endDeg * Math.PI / 180
1771                 var cx = 16, cy = 16, r = 10.5
1772
1773                 ctx.beginPath()
1774                 ctx.arc(cx, cy, r, s, e, false)
1775                 ctx.stroke()
1776
1777                 var ex = cx + r * Math.cos(e)
1778                 var ey = cy + r * Math.sin(e)
1779                 var tx = -Math.sin(e)
1780                 var ty = Math.cos(e)
1781                 var px = -ty
1782                 var py = tx
1783                 var geri = 6.2
1784                 var yaricap = 3.6
1785                 var bx = ex - tx * geri
1786                 var by = ey - ty * geri
1787
1788                 ctx.beginPath()
1789                 ctx.moveTo(ex, ey)
1790                 ctx.lineTo(bx + px * yaricap, by + py * yaricap)
```

```
1791         ctx.lineTo(bx - px * yaricap, by - py * yaricap)
1792         ctx.closePath()
1793         ctx.fill()
1794     }
1795
1796     yay0ku(195, 345)
1797     yay0ku(15, 165)
1798 }
1799 }
1800
1801 Text {
1802     width: parent.width
1803     horizontalAlignment: Text.AlignHCenter
1804     wrapMode: Text.WordWrap
1805     text: "Yeniden Kalibre Et"
1806     color: "#dce8f5"
1807     font.family: "Segoe UI"
1808     font.pixelSize: 13
1809     font.bold: true
1810 }
1811
1812 Text {
1813     width: parent.width
1814     horizontalAlignment: Text.AlignHCenter
1815     wrapMode: Text.WordWrap
1816     text: "Sıfırdan yeni ölçüm al"
1817     color: "#6b7280"
1818     font.family: "Segoe UI"
1819     font.pixelSize: 11
1820 }
```

```
1821         }
1822
1823         MouseArea {
1824             anchors.fill: parent
1825             hoverEnabled: true
1826             cursorShape: Qt.PointingHandCursor
1827             onEntered: mesafeYenidenKarti.hovered = true
1828             onExited: mesafeYenidenKarti.hovered = false
1829             onClicked: mesafeYenidenOnayGoster = true
1830         }
1831     }
1832 }
1833 }
1834
1835 Column {
1836     anchors.centerIn: parent
1837     spacing: 16
1838     width: 340
1839     visible: mesafeGoruntuleModu
1840
1841     Text {
1842         text: "KAYITLI KALİBRASYON NOKTALARI"
1843         color: "#6b7280"
1844         font.family: "Segoe UI"
1845         font.pixelSize: 11
1846         font.bold: true
1847         font.letterSpacing: 1
1848     }
1849
1850     Rectangle {
1851         width: parent.width
```



```
1852             height: 320
1853             radius: 14
1854             color: "#0f1420"
1855             border.color: "#1e2a3f"
1856             border.width: 1
1857
1858         ListView {
1859             anchors.fill: parent
1860             anchors.margins: 16
1861             clip: true
1862             spacing: 8
1863
1864             model: mesafeNoktalari
1865
1866             delegate: Rectangle {
1867                 width: ListView.view.width
1868                 height: 48
1869                 radius: 8
1870                 color: "#0a0e17"
1871                 border.color: "#1e2a3f"
1872                 border.width: 1
1873
1874                 Row {
1875                     anchors.fill: parent
1876                     anchors.leftMargin: 12
1877                     anchors.rightMargin: 12
1878                     spacing: 8
1879
1880                     Rectangle {
1881                         width: 22
```

```
1882         height: 22
1883
1884         radius: 11
1885
1886         anchors.verticalCenter: parent.verticalCenter
1887
1888         color: "#241a38"
1889
1890         border.color: "#9333ea"
1891
1892         border.width: 1
1893
1894
1895
1896
1897
1898
1899
1900
1901
1902
1903
1904
1905
1906
1907
1908
1909
1910
1911
1912
```

```
Text {
    anchors.centerIn: parent
    text: index + 1
    color: "#c084fc"
    font.pixelSize: 10
    font.bold: true
}

}

Column {
    anchors.verticalCenter: parent.verticalCenter
    spacing: 1

    Text {
        text: (modelData.hedefMm / 10).toFixed(1) + " cm"
        color: "#dce8f5"
        font.family: "Segoe UI"
        font.pixelSize: 13
        font.bold: true
    }

    Text {
        text: "ADC: " + modelData.hamDeger.toFixed(0)
        color: "#6b7280"
```

```
1913         font.family: "Segoe UI"
1914         font.pixelSize: 10
1915     }
1916 }
1917 }
1918 }
1919 }
1920 }
1921
1922 Button {
1923     width: parent.width
1924     height: 46
1925     text: "Düzenle"
1926     font.pixelSize: 14
1927     font.bold: true
1928
1929     onClicked: {
1930         mesafeToplamNokta = mesafeNoktalari.length
1931         mesafeGoruntuleModu = false
1932         mesafeTamDuzenlemeModu = true
1933         mesafeSayiSecildi = true
1934         mesafeDuzenlemeIndex = -1
1935     }
1936
1937     background: Rectangle {
1938         radius: 8
1939         color: parent.hovered ? "#a855f7" : "#9333ea"
1940         Behavior on color { ColorAnimation { duration: 120 } }
1941     }
1942 }
```

```
1943         contentItem: Text {
1944             text: parent.text
1945             color: "#ffffff"
1946             font: parent.font
1947             horizontalAlignment: Text.AlignHCenter
1948             verticalAlignment: Text.AlignVCenter
1949         }
1950     }
1951 }
1952
1953 Rectangle {
1954     anchors.fill: parent
1955     color: "#000000b3"
1956     visible: mesafeYenidenOnayGoster
1957     z: 200
1958
1959     MouseArea { anchors.fill: parent }
1960
1961     Rectangle {
1962         anchors.centerIn: parent
1963         width: 360
1964         height: 200
1965         radius: 14
1966         color: "#0f1420"
1967         border.color: "#1e2a3f"
1968         border.width: 1
1969
1970         Column {
1971             anchors.centerIn: parent
1972             spacing: 18
1973             width: parent.width - 40
```

```
1974
1975     Text {
1976         width: parent.width
1977         horizontalAlignment: Text.AlignHCenter
1978         wrapMode: Text.WordWrap
1979         text: "Mevcut kalibrasyon silinecek, emin misiniz?"
1980         color: "#dce8f5"
1981         font.family: "Segoe UI"
1982         font.pixelSize: 14
1983         font.bold: true
1984     }
1985
1986     Row {
1987         anchors.horizontalCenter: parent.horizontalCenter
1988         spacing: 12
1989
1990         Button {
1991             width: 150
1992             height: 42
1993             text: "Vazgeç"
1994
1995             onClicked: mesafeYenidenOnayGoster = false
1996
1997             background: Rectangle {
1998                 radius: 8
1999                 color: parent.hovered ? "#1e2a3f" : "transparent"
2000                 border.color: "#1e2a3f"
2001                 border.width: 1
2002                 Behavior on color { ColorAnimation { duration: 120 } }
2003             }
```

```
2004
2005         contentItem: Text {
2006             text: parent.text
2007             color: "#9ca3af"
2008             font: parent.font
2009             horizontalAlignment: Text.AlignHCenter
2010             verticalAlignment: Text.AlignVCenter
2011         }
2012     }
2013
2014     Button {
2015         width: 150
2016         height: 42
2017         text: "Evet, Yeniden Başla"
2018         font.bold: true
2019
2020         onClicked: {
2021             mesafeYenidenOnayGoster = false
2022             mesafeAraEkranGoster = false
2023             mesafeGoruntuleModu = false
2024             mesafeTamDuzenlemeModu = false
2025             mesafeNoktalarI = []
2026             mesafeSayiSecildi = false
2027             mesafeDuzenlemeIndex = -1
2028         }
2029
2030         background: Rectangle {
2031             radius: 8
2032             color: parent.hovered ? "#b91c1c" : "#991b1b"
2033             Behavior on color { ColorAnimation { duration: 120 } }
2034         }
```

```
2035
2036             contentItem: Text {
2037                 text: parent.text
2038                 color: "#ffffff"
2039                 font: parent.font
2040                 horizontalAlignment: Text.AlignHCenter
2041                 verticalAlignment: Text.AlignVCenter
2042             }
2043         }
2044     }
2045 }
2046 }
2047 }
2048
2049 Rectangle {
2050     anchors.fill: parent
2051     color: "#000000b3"
2052     visible: mesafeTamamlandiOnayGoster
2053     z: 200
2054
2055     MouseArea { anchors.fill: parent }
2056
2057     Rectangle {
2058         anchors.centerIn: parent
2059         width: 380
2060         height: 220
2061         radius: 14
2062         color: "#0f1420"
2063         border.color: "#16a34a"
2064         border.width: 1
```

```
2065
2066         Column {
2067             anchors.centerIn: parent
2068             spacing: 18
2069             width: parent.width - 40
2070
2071         Column {
2072             width: parent.width
2073             spacing: 6
2074
2075             Text {
2076                 width: parent.width
2077                 horizontalAlignment: Text.AlignHCenter
2078                 text: "Kalibrasyon Tamamlandı"
2079                 color: "#4ade80"
2080                 font.family: "Segoe UI"
2081                 font.pixelSize: 17
2082                 font.bold: true
2083             }
2084
2085             Text {
2086                 width: parent.width
2087                 horizontalAlignment: Text.AlignHCenter
2088                 wrapMode: Text.WordWrap
2089                 text: "Toplam " + mesafeToplamNokta + " nokta girildi. Bu
kalibrasyonu kaydetmek istiyor musunuz?"
2090                 color: "#9ca3af"
2091                 font.family: "Segoe UI"
2092                 font.pixelSize: 13
2093             }
2094         }
```



```

2095
2096         Row {
2097             anchors.horizontalCenter: parent.horizontalCenter
2098             spacing: 12
2099
2100             Button {
2101                 width: 150
2102                 height: 42
2103                 text: "Noktaları Kontrol Et"
2104
2105                 onClicked: mesafeTamamlandiOnayGoster = false
2106
2107                 background: Rectangle {
2108                     radius: 8
2109                     color: parent.hovered ? "#1e2a3f" : "transparent"
2110                     border.color: "#1e2a3f"
2111                     border.width: 1
2112                     Behavior on color { ColorAnimation { duration: 120 } }
2113                 }
2114
2115                 contentItem: Text {
2116                     text: parent.text
2117                     color: "#9ca3af"
2118                     font: parent.font
2119                     horizontalAlignment: Text.AlignHCenter
2120                     verticalAlignment: Text.AlignVCenter
2121                 }
2122             }
2123
2124             Button {
2125                 width: 150

```

```

2126             height: 42
2127             text: "Kaydet ve Bitir"
2128             font.bold: true
2129
2130             onClicked: {
2131                 mesafeTamamlandiOnayGoster = false
2132                 mesafeKaydetVeBildir()
2133             }
2134
2135             background: Rectangle {
2136                 radius: 8
2137                 color: parent.hovered ? "#22c55e" : "#16a34a"
2138                 Behavior on color { ColorAnimation { duration: 120 } }
2139             }
2140
2141             contentItem: Text {
2142                 text: parent.text
2143                 color: "#ffffff"
2144                 font: parent.font
2145                 horizontalAlignment: Text.AlignHCenter
2146                 verticalAlignment: Text.AlignVCenter
2147             }
2148         }
2149     }
2150 }
2151 }
2152 }
2153
2154 Column {
2155     anchors.centerIn: parent

```

```
2156         spacing: 24
2157
2158         width: 360
2159
2160         visible: !mesafeSayiSecildi && !mesafeAraEkranGoster && !mesafeGoruntuleModu
2161
2162         Text {
2163             width: parent.width
2164             horizontalAlignment: Text.AlignHCenter
2165             wrapMode: Text.WordWrap
2166             text: "Kaç noktalı kalibrasyon yapmak istersiniz?"
2167             color: "#dce8f5"
2168             font.family: "Segoe UI"
2169             font.pixelSize: 16
2170             font.bold: true
2171         }
2172
2173         Text {
2174             width: parent.width
2175             horizontalAlignment: Text.AlignHCenter
2176             wrapMode: Text.WordWrap
2177             text: "Daha fazla nokta, daha hassas kalibrasyon sağlar"
2178             color: "#6b7280"
2179             font.family: "Segoe UI"
2180             font.pixelSize: 12
2181         }
2182
2183         Grid {
2184             anchors.horizontalCenter: parent.horizontalCenter
2185             columns: 4
2186             spacing: 12
2187
2188             Repeater {
```

```
2187         model: [4, 6, 8, 10]
2188
2189     Rectangle {
2190         id: mesafeSecKarti
2191         width: 72
2192         height: 72
2193         radius: 12
2194         color: mesafeSecKarti.hovered ? "#241a38" : "#0f1420"
2195         border.color: "#9333ea"
2196         border.width: 1
2197         property bool hovered: false
2198         Behavior on color { ColorAnimation { duration: 120 } }
2199
2200     Text {
2201         anchors.centerIn: parent
2202         text: modelData
2203         color: "#dce8f5"
2204         font.family: "Segoe UI"
2205         font.pixelSize: 26
2206         font.bold: true
2207     }
2208
2209     MouseArea {
2210         anchors.fill: parent
2211         hoverEnabled: true
2212         cursorShape: Qt.PointingHandCursor
2213         onEntered: mesafeSecKarti.hovered = true
2214         onExited: mesafeSecKarti.hovered = false
2215         onClicked: {
2216             mesafeToplamNokta = modelData
```

```
2217         mesafeNoktalari = []
2218         mesafeDuzenlemeIndex = -1
2219         mesafeSayiSecildi = true
2220     }
2221 }
2222 }
2223 }
2224 }
2225 }
2226
2227 Row {
2228     anchors.centerIn: parent
2229     spacing: 24
2230     visible: mesafeSayiSecildi
2231
2232     Rectangle {
2233         width: 340
2234         height: 500
2235         radius: 14
2236         color: "#0f1420"
2237         border.color: "#1e2a3f"
2238         border.width: 1
2239
2240         Column {
2241             anchors.centerIn: parent
2242             spacing: 16
2243             width: parent.width - 40
2244
2245             Rectangle {
2246                 anchors.horizontalCenter: parent.horizontalCenter
2247                 visible: mesafeOlcumKayitliVar && !mesafeTamDuzenlemeModu
```

```

2248         width: mesafeSonKaliMetni.implicitWidth + 20
2249
2250         height: 24
2251
2252         radius: 12
2253
2254         color: "#0a0e17"
2255
2256         border.color: "#1e2a3f"
2257
2258         border.width: 1
2259
2260
2261     Text {
2262
2263         id: mesafeSonKaliMetni
2264
2265         anchors.centerIn: parent
2266
2267         text: "Kayıtlı: " + mesafeOlcumNoktaSayisi + " nokta"
2268
2269         color: "#6b7280"
2270
2271         font.pixelSize: 11
2272
2273     }
2274 }
2275
2276 Text {
2277
2278     anchors.horizontalCenter: parent.horizontalCenter
2279
2280     text: mesafeTamDuzenlemeModu
2281
2282         ? "Kalibrasyonu Düzenle"
2283
2284         : "Nokta " + (mesafeNoktalari.length + 1) + " / " +
mesafeToplamNokta
2285
2286     color: "#9333ea"
2287
2288     font.family: "Segoe UI"
2289
2290     font.pixelSize: 13
2291
2292     font.bold: true
2293
2294     font.letterSpacing: 1
2295
2296 }
2297
2298 Rectangle {
2299
2300     width: parent.width

```

```

2278         height: 6
2279
2280         radius: 3
2281
2282         color: "#1e2a3f"
2283
2284
2285         Rectangle {
2286
2287             width: parent.width * (mesafeNoktalari.length / mesafeToplamNokta)
2288
2289             height: parent.height
2290
2291             radius: 3
2292
2293             color: "#9333ea"
2294
2295             Behavior on width { NumberAnimation { duration: 150 } }
2296
2297         }
2298     }
2299
2300
2301     Text {
2302
2303         width: parent.width
2304
2305         horizontalAlignment: Text.AlignHCenter
2306
2307         wrapMode: Text.WordWrap
2308
2309         text: mesafeDuzenlemeIndex >= 0
2310
2311             ? "Nokta " + (mesafeDuzenlemeIndex + 1) + " düzenleniyor"
2312
2313             : (mesafeTamDuzenlemeModu
2314
2315                 ? "Düzenlemek istediğiniz noktaya sağdaki listeden tıklayın"
2316
2317                 : "Hedef mesafeyi gir, sonra boruyu o mesafeye getir")
2318
2319         color: mesafeDuzenlemeIndex >= 0 ? "#9333ea" : "#9ca3af"
2320
2321         font.family: "Segoe UI"
2322
2323         font.bold: mesafeDuzenlemeIndex >= 0
2324
2325         font.pixelSize: 13
2326
2327     }
2328
2329
2330     TextField {
2331
2332         id: hedefCmKutusu

```

```

2308         anchors.horizontalCenter: parent.horizontalCenter
2309
2310         width: 180
2311
2312         height: 44
2313
2314         placeholderText: "cm"
2315
2316         placeholderTextColor: "#4b5563"
2317
2318         color: "ffffff"
2319
2320         font.pixelSize: 16
2321
2322         horizontalAlignment: TextInput.AlignHCenter
2323
2324         validator: DoubleValidator { bottom: 0; top: 300; decimals: 1 }
2325
2326         background: Rectangle {
2327
2328             color: "#0a0e17"
2329
2330             radius: 10
2331
2332             border.color: hedefCmKutusu.activeFocus ? "#9333ea" : "#1e2a3f"
2333
2334             border.width: 1
2335
2336         }
2337     }
2338
2339     Rectangle {
2340
2341         anchors.horizontalCenter: parent.horizontalCenter
2342
2343         width: parent.width
2344
2345         height: 64
2346
2347         radius: 12
2348
2349         color: "#132335"
2350
2351         border.color: wifiManager.baglandi ? "#9333ea" : "#dc2626"
2352
2353         border.width: 1
2354
2355         Behavior on border.color { ColorAnimation { duration: 120 } }
2356
2357         Text {
2358
2359             anchors.centerIn: parent
2360
2361             text: "Ham değeri: " + sensorManager.hamMesafe.toFixed(0)
2362
2363             color: wifiManager.baglandi ? "#c084fc" : "#f87171"

```



```

2339             font.family: "Segoe UI"
2340
2341             font.pixelSize: 20
2342
2343             font.bold: true
2344
2345         }
2346     }
2347
2348     Text {
2349
2350         width: parent.width
2351
2352         horizontalAlignment: Text.AlignHCenter
2353
2354         wrapMode: Text.WordWrap
2355
2356         visible: !wifiManager.baglandi
2357
2358         text: "Cihaz bağlı değil – ham değer 0 olarak kaydedilecek"
2359
2360         color: "#f87171"
2361
2362         font.family: "Segoe UI"
2363
2364         font.pixelSize: 11
2365     }
2366
2367     Button {
2368
2369         id: mesafeNoktaKaydetButonu
2370
2371         width: parent.width
2372
2373         height: 46
2374
2375         property bool tumNoktalarGirildi: !mesafeTamDuzenlemeModu &&
mesafeDuzenlemeIndex < 0 && mesafeNoktalar.length >= mesafeToplamNokta
2376
2377         text: mesafeDuzenlemeIndex >= 0
2378
2379             ? "Noktayı Güncelle"
2380
2381             : (tumNoktalarGirildi
2382
2383                 ? "Kalibrasyonu Tamamla"
2384
2385                 : (mesafeNoktalar.length >= mesafeToplamNokta - 1 ? "Son
Noktayı Kaydet" : "Bu Noktayı Kaydet"))
2386
2387         font.pixelSize: 14
2388
2389         font.bold: true

```

```

2368             enabled: tumNoktalarGirildi || (hedefCmKutusu.text.length > 0 &&
(mesafeDuzenlemeIndex >= 0 || mesafeNoktalari.length < mesafeToplamNokta))

2369
2370         onClicked: {
2371             if (tumNoktalarGirildi) {
2372                 mesafeTamamlandiOnayGoster = true
2373                 return
2374             }
2375
2376             var yeniNokta = { hedefMm: parseFloat(hedefCmKutusu.text) * 10,
hamDeger: sensorManager.hamMesafe }
2377
2378             if (mesafeDuzenlemeIndex >= 0) {
2379                 // Var olan bir noktayı yerinde güncelliyoruz, diğer noktalar
etkilenmiyor
2380                 var guncelDizi = mesafeNoktalari.slice()
2381                 guncelDizi[mesafeDuzenlemeIndex] = yeniNokta
2382                 mesafeNoktalari = guncelDizi
2383                 mesafeDuzenlemeIndex = -1
2384                 hedefCmKutusu.text = ""
2385                 return
2386             }
2387
2388             var yeniDizi = mesafeNoktalari.slice()
2389             yeniDizi.push(yeniNokta)
2390             mesafeNoktalari = yeniDizi
2391             hedefCmKutusu.text = ""
2392
2393             if (mesafeNoktalari.length >= mesafeToplamNokta) {
2394                 mesafeTamamlandiOnayGoster = true
2395             }
2396         }

```

```

2397
2398         background: Rectangle {
2399             radius: 10
2400             color: parent.enabled ? (parent.hovered ? "#a855f7" : "#9333ea") :
"#1e2a3f"
2401             Behavior on color { ColorAnimation { duration: 120 } }
2402         }
2403
2404         contentItem: Text {
2405             text: parent.text
2406             color: "#ffffff"
2407             font: parent.font
2408             horizontalAlignment: Text.AlignHCenter
2409             verticalAlignment: Text.AlignVCenter
2410         }
2411     }
2412
2413     Button {
2414         width: parent.width
2415         height: 46
2416         visible: mesafeTamDuzenlemeModu
2417         text: "Değişiklikleri Kaydet"
2418         font.pixelSize: 14
2419         font.bold: true
2420
2421         onClicked: mesafeKaydetVeBildir()
2422
2423         background: Rectangle {
2424             radius: 10
2425             color: parent.hovered ? "#22c55e" : "#16a34a"
2426             Behavior on color { ColorAnimation { duration: 120 } }

```

```

2427         }
2428
2429         contentItem: Text {
2430             text: parent.text
2431             color: "#ffffff"
2432             font: parent.font
2433             horizontalAlignment: Text.AlignHCenter
2434             verticalAlignment: Text.AlignVCenter
2435         }
2436     }
2437
2438     Button {
2439         width: parent.width
2440         height: 38
2441         visible: mesafeDuzenlemeIndex >= 0
2442         text: "Vazgeç"
2443         font.pixelSize: 13
2444
2445         onClicked: {
2446             mesafeDuzenlemeIndex = -1
2447             hedefCmKutusu.text = ""
2448         }
2449
2450         background: Rectangle {
2451             radius: 8
2452             color: parent.hovered ? "#1e2a3f" : "transparent"
2453             border.color: "#1e2a3f"
2454             border.width: 1
2455             Behavior on color { ColorAnimation { duration: 120 } }
2456         }
2457

```

```
2458             contentItem: Text {
2459                 text: parent.text
2460                 color: "#9ca3af"
2461                 font: parent.font
2462                 horizontalAlignment: Text.AlignHCenter
2463                 verticalAlignment: Text.AlignVCenter
2464             }
2465         }
2466     }
2467 }
2468
2469 Rectangle {
2470     width: 300
2471     height: 500
2472     radius: 14
2473     color: "#0f1420"
2474     border.color: "#1e2a3f"
2475     border.width: 1
2476
2477     Column {
2478         anchors.top: parent.top
2479         anchors.left: parent.left
2480         anchors.right: parent.right
2481         anchors.margins: 20
2482         spacing: 4
2483
2484         Text {
2485             text: "KAYDEDİLEN NOKTALAR"
2486             color: "#6b7280"
2487             font.family: "Segoe UI"
```

```
2488         font.pixelSize: 11
2489         font.bold: true
2490         font.letterSpacing: 1
2491     }
2492
2493     Text {
2494         text: "Düzeltilmek için bir noktaya tıkla"
2495         color: "#4b5563"
2496         font.family: "Segoe UI"
2497         font.pixelSize: 10
2498     }
2499 }
2500
2501 ListView {
2502     anchors.top: parent.top
2503     anchors.left: parent.left
2504     anchors.right: parent.right
2505     anchors.bottom: parent.bottom
2506     anchors.margins: 16
2507     anchors.topMargin: 60
2508     clip: true
2509     spacing: 8
2510
2511     model: mesafeNoktalari
2512
2513     delegate: Rectangle {
2514         width: ListView.view.width
2515         height: 48
2516         radius: 8
2517         color: mesafeSatirAlani.containsMouse ? "#241a38" : "#0a0e17"
2518         border.color: index === mesafeDuzenlemeIndex ? "#9333ea" : "#1e2a3f"
```

```

2519         border.width: index === mesafeDuzenlemeIndex ? 2 : 1
2520     Behavior on color { ColorAnimation { duration: 120 } }
2521
2522     Row {
2523         anchors.fill: parent
2524         anchors.leftMargin: 12
2525         anchors.rightMargin: 12
2526         spacing: 8
2527
2528         Rectangle {
2529             width: 22
2530             height: 22
2531             radius: 11
2532             anchors.verticalCenter: parent.verticalCenter
2533             color: "#241a38"
2534             border.color: "#9333ea"
2535             border.width: 1
2536
2537             Text {
2538                 anchors.centerIn: parent
2539                 text: index + 1
2540                 color: "#c084fc"
2541                 font.pixelSize: 10
2542                 font.bold: true
2543             }
2544         }
2545
2546         Column {
2547             anchors.verticalCenter: parent.verticalCenter
2548             spacing: 1

```

```
2549
2550         Text {
2551             text: (modelData.hedefMm / 10).toFixed(1) + " cm"
2552             color: "#dce8f5"
2553             font.family: "Segoe UI"
2554             font.pixelSize: 13
2555             font.bold: true
2556         }
2557
2558         Text {
2559             text: "ADC: " + modelData.hamDeger.toFixed(0)
2560             color: "#6b7280"
2561             font.family: "Segoe UI"
2562             font.pixelSize: 10
2563         }
2564     }
2565 }
2566
2567 Text {
2568     anchors.right: parent.right
2569     anchors.verticalCenter: parent.verticalCenter
2570     anchors.rightMargin: 12
2571     text: "↖"
2572     color: "#6b7280"
2573     font.pixelSize: 13
2574     visible: mesafeSatirAlani.containsMouse
2575 }
2576
2577 MouseArea {
2578     id: mesafeSatirAlani
2579     anchors.fill: parent
```



```

2580             hoverEnabled: true
2581             cursorShape: Qt.PointingHandCursor
2582             onClicked: {
2583                 hedefCmKutusu.text = (modelData.hedefMm / 10).toString()
2584                 mesafeDuzenlemeIndex = index
2585             }
2586         }
2587     }
2588 }
2589 }
2590 }
2591 }

```

```

2592
2593 Rectangle {
2594     id: mesafeBildirimKutusu
2595     anchors.top: parent.top
2596     anchors.horizontalCenter: parent.horizontalCenter
2597     anchors.topMargin: 12
2598     z: 100
2599     width: mesafeBildirimMetni.implicitWidth + 24
2600     height: 32
2601     radius: 16
2602     color: mesafeBildirimBasarili ? "#123321" : "#3a1414"
2603     opacity: mesafeBildirimGoster ? 1 : 0
2604     visible: opacity > 0
2605     Behavior on opacity { NumberAnimation { duration: 200 } }
2606
2607     Text {
2608         id: mesafeBildirimMetni
2609         anchors.centerIn: parent

```

```
2610         text: mesafeBildirimMesaj
2611         color: mesafeBildirimBasarili ? "#4ade80" : "#f87171"
2612         font.family: "Segoe UI"
2613         font.pixelSize: 12
2614         font.bold: true
2615     }
2616 }
2617 }
2618 }
2619 }
2620 }
```

qml/pages/LoginPage.qml

```
1 import QtQuick 6.7

2 import QtQuick.Controls 6.7

3

4 Rectangle {

5     id: sayfa

6

7     property bool sifreGorunur: false

8     property bool beniHatirla: false

9

10    signal girisBasarili()

11

12    gradient: Gradient {

13        GradientStop { position: 0.0; color: "#101c33" }

14        GradientStop { position: 1.0; color: "#0a1220" }

15    }

16

17    Row {

18        anchors.top: parent.top

19        anchors.right: parent.right

20        anchors.margins: 24

21        spacing: 8

22

23        Rectangle {

24            width: 44

25            height: 30

26            radius: 6

27            color: "#0f1420"

28            border.color: "#3b82f6"

29            border.width: 1

30
```

```
31         Text {
32             anchors.centerIn: parent
33             text: "TR"
34             color: "#ffffff"
35             font.pixelSize: 12
36             font.bold: true
37         }
38     }
39
40     Rectangle {
41         width: 44
42         height: 30
43         radius: 6
44         color: "#0f1420"
45         border.color: "#1e2a3f"
46         border.width: 1
47
48         Text {
49             anchors.centerIn: parent
50             text: "EN"
51             color: "#6b7280"
52             font.pixelSize: 12
53         }
54     }
55 }
56
57 Rectangle {
58     anchors.centerIn: kart
59     width: kart.width + 48
60     height: kart.height + 48
```

```
61         radius: 28
62         color: "#18000000"
63     }
64
65     Rectangle {
66         anchors.centerIn: kart
67         width: kart.width + 32
68         height: kart.height + 32
69         radius: 24
70         color: "#25000000"
71     }
72
73     Rectangle {
74         anchors.centerIn: kart
75         width: kart.width + 18
76         height: kart.height + 18
77         radius: 20
78         color: "#35000000"
79     }
80
81     Rectangle {
82         anchors.centerIn: kart
83         width: kart.width + 8
84         height: kart.height + 8
85         radius: 18
86         color: "#45000000"
87     }
88
89     Rectangle {
90         id: kart
91         anchors.centerIn: parent
```

```
92     width: 400
93     color: "#0f1420"
94     radius: 16
95     border.color: "#4b5563"
96     border.width: 1
97     height: govde.implicitHeight + 48
98
99     Column {
100         id: govde
101         anchors.top: parent.top
102         anchors.horizontalCenter: parent.horizontalCenter
103         anchors.topMargin: 32
104         width: parent.width - 48
105         spacing: 20
106
107         Image {
108             anchors.horizontalCenter: parent.horizontalCenter
109             source: "../assets/logo.png"
110             width: 144
111             height: 144
112             fillMode: Image.PreserveAspectFit
113         }
114
115         Rectangle {
116             width: parent.width
117             height: 1
118             color: "#1e2a3f"
119         }
120
121         Column {
```

```
122         width: parent.width
123
124         spacing: 4
125
126         Text {
127             anchors.horizontalCenter: parent.horizontalCenter
128             text: "Sisteme Giriş"
129             color: "#ffffff"
130             font.family: "Segoe UI"
131             font.pixelSize: 22
132             font.bold: true
133         }
134
135         Text {
136             anchors.horizontalCenter: parent.horizontalCenter
137             text: "Yetkili hesabınızla devam edin"
138             color: "#6b7280"
139             font.family: "Segoe UI"
140             font.pixelSize: 12
141         }
142     }
143
144     Column {
145         width: parent.width
146
147         spacing: 6
148
149         Text {
150             text: "KULLANICI ADI"
151             color: "#6b7280"
152             font.family: "Segoe UI"
153             font.pixelSize: 10
154             font.letterSpacing: 1
```

```
153         }
154
155         TextField {
156             id: kullaniciAdiKutusu
157             width: parent.width
158             height: 42
159             placeholderText: "kullanici_adi"
160             placeholderTextColor: "#4b5563"
161             color: "#ffffff"
162             font.pixelSize: 13
163             leftPadding: 12
164             verticalAlignment: TextInput.AlignVCenter
165             background: Rectangle {
166                 color: "#0a0e17"
167                 radius: 6
168                 border.color: kullaniciAdiKutusu.activeFocus ? "#3b82f6" : "#1e2a3f"
169                 border.width: 1
170             }
171         }
172     }
173
174     Column {
175         width: parent.width
176         spacing: 6
177
178         Text {
179             text: "ŞİFRE"
180             color: "#6b7280"
181             font.family: "Segoe UI"
182             font.pixelSize: 10
```



```
183         font.letterSpacing: 1
184     }
185
186     TextField {
187         id: sifreKutusu
188         width: parent.width
189         height: 42
190         placeholderText: "*****"
191         placeholderTextColor: "#4b5563"
192         echoMode: sayfa.sifreGorunur ? TextInput.Normal : TextInput.Password
193         color: "#ffffff"
194         font.pixelSize: 13
195         leftPadding: 12
196         rightPadding: 48
197         verticalAlignment: TextInput.AlignVCenter
198         background: Rectangle {
199             color: "#0a0e17"
200             radius: 6
201             border.color: sifreKutusu.activeFocus ? "#3b82f6" : "#1e2a3f"
202             border.width: 1
203         }
204
205         Rectangle {
206             id: sifreGosterButonu
207             anchors.right: parent.right
208             anchors.verticalCenter: parent.verticalCenter
209             anchors.rightMargin: 8
210             width: 28
211             height: 28
212             radius: 6
```

```
213         color: sifreGosterMouse.pressed ? "#162033" : (sifreGosterMouse.containsMouse ?
"#111827" : "transparent")

214         border.color: sayfa.sifreGorunur ? "#334155" : "transparent"

215         border.width: 1

216

217         property color ikonRengi: sifreGosterMouse.containsMouse || sayfa.sifreGorunur ?
"#cbd5e1" : "#6b7280"

218

219         Canvas {

220             id: sifreIkonu

221             anchors.centerIn: parent

222             width: 18

223             height: 18

224

225             onPaint: {

226                 var ctx = getContext("2d")

227                 ctx.clearRect(0, 0, width, height)

228                 ctx.lineWidth = 1.7

229                 ctx.lineCap = "round"

230                 ctx.lineJoin = "round"

231                 ctx.strokeStyle = sifreGosterButonu.ikonRengi

232

233                 ctx.beginPath()

234                 ctx.moveTo(1.5, 9)

235                 ctx.bezierCurveTo(4.2, 4.8, 7.0, 3.8, 9, 3.8)

236                 ctx.bezierCurveTo(11.0, 3.8, 13.8, 4.8, 16.5, 9)

237                 ctx.bezierCurveTo(13.8, 13.2, 11.0, 14.2, 9, 14.2)

238                 ctx.bezierCurveTo(7.0, 14.2, 4.2, 13.2, 1.5, 9)

239                 ctx.stroke()

240

241                 ctx.beginPath()

242                 ctx.arc(9, 9, 2.5, 0, Math.PI * 2, false)
```

```
243         ctx.stroke()
244
245         if (sayfa.sifreGorunur) {
246             ctx.beginPath()
247             ctx.moveTo(3, 15)
248             ctx.lineTo(15, 3)
249             ctx.stroke()
250         }
251     }
252
253     Connections {
254         target: sayfa
255         function onSifreGorunurChanged() {
256             sifreIkonu.requestPaint()
257         }
258     }
259
260     Connections {
261         target: sifreGosterButonu
262         function onIkonRengiChanged() {
263             sifreIkonu.requestPaint()
264         }
265     }
266 }
267
268 MouseArea {
269     id: sifreGosterMouse
270     anchors.fill: parent
271     hoverEnabled: true
272     cursorShape: Qt.PointingHandCursor
```

```
273         onClicked: sayfa.sifreGorunur = !sayfa.sifreGorunur
274     }
275 }
276
277     Text {
278         anchors.right: parent.right
279         anchors.verticalCenter: parent.verticalCenter
280         anchors.rightMargin: 12
281         text: sayfa.sifreGorunur ? "👁" : "👁"
282         color: "#6b7280"
283         font.pixelSize: 14
284         visible: false
285
286         MouseArea {
287             anchors.fill: parent
288             anchors.margins: -8
289             cursorShape: Qt.PointingHandCursor
290             onClicked: sayfa.sifreGorunur = !sayfa.sifreGorunur
291         }
292     }
293 }
294 }
295
296     Button {
297         width: parent.width
298         height: 44
299         text: "Giriş Yap"
300         font.family: "Segoe UI"
301         font.pixelSize: 14
302         font.bold: true
303     }
```

```
304         onClicked: {
305             var sonuc = backend.girisYap(kullaniciAdiKutusu.text, sifreKutusu.text)
306
307             if (sonuc) {
308                 sayfa.girisBasarili()
309             } else {
310                 console.log("Kullanici adi veya sifre hatali")
311             }
312         }
313
314         background: Rectangle {
315             radius: 8
316             color: parent.pressed ? "#2563eb" : (parent.hovered ? "#4f8cf7" : "#3b82f6")
317             Behavior on color { ColorAnimation { duration: 120 } }
318         }
319
320         contentItem: Text {
321             text: parent.text
322             color: "#ffffff"
323             font: parent.font
324             horizontalAlignment: Text.AlignHCenter
325             verticalAlignment: Text.AlignVCenter
326         }
327     }
328
329     Rectangle {
330         width: parent.width
331         height: 1
332         color: "#1e2a3f"
333     }
```

```
334
335     Text {
336         anchors.horizontalCenter: parent.horizontalCenter
337         text: "\u00A9 2026 Liya Test A.Ş. SLIPER Analiz Yazılımı"
338         color: "#4b5563"
339         font.family: "Segoe UI"
340         font.pixelSize: 10
341     }
342 }
343 }
344 }
```

```
1 import QtQuick 6.7
2 import QtQuick.Controls 6.7
3 import QtCharts 6.7
4
5 Rectangle {
6     color: "#0a0e17"
7     property real zamanSayaci: 0
8     property int aktifOlcumId: -1
9     property string uyariMesaji: ""
10
11     property var basincGecmis: []
12     property var konumGecmis: []
13     property var hizGecmis: []
14     property var debiGecmis: []
15
16     // TS EN 206 / TS 13515 normal beton dayanım sınıfları (C sınıfı) – çimento dozajı
17     // ve azami su/çimento oranı, ilgili sınıfın minimum bağlayıcı gereksinimine göre.
18     readonly property var recetelListesi: [
19         { sinif: "Reçete Seçin...", cimento: "", suCimento: "", aciklama: "" },
20         { sinif: "C16/20", cimento: "min. 260 kg/m³", suCimento: "maks. 0.65", aciklama: "Hafif yükte yalın/donatılı beton" },
21         { sinif: "C20/25", cimento: "min. 280 kg/m³", suCimento: "maks. 0.60", aciklama: "Standart betonarme" },
22         { sinif: "C25/30", cimento: "min. 300 kg/m³", suCimento: "maks. 0.55", aciklama: "Standart betonarme (TS EN 206)" },
23         { sinif: "C30/37", cimento: "min. 320 kg/m³", suCimento: "maks. 0.50", aciklama: "Standart betonarme" },
24         { sinif: "C30/37 SCC", cimento: "min. 380 kg/m³", suCimento: "maks. 0.45", aciklama: "Kendiliğinden yerleşen beton (SF2)" },
25         { sinif: "C35/45", cimento: "min. 340 kg/m³", suCimento: "maks. 0.45", aciklama: "Orta-yüksek dayanım" },
26         { sinif: "C40/50", cimento: "min. 360 kg/m³", suCimento: "maks. 0.40", aciklama: "Yüksek dayanım" },
27         { sinif: "C45/55", cimento: "min. 380 kg/m³", suCimento: "maks. 0.38", aciklama: "Yüksek dayanım" },
```

```
28      { sinif: "C50/60", cimento: "min. 400 kg/m³", suCimento: "maks. 0.35", aciklama: "Yüksek dayanım (TS
EN 206 normal beton sınırı)" }

29  ]

30

31  signal olcumTamamlandi(int id)

32

33  function resetMeasurementScreen() {

34      aktifOlcumId = -1

35      uyarıMesaji = ""

36      musterıKutusu.text = ""

37      receteKutusu.currentIndex = 0

38      agirlikKutusu.text = ""

39      calculator.sifirla()

40      grafikleriSifirla()

41  }

42

43  function grafikleriSifirla() {

44      zamanSayaci = 0

45      basincSerisi.clear()

46      konumSerisi.clear()

47      hizSerisi.clear()

48      debiSerisi.clear()

49      basincGecmis = []

50      konumGecmis = []

51      hizGecmis = []

52      debiGecmis = []

53      xEkseni.min = 0

54      xEkseni.max = 60

55      yEkseni.min = 0

56      yEkseni.max = 1100

57      konumXEkseni.min = 0
```



```
58     konumXEkseni.max = 60
59     konumYEkseni.max = 500
60     hizXEkseni.min = 0
61     hizXEkseni.max = 60
62     hizYEkseni.max = 4
63     debiXEkseni.min = 0
64     debiXEkseni.max = 60
65     debiYEkseni.max = 180
66 }
67
68 function eksenTavaniHesapla(dizi, tavan, taban) {
69     if (dizi.length === 0) return taban
70     var maxDeger = Math.max.apply(null, dizi)
71     var hedef = maxDeger * 1.25
72     if (hedef < taban) hedef = taban
73     if (hedef > tavan) hedef = tavan
74     return hedef
75 }
76
77 // Basınç (mbar) atmosferik seviyede sabit bir taban etrafında dalgalanır,
78 // bu yüzden sabit tavanlı ölçekleme yerine veriye göre kayan min/max kullanılır.
79 function basincEkseniHesapla(dizi) {
80     if (dizi.length === 0) return { min: 0, max: 1100 }
81     var maxDeger = Math.max.apply(null, dizi)
82     var minDeger = Math.min.apply(null, dizi)
83     var pad = Math.max((maxDeger - minDeger) * 0.2, maxDeger * 0.02, 5)
84     return { min: Math.max(0, minDeger - pad), max: maxDeger + pad }
85 }
86
87 Connections {
```

```
88     target: sensorManager
89     function onVeriGuncellendi() {
90         if (!sensorManager.veriGecerli || aktifOlcumId <= 0 || calculator.duraklatildi) return
91
92         zamanSayaci += 0.2
93
94         basincSerisi.append(zamanSayaci, sensorManager.basinc)
95         konumSerisi.append(zamanSayaci, sensorManager.konum)
96         hizSerisi.append(zamanSayaci, sensorManager.hiz)
97         debiSerisi.append(zamanSayaci, sensorManager.debi)
98
99         basincGecmis.push(sensorManager.basinc)
100        konumGecmis.push(sensorManager.konum)
101        hizGecmis.push(sensorManager.hiz)
102        debiGecmis.push(sensorManager.debi)
103
104        if (zamanSayaci > 60) {
105            basincSerisi.remove(0)
106            konumSerisi.remove(0)
107            hizSerisi.remove(0)
108            debiSerisi.remove(0)
109
110            basincGecmis.shift()
111            konumGecmis.shift()
112            hizGecmis.shift()
113            debiGecmis.shift()
114
115            xEkseni.min = zamanSayaci - 60
116            xEkseni.max = zamanSayaci
117            konumXEkseni.min = zamanSayaci - 60
118            konumXEkseni.max = zamanSayaci
```

```
119         hizXEkseni.min = zamanSayaci - 60
120         hizXEkseni.max = zamanSayaci
121         debiXEkseni.min = zamanSayaci - 60
122         debiXEkseni.max = zamanSayaci
123     }
124
125     var basincAralik = basincEkseniHesapla(basincGecmis)
126     yEkseni.min = basincAralik.min
127     yEkseni.max = basincAralik.max
128     konumYEkseni.max = eksenTavaniHesapla(konumGecmis, 500, 50)
129     hizYEkseni.max = eksenTavaniHesapla(hizGecmis, 4, 0.5)
130     debiYEkseni.max = eksenTavaniHesapla(debiGecmis, 180, 10)
131
132     calculator.konumGuncelle(sensorManager.konum)
133 }
134 }
135
136 Row {
137     anchors.fill: parent
138     spacing: 0
139
140     Rectangle {
141         id: testBilgileriPaneli
142         width: 280
143         height: parent.height
144         color: "#0f1420"
145
146         Text {
147             anchors.top: parent.top
148             anchors.left: parent.left
```

```
149         anchors.margins: 20
150         text: "TEST BİLGİLERİ"
151         color: "#6b7280"
152         font.family: "Segoe UI"
153         font.pixelSize: 12
154         font.bold: true
155         font.letterSpacing: 1
156     }
157
158     Column {
159         id: formAlanlari
160         anchors.top: parent.top
161         anchors.left: parent.left
162         anchors.right: parent.right
163         anchors.margins: 20
164         anchors.topMargin: 60
165         spacing: 6
166
167         Text {
168             text: "Müşteri"
169             color: "#9ca3af"
170             font.family: "Segoe UI"
171             font.pixelSize: 12
172         }
173
174         TextField {
175             id: musterikutusu
176             width: parent.width
177             height: 38
178             placeholderText: "Musteri adi girin..."
179             placeholderTextColor: "#4b5563"
```

```
180         color: "#ffffff"
181
182         font.pixelSize: 13
183
184         leftPadding: 10
185
186         verticalAlignment: TextInput.AlignVCenter
187
188         background: Rectangle {
189             color: "#0a0e17"
190
191             radius: 6
192
193             border.color: musterikutusu.activeFocus ? "#3b82f6" : "#1e2a3f"
194
195             border.width: 1
196         }
197     }
198
199     Text {
200         text: "Beton Reçetesi"
201
202         color: "#9ca3af"
203
204         font.family: "Segoe UI"
205
206         font.pixelSize: 12
207     }
208
209     ComboBox {
210         id: receteKutusu
211
212         width: parent.width
213
214         height: 38
215
216         model: recetelistesi
217
218         textRole: "sinif"
219
220         background: Rectangle {
221             color: "#0a0e17"
222
223             radius: 6
224
225             border.color: receteKutusu.activeFocus ? "#3b82f6" : "#1e2a3f"
```

```

210         border.width: 1
211     }
212
213     contentItem: Text {
214         text: receteKutusu.displayText
215         color: "#9ca3af"
216         font.pixelSize: 13
217         leftPadding: 10
218         verticalAlignment: Text.AlignVCenter
219     }
220
221     delegate: ItemDelegate {
222         id: receteDelege
223         width: receteKutusu.width
224         height: modelData.cimento.length > 0 ? 52 : 36
225         highlighted: receteKutusu.highlightedIndex === index
226
227         background: Rectangle {
228             color: receteDelege.highlighted ? "#182644" : "#0a0e17"
229         }
230
231         contentItem: Column {
232             anchors.verticalCenter: parent.verticalCenter
233             anchors.left: parent.left
234             anchors.leftMargin: 10
235             anchors.right: parent.right
236             anchors.rightMargin: 10
237             spacing: 2
238
239             Text {
240                 text: modelData.sinif

```

```

241                color: "#dce8f5"
242                font.family: "Segoe UI"
243                font.pixelSize: 13
244                font.bold: true
245            }
246
247            Text {
248                visible: modelData.cimento.length > 0
249                text: "Çimento " + modelData.cimento + " • Su/Çimento " +
modelData.suCimento
250                color: "#6b7280"
251                font.family: "Segoe UI"
252                font.pixelSize: 10
253            }
254        }
255    }
256 }
257
258 Text {
259     text: "Eklenen Ağırlık (kg)"
260     color: "#9ca3af"
261     font.family: "Segoe UI"
262     font.pixelSize: 12
263 }
264
265 TextField {
266     id: agirlikKutusu
267     width: parent.width
268     height: 38
269     placeholderText: "örn: 4.8"
270     placeholderTextColor: "#4b5563"

```

```
271         color: "#ffffff"
272
273         font.pixelSize: 13
274
275         leftPadding: 10
276
277         verticalAlignment: TextInput.AlignVCenter
278
279         validator: DoubleValidator { bottom: 0; top: 20; decimals: 1 }
280
281         background: Rectangle {
282             color: "#0a0e17"
283
284             radius: 6
285
286             border.color: agirlikKutusu.activeFocus ? "#3b82f6" : "#1e2a3f"
287
288             border.width: 1
289         }
290     }
291
292     Item { width: parent.width; height: 10 }
293
294     Button {
295         id: baslatButonu
296
297         width: parent.width
298
299         height: 44
300
301         text: "▶ Ölçümü Başlat"
302
303         font.family: "Segoe UI"
304
305         font.pixelSize: 14
306
307         font.bold: true
308
309         enabled: aktifOlcumId <= 0
310
311         onClicked: {
312             if (musteriKutusu.text.trim().length === 0) {
313                 uyariMesaji = "Lütfen müşteri adını girin."
314                 return
315             }
316         }
317     }
```



```
301
302         if (receteKutusu.currentIndex <= 0) {
303             uyarıMesajı = "Lütfen bir beton reçetesi seçin."
304             return
305         }
306
307         if (!sensorManager.verıGecerli) {
308             console.warn("Gerçek sensor verisi yok, ölçüm baslatılmadı.")
309             uyarıMesajı = "Cihaz bağlı değil. Lütfen önce sol alttan SLİPER-ESP32'ye
bağlanın."
310             return
311         }
312
313         uyarıMesajı = ""
314         grafikleriSıfırla()
315
316         calculator.sıfırla()
317         aktifÖlçümId = database.ölçümBaslat(
318             müşteriKutusu.text,
319             receteKutusu.currentText,
320             parseFloat(ağırlıkKutusu.text)
321         )
322         console.log("Aktif ölçüm id:", aktifÖlçümId)
323     }
324
325     background: Rectangle {
326         radius: 8
327         color: !baslatButonu.enabled ? "#14532d" : (baslatButonu.pressed ? "#15803d" :
(baslatButonu.hovered ? "#22c55e" : "#16a34a"))
328         opacity: baslatButonu.enabled ? 1.0 : 0.5
329         Behavior on color { ColorAnimation { duration: 120 } }
330     }
```

```

331
332         contentItem: Text {
333             text: baslatButonu.text
334             color: "#ffffff"
335             opacity: baslatButonu.enabled ? 1.0 : 0.6
336             font: baslatButonu.font
337             horizontalAlignment: Text.AlignHCenter
338             verticalAlignment: Text.AlignVCenter
339         }
340     }
341
342     Text {
343         width: parent.width
344         visible: uyariMesaji.length > 0
345         text: uyariMesaji
346         color: "#f87171"
347         font.family: "Segoe UI"
348         font.pixelSize: 11
349         wrapMode: Text.WordWrap
350     }
351
352     Row {
353         width: parent.width
354         spacing: 10
355
356         Button {
357             id: duraklatButonu
358             width: (parent.width - parent.spacing) / 2
359             height: 40
360             text: calculator.duraklatildi ? "► Devam Et" : "|| Duraklat"

```

```

361         font.family: "Segoe UI"
362         font.pixelSize: 13
363
364         onClicked: {
365             if (calculator.duraklatildi) {
366                 calculator.devamEt()
367             } else {
368                 calculator.duraklat()
369             }
370         }
371
372         background: Rectangle {
373             radius: 8
374             color: duraklatButonu.pressed ? "#78350f" : (duraklatButonu.hovered ? "#a16207" :
"#92400e")
375             border.color: "#d97706"
376             border.width: 1
377         }
378
379         contentItem: Text {
380             text: duraklatButonu.text
381             color: "#ffffff"
382             font: duraklatButonu.font
383             horizontalAlignment: Text.AlignHCenter
384             verticalAlignment: Text.AlignVCenter
385         }
386     }
387
388     Button {
389         id: bitirButonu
390         width: (parent.width - parent.spacing) / 2

```

```

391         height: 40
392         text: "■ Bitir"
393         font.family: "Segoe UI"
394         font.pixelSize: 13
395
396         onClicked: {
397             if (aktifOlcumId > 0) {
398                 bitirOnayPopup.open()
399             } else {
400                 uyarıMesajı = "Aktif bir ölçüm yok. Önce ölçümü başlatın."
401             }
402         }
403
404         background: Rectangle {
405             radius: 8
406             color: bitirButonu.pressed ? "#7f1d1d" : (bitirButonu.hovered ? "#b91c1c" :
"#991b1b")
407             border.color: "#dc2626"
408             border.width: 1
409         }
410
411         contentItem: Text {
412             text: bitirButonu.text
413             color: "#ffffff"
414             font: bitirButonu.font
415             horizontalAlignment: Text.AlignHCenter
416             verticalAlignment: Text.AlignVCenter
417         }
418     }
419 }
420 }
```

```
421
422     Rectangle {
423         id: durumKarti
424         anchors.top: formAlanlari.bottom
425         anchors.left: parent.left
426         anchors.right: parent.right
427         anchors.topMargin: 20
428         anchors.leftMargin: 20
429         anchors.rightMargin: 20
430         height: 52
431         radius: 10
432         color: "#0a0e17"
433         border.color: "#1e2a3f"
434         border.width: 1
435
436     Row {
437         anchors.left: parent.left
438         anchors.verticalCenter: parent.verticalCenter
439         anchors.leftMargin: 14
440         spacing: 10
441
442         Rectangle {
443             id: trafikIsigi
444             width: 12
445             height: 12
446             radius: 6
447             anchors.verticalCenter: parent.verticalCenter
448             color: {
449                 if (!sensorManager.veriGecerli) return "#dc2626"
450                 if (calculator.duraklatildi) return "#6b7280"
451                 if (calculator.durum === "YUKARIDA") return "#16a34a"
```

```
452         if (calculator.durum === "INIYOR") return "#f59e0b"
453         return "#dc2626"
454     }
455
456     Rectangle {
457         anchors.centerIn: parent
458         width: parent.width + 8
459         height: parent.height + 8
460         radius: width / 2
461         color: "transparent"
462         border.color: parent.color
463         border.width: 1
464         opacity: 0.35
465     }
466 }
467
468 Text {
469     anchors.verticalCenter: parent.verticalCenter
470     text: {
471         if (!sensorManager.veriGecerli) return "VERI YOK"
472         if (calculator.duraklatildi) return "DURAKLATILDI"
473         return calculator.durum
474     }
475     color: "#dce8f5"
476     font.family: "Segoe UI"
477     font.pixelSize: 13
478     font.bold: true
479 }
480 }
481
```

```

482         Row {
483             anchors.right: parent.right
484             anchors.verticalCenter: parent.verticalCenter
485             anchors.rightMargin: 14
486             spacing: 6
487
488             Text {
489                 anchors.verticalCenter: parent.verticalCenter
490                 text: "STROKE"
491                 color: "#6b7280"
492                 font.family: "Segoe UI"
493                 font.pixelSize: 10
494                 font.letterSpacing: 1
495             }
496
497             Text {
498                 anchors.verticalCenter: parent.verticalCenter
499                 text: calculator.strokeSayisi
500                 color: "#e8a020"
501                 font.family: "Segoe UI"
502                 font.pixelSize: 16
503                 font.bold: true
504             }
505         }
506     }
507
508     Text {
509         id: anlikDegerlerBaslik
510         anchors.top: durumKarti.bottom
511         anchors.left: parent.left
512         anchors.topMargin: 24

```

```
513         anchors.leftMargin: 20
514         text: "ANLIK DEĞERLER"
515         color: "#6b7280"
516         font.family: "Segoe UI"
517         font.pixelSize: 12
518         font.bold: true
519         font.letterSpacing: 1
520     }
521
522     Grid {
523         anchors.top: anlikDegerlerBaslik.bottom
524         anchors.left: parent.left
525         anchors.topMargin: 12
526         anchors.leftMargin: 20
527         columns: 2
528         spacing: 10
529
530         Rectangle {
531             id: basincKarti
532             width: 115
533             height: 90
534             radius: 10
535             color: "#0a0e17"
536             border.color: "#1e2a3f"
537             border.width: 1
538             clip: true
539
540             Rectangle { width: 4; height: parent.height; color: "#3b82f6" }
541
542             Column {
```



```
543         anchors.top: parent.top
544         anchors.right: parent.right
545         anchors.bottom: parent.bottom
546         anchors.left: parent.left
547         anchors.topMargin: 10
548         anchors.rightMargin: 10
549         anchors.bottomMargin: 10
550         anchors.leftMargin: 16
551         spacing: 6
552
553     Text {
554         text: "BASINÇ"
555         color: "#6b7280"
556         font.family: "Segoe UI"
557         font.pixelSize: 10
558         font.letterSpacing: 1
559     }
560
561     Text {
562         text: sensorManager.veriGecerli ? sensorManager.basinc.toFixed(1) : "--"
563         color: "#3b82f6"
564         font.family: "Segoe UI"
565         font.pixelSize: 22
566         font.bold: true
567     }
568
569     Text {
570         text: "mbar"
571         color: "#4b5563"
572         font.family: "Segoe UI"
573         font.pixelSize: 10
```

```
574         }
575     }
576 }
577
578 Rectangle {
579     width: 115
580     height: 90
581     radius: 10
582     color: "#0a0e17"
583     border.color: "#1e2a3f"
584     border.width: 1
585     clip: true
586
587     Rectangle { width: 4; height: parent.height; color: "#9333ea" }
588
589     Column {
590         anchors.top: parent.top
591         anchors.right: parent.right
592         anchors.bottom: parent.bottom
593         anchors.left: parent.left
594         anchors.topMargin: 10
595         anchors.rightMargin: 10
596         anchors.bottomMargin: 10
597         anchors.leftMargin: 16
598         spacing: 6
599
600         Text {
601             text: "KONUM"
602             color: "#6b7280"
603             font.family: "Segoe UI"
```

```
604         font.pixelSize: 10
605         font.letterSpacing: 1
606     }
607
608     Text {
609         text: sensorManager.veriGecerli ? sensorManager.konum.toFixed(1) : "--"
610         color: "#9333ea"
611         font.family: "Segoe UI"
612         font.pixelSize: 22
613         font.bold: true
614     }
615
616     Text {
617         text: "mm"
618         color: "#4b5563"
619         font.family: "Segoe UI"
620         font.pixelSize: 10
621     }
622 }
623
624
625 Rectangle {
626     width: 115
627     height: 90
628     radius: 10
629     color: "#0a0e17"
630     border.color: "#1e2a3f"
631     border.width: 1
632     clip: true
633
634     Rectangle { width: 4; height: parent.height; color: "#f59e0b" }
```

```
635
636         Column {
637             anchors.top: parent.top
638             anchors.right: parent.right
639             anchors.bottom: parent.bottom
640             anchors.left: parent.left
641             anchors.topMargin: 10
642             anchors.rightMargin: 10
643             anchors.bottomMargin: 10
644             anchors.leftMargin: 16
645             spacing: 6
646
647             Text {
648                 text: "HIZ"
649                 color: "#6b7280"
650                 font.family: "Segoe UI"
651                 font.pixelSize: 10
652                 font.letterSpacing: 1
653             }
654
655             Text {
656                 text: sensorManager.veriGecerli ? sensorManager.hiz.toFixed(1) : "--"
657                 color: "#f59e0b"
658                 font.family: "Segoe UI"
659                 font.pixelSize: 22
660                 font.bold: true
661             }
662
663             Text {
664                 text: "m/s"
```

```
665         color: "#4b5563"
666         font.family: "Segoe UI"
667         font.pixelSize: 10
668     }
669 }
670 }
671
672 Rectangle {
673     width: 115
674     height: 90
675     radius: 10
676     color: "#0a0e17"
677     border.color: "#1e2a3f"
678     border.width: 1
679     clip: true
680
681     Rectangle { width: 4; height: parent.height; color: "#16a34a" }
682
683     Column {
684         anchors.top: parent.top
685         anchors.right: parent.right
686         anchors.bottom: parent.bottom
687         anchors.left: parent.left
688         anchors.topMargin: 10
689         anchors.rightMargin: 10
690         anchors.bottomMargin: 10
691         anchors.leftMargin: 16
692         spacing: 6
693
694         Text {
695             text: "DEBI"
```

```
696         color: "#6b7280"
697         font.family: "Segoe UI"
698         font.pixelSize: 10
699         font.letterSpacing: 1
700     }
701
702     Text {
703         text: sensorManager.veriGecerli ? sensorManager.debi.toFixed(1) : "--"
704         color: "#16a34a"
705         font.family: "Segoe UI"
706         font.pixelSize: 22
707         font.bold: true
708     }
709
710     Text {
711         text: "m3/h"
712         color: "#4b5563"
713         font.family: "Segoe UI"
714         font.pixelSize: 10
715     }
716 }
717 }
718 }
719 }
720
721 Rectangle {
722     id: panelAyraci
723     width: 3
724     height: parent.height
725     color: "#05070b"
```

```
726
727     Rectangle {
728         anchors.right: parent.right
729         anchors.top: parent.top
730         anchors.bottom: parent.bottom
731         width: 1
732         color: "#403b82f6"
733     }
734 }
735
736 Item {
737     width: parent.width - testBilgileriPaneli.width - panelAyraci.width
738     height: parent.height
739
740     Grid {
741         anchors.fill: parent
742         anchors.margins: 16
743         columns: 2
744         spacing: 16
745
746         Rectangle {
747             width: (parent.width - parent.spacing) / 2
748             height: (parent.height - parent.spacing) / 2
749             radius: 10
750             color: "#0f1420"
751             border.color: "#17263d"
752             border.width: 1
753
754             Item {
755                 id: basincBaslik
756                 anchors.top: parent.top
```

```

757         anchors.left: parent.left
758         anchors.right: parent.right
759         anchors.margins: 14
760         height: 20
761
762         Rectangle {
763             id: basincNoktasi
764             width: 8
765             height: 8
766             radius: 4
767             color: "#3b82f6"
768             anchors.left: parent.left
769             anchors.verticalCenter: parent.verticalCenter
770         }
771
772         Text {
773             anchors.left: basincNoktasi.right
774             anchors.leftMargin: 8
775             anchors.verticalCenter: parent.verticalCenter
776             text: "Basınc - Zaman"
777             color: "#9ca3af"
778             font.family: "Segoe UI"
779             font.pixelSize: 13
780             font.bold: true
781         }
782
783         Text {
784             anchors.right: parent.right
785             anchors.verticalCenter: parent.verticalCenter
786             text: sensorManager.veriGecerli ? sensorManager.basinc.toFixed(1) + " mbar" : "--
mbar"

```



```
787         color: "#3b82f6"
788         font.family: "Segoe UI"
789         font.pixelSize: 13
790         font.bold: true
791     }
792 }
793
794 ChartView {
795     anchors.top: basincBaslik.bottom
796     anchors.left: parent.left
797     anchors.right: parent.right
798     anchors.bottom: parent.bottom
799     anchors.topMargin: 4
800     anchors.bottomMargin: 4
801     backgroundColor: "transparent"
802     legend.visible: false
803     antialiasing: true
804
805     ValueAxis {
806         id: xEkseni
807         min: 0
808         max: 60
809         gridLineColor: "#182131"
810         labelsColor: "#4b5563"
811         labelsFont.pixelSize: 9
812         lineVisible: false
813         minorGridVisible: false
814     }
815
816     ValueAxis {
```

```

817             id: yEkseni
818             min: 0
819             max: 1100
820             gridLineColor: "#182131"
821             labelsColor: "#4b5563"
822             labelsFont.pixelSize: 9
823             lineVisible: false
824             minorGridVisible: false
825         }
826
827         LineSeries {
828             id: basincSerisi
829             axisX: xEkseni
830             axisY: yEkseni
831             color: "#3b82f6"
832             width: 2
833         }
834     }
835 }
836
837 Rectangle {
838     width: (parent.width - parent.spacing) / 2
839     height: (parent.height - parent.spacing) / 2
840     radius: 10
841     color: "#0f1420"
842     border.color: "#241a38"
843     border.width: 1
844
845     Item {
846         id: konumBaslik
847         anchors.top: parent.top

```

```

848         anchors.left: parent.left
849         anchors.right: parent.right
850         anchors.margins: 14
851         height: 20
852
853         Rectangle {
854             id: konumNoktasi
855             width: 8
856             height: 8
857             radius: 4
858             color: "#9333ea"
859             anchors.left: parent.left
860             anchors.verticalCenter: parent.verticalCenter
861         }
862
863         Text {
864             anchors.left: konumNoktasi.right
865             anchors.leftMargin: 8
866             anchors.verticalCenter: parent.verticalCenter
867             text: "Konum - Zaman"
868             color: "#9ca3af"
869             font.family: "Segoe UI"
870             font.pixelSize: 13
871             font.bold: true
872         }
873
874         Text {
875             anchors.right: parent.right
876             anchors.verticalCenter: parent.verticalCenter
877             text: sensorManager.veriGecerli ? sensorManager.konum.toFixed(1) + " mm" : "--
mm"

```

```
878             color: "#9333ea"
879             font.family: "Segoe UI"
880             font.pixelSize: 13
881             font.bold: true
882         }
883     }
884
885     ChartView {
886         anchors.top: konumBaslik.bottom
887         anchors.left: parent.left
888         anchors.right: parent.right
889         anchors.bottom: parent.bottom
890         anchors.topMargin: 4
891         anchors.bottomMargin: 4
892         backgroundColor: "transparent"
893         legend.visible: false
894         antialiasing: true
895
896         ValueAxis {
897             id: konumXEkseni
898             min: 0
899             max: 60
900             gridLineColor: "#182131"
901             labelsColor: "#4b5563"
902             labelsFont.pixelSize: 9
903             lineVisible: false
904             minorGridVisible: false
905         }
906
907         ValueAxis {
```

```

908         id: konumYEkseni
909         min: 0
910         max: 500
911         gridLineColor: "#182131"
912         labelsColor: "#4b5563"
913         labelsFont.pixelSize: 9
914         lineVisible: false
915         minorGridVisible: false
916     }
917
918     LineSeries {
919         id: konumSerisi
920         axisX: konumXEkseni
921         axisY: konumYEkseni
922         color: "#9333ea"
923         width: 2
924     }
925
926     Connections {
927         target: calculator
928         function onStrokeSayisiChanged() {
929             if (aktifOlcumId > 0 && sensorManager.veriGecerli) {
930                 database.strokeKaydet(aktifOlcumId, sensorManager.basinc,
sensorManager.konum, sensorManager.debi, true)
931             }
932         }
933     }
934 }
935 }
936
937 Rectangle {

```

```
938         width: (parent.width - parent.spacing) / 2
939
940         height: (parent.height - parent.spacing) / 2
941
942         radius: 10
943
944         color: "#0f1420"
945
946         border.color: "#3a2a14"
947
948         border.width: 1
949
950
951
952
953     Item {
954
955         id: hizBaslik
956
957         anchors.top: parent.top
958
959         anchors.left: parent.left
960
961         anchors.right: parent.right
962
963         anchors.margins: 14
964
965         height: 20
966
967
968
969
970         Rectangle {
971
972             id: hizNoktasi
973
974             width: 8
975
976             height: 8
977
978             radius: 4
979
980             color: "#f59e0b"
981
982             anchors.left: parent.left
983
984             anchors.verticalCenter: parent.verticalCenter
985
986         }
987
988
989
990
991     Text {
992
993         anchors.left: hizNoktasi.right
994
995         anchors.leftMargin: 8
996
997         anchors.verticalCenter: parent.verticalCenter
998
999         text: "Hız - Zaman"
1000
1001         color: "#9ca3af"
```

```
969         font.family: "Segoe UI"
970         font.pixelSize: 13
971         font.bold: true
972     }
973
974     Text {
975         anchors.right: parent.right
976         anchors.verticalCenter: parent.verticalCenter
977         text: sensorManager.veriGecerli ? sensorManager.hiz.toFixed(1) + " m/s" : "--
m/s"
978         color: "#f59e0b"
979         font.family: "Segoe UI"
980         font.pixelSize: 13
981         font.bold: true
982     }
983 }
984
985 ChartView {
986     anchors.top: hizBaslik.bottom
987     anchors.left: parent.left
988     anchors.right: parent.right
989     anchors.bottom: parent.bottom
990     anchors.topMargin: 4
991     anchors.bottomMargin: 4
992     backgroundColor: "transparent"
993     legend.visible: false
994     antialiasing: true
995
996     ValueAxis {
997         id: hizXEkseni
998         min: 0
```

```
999             max: 60
1000             gridLineColor: "#182131"
1001             labelsColor: "#4b5563"
1002             labelsFont.pixelSize: 9
1003             lineVisible: false
1004             minorGridVisible: false
1005         }
1006
1007         ValueAxis {
1008             id: hizYEkseni
1009             min: 0
1010             max: 4
1011             gridLineColor: "#182131"
1012             labelsColor: "#4b5563"
1013             labelsFont.pixelSize: 9
1014             lineVisible: false
1015             minorGridVisible: false
1016         }
1017
1018         LineSeries {
1019             id: hizSerisi
1020             axisX: hizXEkseni
1021             axisY: hizYEkseni
1022             color: "#f59e0b"
1023             width: 2
1024         }
1025     }
1026 }
1027
1028 Rectangle {
```



```
1029         width: (parent.width - parent.spacing) / 2
1030
1031         height: (parent.height - parent.spacing) / 2
1032
1033         radius: 10
1034
1035         color: "#0f1420"
1036
1037         border.color: "#163321"
1038
1039         border.width: 1
1040
1041
1042     Item {
1043
1044         id: debiBaslik
1045
1046         anchors.top: parent.top
1047
1048         anchors.left: parent.left
1049
1050         anchors.right: parent.right
1051
1052         anchors.margins: 14
1053
1054         height: 20
1055
1056
1057         Rectangle {
1058
1059             id: debiNoktasi
1060
1061             width: 8
1062
1063             height: 8
1064
1065             radius: 4
1066
1067             color: "#16a34a"
1068
1069             anchors.left: parent.left
1070
1071             anchors.verticalCenter: parent.verticalCenter
1072
1073         }
1074
1075
1076         Text {
1077
1078             anchors.left: debiNoktasi.right
1079
1080             anchors.leftMargin: 8
1081
1082             anchors.verticalCenter: parent.verticalCenter
1083
1084             text: "Debi - Zaman"
1085
1086             color: "#9ca3af"
```

```
1060         font.family: "Segoe UI"
1061         font.pixelSize: 13
1062         font.bold: true
1063     }
1064
1065     Text {
1066         anchors.right: parent.right
1067         anchors.verticalCenter: parent.verticalCenter
1068         text: sensorManager.veriGecerli ? sensorManager.debi.toFixed(1) + " m3/h" : "--
m3/h"
1069         color: "#16a34a"
1070         font.family: "Segoe UI"
1071         font.pixelSize: 13
1072         font.bold: true
1073     }
1074 }
1075
1076 ChartView {
1077     anchors.top: debiBaslik.bottom
1078     anchors.left: parent.left
1079     anchors.right: parent.right
1080     anchors.bottom: parent.bottom
1081     anchors.topMargin: 4
1082     anchors.bottomMargin: 4
1083     backgroundColor: "transparent"
1084     legend.visible: false
1085     antialiasing: true
1086
1087     ValueAxis {
1088         id: debiXEkseni
1089         min: 0
```

```
1090             max: 60
1091             gridLineColor: "#182131"
1092             labelsColor: "#4b5563"
1093             labelsFont.pixelSize: 9
1094             lineVisible: false
1095             minorGridVisible: false
1096         }
1097
1098         ValueAxis {
1099             id: debiYEkseni
1100             min: 0
1101             max: 180
1102             gridLineColor: "#182131"
1103             labelsColor: "#4b5563"
1104             labelsFont.pixelSize: 9
1105             lineVisible: false
1106             minorGridVisible: false
1107         }
1108
1109         LineSeries {
1110             id: debiSerisi
1111             axisX: debiXEkseni
1112             axisY: debiYEkseni
1113             color: "#16a34a"
1114             width: 2
1115         }
1116     }
1117 }
1118 }
1119 }
```

```
1120     }
1121
1122     Popup {
1123         id: bitirOnayPopup
1124         modal: true
1125         focus: true
1126         anchors.centerIn: parent
1127         width: 340
1128         padding: 24
1129         closePolicy: Popup.CloseOnEscape | Popup.CloseOnPressOutside
1130
1131         background: Rectangle {
1132             color: "#0f1420"
1133             radius: 12
1134             border.color: "#1e2a3f"
1135             border.width: 1
1136         }
1137
1138         Overlay.modal: Rectangle {
1139             color: "#a6000000"
1140         }
1141
1142         contentItem: Column {
1143             spacing: 20
1144
1145             Text {
1146                 width: 292
1147                 text: "Ölçümü bitirmek istediğinize emin misiniz?"
1148                 color: "#dce8f5"
1149                 font.family: "Segoe UI"
1150                 font.pixelSize: 14
```

```
1151         font.bold: true
1152         wrapMode: Text.WordWrap
1153     }
1154
1155     Column {
1156         width: 292
1157         spacing: 10
1158
1159         Button {
1160             id: kaydetBitirButonu
1161             width: parent.width
1162             height: 40
1163             text: "📁 Kaydet ve Bitir"
1164             font.family: "Segoe UI"
1165             font.pixelSize: 13
1166
1167             onClicked: {
1168                 bitirOnayPopup.close()
1169                 var bitenId = aktifOlcumId
1170                 console.log("Olcum kaydedilip sonlandirildi, id:", bitenId)
1171                 resetMeasurementScreen()
1172                 olcumTamamlandi(bitenId)
1173             }
1174
1175             background: Rectangle {
1176                 radius: 8
1177                 color: kaydetBitirButonu.pressed ? "#14532d" : (kaydetBitirButonu.hovered ? "#15803d"
: "#16a34a")
1178                 border.color: "#22c55e"
1179                 border.width: 1
1180             }
```

```

1181
1182         contentItem: Text {
1183             text: kaydetBitirButonu.text
1184             color: "#ffffff"
1185             font: kaydetBitirButonu.font
1186             horizontalAlignment: Text.AlignHCenter
1187             verticalAlignment: Text.AlignVCenter
1188         }
1189     }
1190
1191     Button {
1192         id: silBitirButonu
1193         width: parent.width
1194         height: 40
1195         text: "🗑 Testi Sil ve Bitir"
1196         font.family: "Segoe UI"
1197         font.pixelSize: 13
1198
1199         onClicked: {
1200             bitirOnayPopup.close()
1201             var silinecekId = aktifOlcumId
1202             var basarili = database.olcumSil(silinecekId)
1203             console.log("Olcum silinip sonlandirildi, id:", silinecekId, "basarili:", basarili)
1204             resetMeasurementScreen()
1205         }
1206
1207         background: Rectangle {
1208             radius: 8
1209             color: silBitirButonu.pressed ? "#7f1d1d" : (silBitirButonu.hovered ? "#b91c1c" :
"#991b1b")
1210             border.color: "#dc2626"

```

```
1211         border.width: 1
1212     }
1213
1214     contentItem: Text {
1215         text: silBitirButonu.text
1216         color: "#ffffff"
1217         font: silBitirButonu.font
1218         horizontalAlignment: Text.AlignHCenter
1219         verticalAlignment: Text.AlignVCenter
1220     }
1221 }
1222
1223 Button {
1224     id: vazgecButonu
1225     width: parent.width
1226     height: 40
1227     text: "Vazgeç, Teste Devam Et"
1228     font.family: "Segoe UI"
1229     font.pixelSize: 13
1230
1231     onClicked: bitirOnayPopup.close()
1232
1233     background: Rectangle {
1234         radius: 8
1235         color: vazgecButonu.pressed ? "#1e2a3f" : (vazgecButonu.hovered ? "#243349" :
"#182131")
1236         border.color: "#2c3b52"
1237         border.width: 1
1238     }
1239
1240     contentItem: Text {
```

```
1241         text: vazgecButonu.text
1242         color: "#dce8f5"
1243         font: vazgecButonu.font
1244         horizontalAlignment: Text.AlignHCenter
1245         verticalAlignment: Text.AlignVCenter
1246     }
1247 }
1248 }
1249 }
1250 }
1251 }
```


qml/pages/SonuclarPage.qml

```
1 import QtQuick 6.7
2 import QtQuick.Controls 6.7
3 import QtCharts 6.7
4
5 Rectangle {
6     color: "#0a0e17"
7
8     property int olcumId: -1
9     property real hesaplananTau0: 0
10    property real hesaplananMu: 0
11    property int playbackAdimi: -1
12    property bool oynatiliyor: false
13    property int playbackHizi: 1
14    property string musteriAdi: ""
15    property string receteAdi: ""
16    property real agirlikDegeri: 0
17    property string olcumTarihi: ""
18
19    onOlcumIdChanged: verileriYukle()
20
21    function verileriYukle() {
22        if (olcumId <= 0) {
23            return
24        }
25
26        playbackAdimi = -1
27        oynatiliyor = false
28        playbackTimer.stop()
29
30        var bilgi = database.olcumBilgisiGetir(olcumId)
```

```
31     muster Adi = bilgi.bulundu ? bilgi.musteri : "Bilinmiyor"
32     receteAdi = bilgi.bulundu ? bilgi.recete : "Bilinmiyor"
33     agirlikDegeri = bilgi.bulundu ? bilgi.agirlik : 0
34     olcumTarihi = bilgi.bulundu ? bilgi.tarih : ""
35
36     var sonuc = database.binghamHesapla(olcumId)
37
38     hesaplananTau0 = sonuc.tau0
39     hesaplananMu = sonuc.mu
40
41     tau0Metni.text = sonuc.tau0.toFixed(2) + " mbar"
42     muMetni.text = sonuc.mu.toFixed(2) + " mbar·h/m³"
43     r2Metni.text = sonuc.r2.toFixed(2)
44
45     durumKutusu.durumIyiMi = sonuc.yeterliVeri && sonuc.tau0 < 5 && sonuc.mu < 10
46     durumKutusu.tau0Yuksek = sonuc.tau0 >= 5
47     durumKutusu.muYuksek = sonuc.mu >= 10
48
49     pqSerisi.clear()
50     regresyonCizgisi.clear()
51     oynatmaSerisi.clear()
52     strokeModeli.clear()
53
54     var strokeListesi = database.strokeVerileriGetir(olcumId)
55     var pMaks = 0
56     var qMaks = 0
57     for (var i = 0; i < strokeListesi.length; i++) {
58         var s = strokeListesi[i]
59         pqSerisi.append(s.debi, s.basinc)
60         strokeModeli.append({ stroke: s.stroke, p: s.basinc, q: s.debi, gecerli: s.gecerli })
```

```
61         if (s.basinc > pMaks) pMaks = s.basinc
62         if (s.debi > qMaks) qMaks = s.debi
63     }
64
65     // Eksenler gerçek ölçüm verisine göre ayarlanır; sabit aralık kullanılırsa
66     // basınc/debi bu aralığı aştığında noktalar (ve oynatma imleci) grafik
67     // dışına taşıp görünmez olur.
68     qMaks = Math.max(qMaks, 5)
69     regresyonCizgisi.append(0, sonuc.tau0)
70     regresyonCizgisi.append(qMaks, sonuc.tau0 + sonuc.mu * qMaks)
71
72     pMaks = Math.max(pMaks, sonuc.tau0 + sonuc.mu * qMaks, 10)
73     pEkseni.max = pMaks * 1.15
74     qEkseni.max = qMaks * 1.15
75 }
76
77 Timer {
78     id: playbackTimer
79     interval: 1000 / playbackHizi
80     repeat: true
81     running: false
82
83     onTriggered: {
84         if (playbackAdimi < strokeModeli.count - 1) {
85             playbackAdimi++
86             var satir = strokeModeli.get(playbackAdimi)
87             oynatmaSerisi.clear()
88             oynatmaSerisi.append(satir.q, satir.p)
89         } else {
90             oynatiliyor = false
91             playbackTimer.stop()
```

```
92         }
93     }
94 }
95
96 Row {
97     anchors.fill: parent
98     spacing: 0
99
100     Rectangle {
101         id: ozetPaneli
102         width: 280
103         height: parent.height
104         color: "#0f1420"
105
106         Text {
107             anchors.top: parent.top
108             anchors.left: parent.left
109             anchors.margins: 20
110             text: "SONUÇ ÖZETİ"
111             color: "#6b7280"
112             font.family: "Segoe UI"
113             font.pixelSize: 12
114             font.bold: true
115             font.letterSpacing: 1
116         }
117
118         Flickable {
119             anchors.top: parent.top
120             anchors.left: parent.left
121             anchors.right: parent.right
```

```

122         anchors.bottom: parent.bottom
123
124         anchors.margins: 20
125
126         anchors.topMargin: 60
127
128         contentWidth: width
129
130         contentHeight: sonucKartlari.height
131
132         clip: true
133
134     Column {
135
136         id: sonucKartlari
137
138         width: parent.width
139
140         spacing: 10
141
142
143         Rectangle {
144
145             width: parent.width
146
147             height: 70
148
149             radius: 10
150
151             color: "#0a0e17"
152
153             border.color: "#17263d"
154
155             border.width: 1
156
157             clip: true
158
159
160             Rectangle { width: 4; height: parent.height; color: "#3b82f6" }
161
162
163             Column {
164
165                 anchors.left: parent.left
166
167                 anchors.verticalCenter: parent.verticalCenter
168
169                 anchors.leftMargin: 20
170
171                 spacing: 2
172
173
174                 Text {
175
176                     text: "AKMA GERİLMESİ ( $\tau_0$ )"

```

```
153         color: "#6b7280"
154
155         font.family: "Segoe UI"
156
157         font.pixelSize: 10
158
159         font.letterSpacing: 1
160
161     }
162
163     Text {
164
165         id: tau0Metni
166
167         text: "-"
168
169         color: "#3b82f6"
170
171         font.family: "Segoe UI"
172
173         font.pixelSize: 20
174
175         font.bold: true
176
177     }
178
179 }
180
181
182 Rectangle {
183
184     width: parent.width
185
186     height: 70
187
188     radius: 10
189
190     color: "#0a0e17"
191
192     border.color: "#241a38"
193
194     border.width: 1
195
196     clip: true
197
198
199     Rectangle { width: 4; height: parent.height; color: "#9333ea" }
200
201
202     Column {
203
204         anchors.left: parent.left
```

```
183         anchors.verticalCenter: parent.verticalCenter
184         anchors.leftMargin: 20
185         spacing: 2
186
187         Text {
188             text: "PLASTİK VISKOZİTE (μ)"
189             color: "#6b7280"
190             font.family: "Segoe UI"
191             font.pixelSize: 10
192             font.letterSpacing: 1
193         }
194
195         Text {
196             id: muMetni
197             text: "-"
198             color: "#9333ea"
199             font.family: "Segoe UI"
200             font.pixelSize: 20
201             font.bold: true
202         }
203     }
204 }
205
206 Rectangle {
207     width: parent.width
208     height: 70
209     radius: 10
210     color: "#0a0e17"
211     border.color: "#163321"
212     border.width: 1
213     clip: true
```

```
214
215         Rectangle { width: 4; height: parent.height; color: "#16a34a" }
216
217         Column {
218             anchors.left: parent.left
219             anchors.verticalCenter: parent.verticalCenter
220             anchors.leftMargin: 20
221             spacing: 2
222
223             Text {
224                 text: "UYUM KALİTESİ (R²)"
225                 color: "#6b7280"
226                 font.family: "Segoe UI"
227                 font.pixelSize: 10
228                 font.letterSpacing: 1
229             }
230
231             Text {
232                 id: r2Metni
233                 text: "-"
234                 color: "#16a34a"
235                 font.family: "Segoe UI"
236                 font.pixelSize: 20
237                 font.bold: true
238             }
239         }
240     }
241
242     Rectangle {
243         width: parent.width
```



```
244         radius: 10
245         color: "#0a0e17"
246         border.color: "#1e2a3f"
247         border.width: 1
248         height: tahminIcerik.implicitHeight + 24
249
250     Column {
251         id: tahminIcerik
252         anchors.top: parent.top
253         anchors.left: parent.left
254         anchors.right: parent.right
255         anchors.margins: 12
256         spacing: 8
257
258         Text {
259             text: "BORU HATTI TAHMİNİ"
260             color: "#6b7280"
261             font.family: "Segoe UI"
262             font.pixelSize: 10
263             font.letterSpacing: 1
264         }
265
266         TextField {
267             id: uzunlukKutusu
268             width: parent.width
269             height: 34
270             placeholderText: "Boru Uzunluğu (m)"
271             placeholderTextColor: "#4b5563"
272             color: "ffffff"
273             font.pixelSize: 12
274             leftPadding: 10
```

```
275         verticalAlignment: TextInput.AlignVCenter
276
277         validator: DoubleValidator { bottom: 0; top: 2000; decimals: 0 }
278
279         background: Rectangle {
280             color: "#060d17"
281
282             radius: 6
283
284             border.color: uzunlukKutusu.activeFocus ? "#3b82f6" : "#1e2a3f"
285
286             border.width: 1
287         }
288     }
289
290     TextField {
291         id: debiKutusu
292
293         width: parent.width
294
295         height: 34
296
297         placeholderText: "Hedef Debi (m³/h)"
298
299         placeholderTextColor: "#4b5563"
300
301         color: "#ffffff"
302
303         font.pixelSize: 12
304
305         leftPadding: 10
306
307         verticalAlignment: TextInput.AlignVCenter
308
309         validator: DoubleValidator { bottom: 0; top: 200; decimals: 1 }
310
311         background: Rectangle {
312             color: "#060d17"
313
314             radius: 6
315
316             border.color: debiKutusu.activeFocus ? "#3b82f6" : "#1e2a3f"
317
318             border.width: 1
319         }
320     }
321
322     Button {
```

```
305         width: parent.width
306
307         height: 36
308
309         text: "Tahmini Hesapla"
310
311         font.pixelSize: 12
312
313         font.bold: true
314
315
316
317         background: Rectangle {
318
319             radius: 6
320
321             color: parent.hovered ? "#4f8cf7" : "#3b82f6"
322
323             Behavior on color { ColorAnimation { duration: 120 } }
324         }
325
326         contentItem: Text {
327
328             text: parent.text
329
330             color: "#ffffff"
331
332             font: parent.font
333
334             horizontalAlignment: Text.AlignHCenter
335
336             verticalAlignment: Text.AlignVCenter
337         }
338
339
340         onClicked: {
341
342             var L = parseFloat(uzunlukKutusu.text)
343
344             var Q = parseFloat(debiKutusu.text)
345
346
347
348             if (isNaN(L) || isNaN(Q) || L <= 0) {
349
350                 tahminSonucu.text = "Geçerli değer girin"
351
352                 return
353             }
354
355
356             var basincMbar = (hesaplananTau0 + hesaplananMu * Q) * (L / 50)
357
358             var basincBar = basincMbar / 1000
```

```
336             tahminSonucu.text = basincBar.toFixed(2) + " bar"
337         }
338     }
339
340     Text {
341         id: tahminSonucu
342         width: parent.width
343         text: "-"
344         color: "#e8a020"
345         font.family: "Segoe UI"
346         font.pixelSize: 18
347         font.bold: true
348         horizontalAlignment: Text.AlignHCenter
349     }
350 }
351 }
352
353 Rectangle {
354     id: durumKutusu
355     width: parent.width
356     height: durumIcerik.implicitHeight + 24
357     radius: 10
358     color: durumIyiMi ? "#0f2417" : "#2a1414"
359     border.color: durumIyiMi ? "#16a34a" : "#dc2626"
360     border.width: 1
361
362     property bool durumIyiMi: true
363     property bool tau0Yuksekk: false
364     property bool muYuksekk: false
365
```

```
366         Column {
367             id: durumIcerik
368             anchors.top: parent.top
369             anchors.horizontalCenter: parent.horizontalCenter
370             anchors.topMargin: 12
371             spacing: 8
372             width: parent.width - 24
373
374         Row {
375             anchors.horizontalCenter: parent.horizontalCenter
376             spacing: 10
377
378             Rectangle {
379                 width: 24
380                 height: 24
381                 radius: 12
382                 anchors.verticalCenter: parent.verticalCenter
383                 color: durumKutusu.durumIyiMi ? "#16a34a" : "#dc2626"
384
385                 Text {
386                     anchors.centerIn: parent
387                     text: durumKutusu.durumIyiMi ? "✓" : "X"
388                     color: "#ffffff"
389                     font.pixelSize: 13
390                     font.bold: true
391                 }
392             }
393
394             Text {
395                 anchors.verticalCenter: parent.verticalCenter
396                 text: durumKutusu.durumIyiMi ? "POMPALANABİLİR" : "POMPALAMA SORUNU"
```

```
397             color: durumKutusu.durumIyiMi ? "#4ade80" : "#f87171"
398
399             font.family: "Segoe UI"
400
401             font.pixelSize: 15
402
403             font.bold: true
404
405         }
406     }
407
408     Text {
409
410         width: parent.width
411
412         horizontalAlignment: Text.AlignHCenter
413
414         visible: !durumKutusu.durumIyiMi
415
416         text: {
417
418             if (durumKutusu.tau0Yuksek && durumKutusu.muYuksek) {
419
420                 return "Hem akma gerilmesi hem plastik viskozite yüksek. Su/çimento
421 oranını artırın ve agrega gradasyonunu gözden geçirin."
422
423             } else if (durumKutusu.tau0Yuksek) {
424
425                 return "Akma gerilmesi yüksek. Su/çimento oranını artırmayı veya
426 süperakışkanlaştırıcı dozajını yükseltmeyi değerlendirin."
427
428             } else if (durumKutusu.muYuksek) {
429
430                 return "Plastik viskozite yüksek. İnce agrega oranını azaltmayı veya
431 su azaltıcı katkı eklemeyi değerlendirin."
432
433             }
434
435             return "Yeterli stroke verisi bulunamadı."
436
437         }
438
439         color: "#9ca3af"
440
441         font.family: "Segoe UI"
442
443         font.pixelSize: 11
444
445         wrapMode: Text.WordWrap
446
447     }
448
449 }
450
451 }
```

```
426         Button {
427             width: parent.width
428             height: 40
429             text: "📄 PDF Rapor Önizle"
430             font.pixelSize: 13
431             font.bold: true
432             enabled: olcumId > 0
433
434             onClicked: {
435                 pdfOnizlemePopup.onizlemeHtml = reportManager.pdfOnizlemeHtml(
436                     olcumId,
437                     musteriAdi,
438                     receteAdi,
439                     hesaplananTau0,
440                     hesaplananMu,
441                     parseFloat(r2Metni.text)
442                 )
443                 pdfOnizlemePopup.open()
444             }
445
446             background: Rectangle {
447                 radius: 8
448                 color: parent.enabled ? (parent.hovered ? "#4f8cf7" : "#3b82f6") : "#1e2a3f"
449                 Behavior on color { ColorAnimation { duration: 120 } }
450             }
451
452             contentItem: Text {
453                 text: parent.text
454                 color: "#ffffff"
455                 font: parent.font
```

```
456             horizontalAlignment: Text.AlignHCenter
457             verticalAlignment: Text.AlignVCenter
458         }
459     }
460
461     Button {
462         width: parent.width
463         height: 40
464         text: "📊 Excel (CSV) Önizle"
465         font.pixelSize: 13
466         font.bold: true
467         enabled: olcumId > 0
468
469         onClicked: {
470             csvOnizlemePopup.satirlar = database.strokeVerileriGetir(olcumId)
471             csvOnizlemePopup.open()
472         }
473
474         background: Rectangle {
475             radius: 8
476             color: parent.enabled ? (parent.hovered ? "#22c55e" : "#16a34a") : "#1e2a3f"
477             Behavior on color { ColorAnimation { duration: 120 } }
478         }
479
480         contentItem: Text {
481             text: parent.text
482             color: "#ffffff"
483             font: parent.font
484             horizontalAlignment: Text.AlignHCenter
485             verticalAlignment: Text.AlignVCenter
486         }
```



```

487         }
488
489         Rectangle {
490             width: parent.width
491             radius: 10
492             color: "#0a0e17"
493             border.color: "#1e2a3f"
494             border.width: 1
495             height: playbackIcerik.implicitHeight + 24
496             visible: olcumId > 0
497
498             Column {
499                 id: playbackIcerik
500                 anchors.top: parent.top
501                 anchors.left: parent.left
502                 anchors.right: parent.right
503                 anchors.margins: 12
504                 spacing: 8
505
506                 Text {
507                     text: "OYNATMA (PLAYBACK)"
508                     color: "#6b7280"
509                     font.family: "Segoe UI"
510                     font.pixelSize: 10
511                     font.letterSpacing: 1
512                 }
513
514                 Text {
515                     text: strokeModeli.count > 0
516
517                     ? "Stroke " + (playbackAdimi + 1) + " / " + strokeModeli.count

```

```

517             : "Oynatılacak stroke yok"
518         color: "#dce8f5"
519         font.family: "Segoe UI"
520         font.pixelSize: 12
521     }
522
523     Flow {
524         width: parent.width
525         spacing: 5
526
527         Button {
528             width: 40
529             height: 34
530             text: oynatiliyor ? "||" : "▶"
531             font.pixelSize: 14
532             enabled: strokeModeli.count > 0
533
534             onClicked: {
535                 if (oynatiliyor) {
536                     oynatiliyor = false
537                     playbackTimer.stop()
538                 } else {
539                     if (playbackAdimi >= strokeModeli.count - 1) {
540                         playbackAdimi = -1
541                     }
542                     oynatiliyor = true
543                     playbackTimer.start()
544                 }
545             }
546
547             background: Rectangle {

```

```
548         radius: 6
549         color: parent.enabled ? (parent.hovered ? "#4f8cf7" : "#3b82f6") :
"#1e2a3f"
550         Behavior on color { ColorAnimation { duration: 120 } }
551     }
552
553     contentItem: Text {
554         text: parent.text
555         color: "#ffffff"
556         font: parent.font
557         horizontalAlignment: Text.AlignHCenter
558         verticalAlignment: Text.AlignVCenter
559     }
560 }
561
562 Repeater {
563     model: [1, 2, 5, 10]
564
565     Button {
566         width: 38
567         height: 34
568         text: modelData + "x"
569         font.pixelSize: 12
570
571         onClicked: playbackHizi = modelData
572
573         background: Rectangle {
574             radius: 6
575             color: playbackHizi === modelData ? "#3b82f6" : "#0f1420"
576             border.color: "#1e2a3f"
577             border.width: 1
```

```

578         }
579
580         contentItem: Text {
581             text: parent.text
582             color: playbackHizi === modelData ? "#ffffff" : "#9ca3af"
583             font: parent.font
584             horizontalAlignment: Text.AlignHCenter
585             verticalAlignment: Text.AlignVCenter
586         }
587     }
588 }
589 }
590 }
591 }
592 }
593 }
594 }
595
596 Rectangle {
597     id: panelAyraci
598     width: 3
599     height: parent.height
600     color: "#05070b"
601
602     Rectangle {
603         anchors.right: parent.right
604         anchors.top: parent.top
605         anchors.bottom: parent.bottom
606         width: 1
607         color: "#403b82f6"

```

```

608         }
609     }
610
611     Item {
612         width: parent.width - ozetPaneli.width - panelAyraci.width
613         height: parent.height
614
615         Rectangle {
616             id: grafikKutusu
617             anchors.top: parent.top
618             anchors.left: parent.left
619             anchors.right: parent.right
620             anchors.margins: 16
621             height: parent.height * 0.55
622             radius: 10
623             color: "#0f1420"
624             border.color: "#1e2a3f"
625             border.width: 1
626
627             Item {
628                 id: grafikBaslik
629                 anchors.top: parent.top
630                 anchors.left: parent.left
631                 anchors.right: parent.right
632                 anchors.margins: 14
633                 height: 20
634
635                 Text {
636                     anchors.left: parent.left
637                     anchors.verticalCenter: parent.verticalCenter
638                     text: "P-Q Dağılım Grafiği"

```

```
639         color: "#9ca3af"
640
641         font.family: "Segoe UI"
642
643         font.pixelSize: 13
644
645         font.bold: true
646     }
647
648
649
650     Row {
651
652         anchors.right: parent.right
653
654         anchors.verticalCenter: parent.verticalCenter
655
656         spacing: 16
657
658
659
660     Row {
661
662         spacing: 6
663
664         anchors.verticalCenter: parent.verticalCenter
665
666
667
668         Rectangle {
669
670             width: 8
671
672             height: 8
673
674             radius: 4
675
676             color: "#3b82f6"
677
678             anchors.verticalCenter: parent.verticalCenter
679         }
680
681
682
683         Text {
684
685             text: "Ölçüm Noktaları"
686
687             color: "#6b7280"
688
689             font.family: "Segoe UI"
690
691             font.pixelSize: 11
692
693             anchors.verticalCenter: parent.verticalCenter
694         }
695     }
```

```
669         }
670
671     Row {
672         spacing: 6
673         anchors.verticalCenter: parent.verticalCenter
674
675         Rectangle {
676             width: 12
677             height: 2
678             color: "#e8a020"
679             anchors.verticalCenter: parent.verticalCenter
680         }
681
682         Text {
683             text: "Regresyon Doğrusu"
684             color: "#6b7280"
685             font.family: "Segoe UI"
686             font.pixelSize: 11
687             anchors.verticalCenter: parent.verticalCenter
688         }
689     }
690
691     Row {
692         spacing: 6
693         anchors.verticalCenter: parent.verticalCenter
694
695         Rectangle {
696             width: 8
697             height: 8
698             radius: 4
699             color: "#e8a020"
```

```

700                anchors.verticalCenter: parent.verticalCenter
701            }
702
703            Text {
704                text: "Oynatma"
705                color: "#6b7280"
706                font.family: "Segoe UI"
707                font.pixelSize: 11
708                anchors.verticalCenter: parent.verticalCenter
709            }
710        }
711    }
712 }
713
714 ChartView {
715     anchors.top: grafikBaslik.bottom
716     anchors.left: parent.left
717     anchors.right: parent.right
718     anchors.bottom: parent.bottom
719     anchors.topMargin: 4
720     anchors.margins: 8
721     backgroundColor: "transparent"
722     legend.visible: false
723     antialiasing: true
724
725     ValueAxis {
726         id: qEkseni
727         min: 0
728         max: 20
729         titleText: "Debi (Q) m³/h"

```



```
730         gridLineColor: "#182131"
731         labelsColor: "#4b5563"
732         labelsFont.pixelSize: 9
733         lineVisible: false
734         minorGridVisible: false
735     }
736
737     ValueAxis {
738         id: pEkseni
739         min: 0
740         max: 100
741         titleText: "Basınç (P) mbar"
742         gridLineColor: "#182131"
743         labelsColor: "#4b5563"
744         labelsFont.pixelSize: 9
745         lineVisible: false
746         minorGridVisible: false
747     }
748
749     ScatterSeries {
750         id: pqSerisi
751         axisX: qEkseni
752         axisY: pEkseni
753         color: "#3b82f6"
754         markerSize: 10
755     }
756
757     LineSeries {
758         id: regresyonCizgisi
759         axisX: qEkseni
760         axisY: pEkseni
```

```
761             color: "#e8a020"
762             width: 2
763         }
764
765         ScatterSeries {
766             id: oynatmaSerisi
767             axisX: qEkseni
768             axisY: pEkseni
769             color: "#e8a020"
770             markerSize: 16
771         }
772     }
773 }
774
775 Rectangle {
776     anchors.top: grafikKutusu.bottom
777     anchors.left: parent.left
778     anchors.right: parent.right
779     anchors.bottom: parent.bottom
780     anchors.margins: 16
781     anchors.topMargin: 8
782     radius: 10
783     color: "#0f1420"
784     border.color: "#1e2a3f"
785     border.width: 1
786
787     Text {
788         anchors.top: parent.top
789         anchors.left: parent.left
790         anchors.margins: 14
```

```

791         text: "Stroke Tablosu"
792         color: "#9ca3af"
793         font.family: "Segoe UI"
794         font.pixelSize: 13
795         font.bold: true
796     }
797
798     Row {
799         id: tabloBasligi
800         anchors.top: parent.top
801         anchors.left: parent.left
802         anchors.right: parent.right
803         anchors.topMargin: 36
804         anchors.leftMargin: 12
805         anchors.rightMargin: 12
806
807         Text { width: parent.width * 0.25; text: "STROKE"; color: "#6b7280"; font.pixelSize: 11;
font.bold: true; font.letterSpacing: 1 }
808
809         Text { width: parent.width * 0.25; text: "P (mbar)"; color: "#6b7280"; font.pixelSize:
11; font.bold: true; font.letterSpacing: 1 }
810
811         Text { width: parent.width * 0.25; text: "Q (m³/h)"; color: "#6b7280"; font.pixelSize:
11; font.bold: true; font.letterSpacing: 1 }
812
813         Text { width: parent.width * 0.25; text: "DURUM"; color: "#6b7280"; font.pixelSize: 11;
font.bold: true; font.letterSpacing: 1 }
814     }
815
816     Rectangle {
817         anchors.top: tabloBasligi.bottom
818         anchors.left: parent.left
819         anchors.right: parent.right
820         anchors.topMargin: 8
821         anchors.leftMargin: 12
822         anchors.rightMargin: 12

```

```

820         height: 1
821         color: "#1e2a3f"
822     }
823
824     ListView {
825         anchors.top: tabloBasligi.bottom
826         anchors.left: parent.left
827         anchors.right: parent.right
828         anchors.bottom: parent.bottom
829         anchors.margins: 12
830         anchors.topMargin: 14
831         clip: true
832
833         model: ListModel {
834             id: strokeModeli
835         }
836
837         delegate: Rectangle {
838             width: parent.width
839             height: 32
840             radius: 6
841             color: index === playbackAdimi ? "#1e2a3f" : (index % 2 === 0 ? "transparent" :
"#0a0e17")
842
843             Row {
844                 anchors.fill: parent
845                 anchors.leftMargin: 6
846
847                 Text { width: parent.width * 0.25; anchors.verticalCenter: parent.verticalCenter;
text: stroke; color: "#dce8f5"; font.pixelSize: 12 }
848
                 Text { width: parent.width * 0.25; anchors.verticalCenter: parent.verticalCenter;
text: p.toFixed(1); color: "#dce8f5"; font.pixelSize: 12 }

```

```

849             Text { width: parent.width * 0.25; anchors.verticalCenter: parent.verticalCenter;
text: q.toFixed(2); color: "#dce8f5"; font.pixelSize: 12 }

850
851             Item {
852                 width: parent.width * 0.25
853                 height: parent.height
854
855                 Rectangle {
856                     anchors.verticalCenter: parent.verticalCenter
857                     width: durumMetni.implicitWidth + 16
858                     height: 20
859                     radius: 10
860                     color: gecerli ? "#123321" : "#2a1414"
861
862                     Text {
863                         id: durumMetni
864                         anchors.centerIn: parent
865                         text: gecerli ? "Geçerli" : "Hatalı"
866                         color: gecerli ? "#4ade80" : "#f87171"
867                         font.pixelSize: 11
868                         font.bold: true
869                     }
870                 }
871             }
872         }
873     }
874 }
875 }
876 }
877 }
878

```

```
879     Rectangle {
880         id: bildirimKutusu
881         property bool hataMi: false
882
883         function goster(mesaj, hata) {
884             bildirimMetni.text = mesaj
885             hataMi = hata
886             opacity = 1
887             bildirimTimer.restart()
888         }
889
890         visible: opacity > 0
891         opacity: 0
892         z: 100
893         anchors.bottom: parent.bottom
894         anchors.horizontalCenter: parent.horizontalCenter
895         anchors.bottomMargin: 24
896         width: bildirimMetni.implicitWidth + 40
897         height: 44
898         radius: 10
899         color: hataMi ? "#7f1d1d" : "#14532d"
900         border.color: hataMi ? "#dc2626" : "#22c55e"
901         border.width: 1
902
903         Behavior on opacity { NumberAnimation { duration: 200 } }
904
905         Text {
906             id: bildirimMetni
907             anchors.centerIn: parent
908             text: ""
```

```
909         color: "#ffffff"
910         font.family: "Segoe UI"
911         font.pixelSize: 13
912         font.bold: true
913     }
914
915     Timer {
916         id: bildirimTimer
917         interval: 3000
918         onTriggered: bildirimKutusu.opacity = 0
919     }
920 }
921
922 Popup {
923     id: pdfOnizlemePopup
924     modal: true
925     focus: true
926     anchors.centerIn: parent
927     width: 640
928     height: Math.min(720, parent.height - 40)
929     padding: 0
930     closePolicy: Popup.CloseOnEscape
931
932     property string onizlemeHtml: ""
933
934     background: Rectangle {
935         color: "#0f1420"
936         radius: 12
937         border.color: "#1e2a3f"
938         border.width: 1
939     }
```

```
940
941     Overlay.modal: Rectangle {
942         color: "#a6000000"
943     }
944
945     contentItem: Column {
946         width: pdfOnizlemePopup.width
947         spacing: 0
948
949         Rectangle {
950             width: parent.width
951             height: 52
952             color: "#0a0e17"
953
954             Text {
955                 anchors.left: parent.left
956                 anchors.leftMargin: 20
957                 anchors.verticalCenter: parent.verticalCenter
958                 text: "PDF Rapor Önizleme"
959                 color: "#dce8f5"
960                 font.family: "Segoe UI"
961                 font.pixelSize: 14
962                 font.bold: true
963             }
964         }
965
966         Rectangle {
967             width: parent.width
968             height: pdfOnizlemePopup.height - 52 - 64
969             color: "#f3f4f6"
```



```
970
971     Flickable {
972         anchors.fill: parent
973         anchors.margins: 20
974         contentWidth: width
975         contentHeight: onizlemeMetni.implicitHeight
976         clip: true
977         boundsBehavior: Flickable.StopAtBounds
978
979         TextEdit {
980             id: onizlemeMetni
981             width: parent.width
982             readOnly: true
983             selectByMouse: true
984             textFormat: TextEdit.RichText
985             wrapMode: Text.WordWrap
986             text: pdfOnizlemePopup.onizlemeHtml
987             color: "#111827"
988             font.pixelSize: 13
989         }
990     }
991 }
992
993 Rectangle {
994     width: parent.width
995     height: 64
996     color: "#0a0e17"
997
998     Row {
999         anchors.centerIn: parent
1000         spacing: 12
```

```
1001
1002         Button {
1003             width: 140
1004             height: 38
1005             text: "📄 İndir (PDF)"
1006             font.pixelSize: 13
1007             font.bold: true
1008
1009             onClicked: {
1010                 var yol = reportManager.pdfOlustur(
1011                     olcumId,
1012                     musteriAdi,
1013                     receteAdi,
1014                     hesaplananTau0,
1015                     hesaplananMu,
1016                     parseFloat(r2Metni.text)
1017                 )
1018                 pdfOnizlemePopup.close()
1019                 if (yol.length > 0) {
1020                     bildirimKutusu.goster("PDF kaydedildi: " + yol, false)
1021                 } else {
1022                     bildirimKutusu.goster("PDF oluşturulamadı.", true)
1023                 }
1024             }
1025
1026             background: Rectangle {
1027                 radius: 8
1028                 color: parent.hovered ? "#4f8cf7" : "#3b82f6"
1029                 Behavior on color { ColorAnimation { duration: 120 } }
1030             }
```

```
1031
1032         contentItem: Text {
1033             text: parent.text
1034             color: "#ffffff"
1035             font: parent.font
1036             horizontalAlignment: Text.AlignHCenter
1037             verticalAlignment: Text.AlignVCenter
1038         }
1039     }
1040
1041     Button {
1042         width: 100
1043         height: 38
1044         text: "Kapat"
1045         font.pixelSize: 13
1046
1047         onClicked: pdfOnizlemePopup.close()
1048
1049         background: Rectangle {
1050             radius: 8
1051             color: parent.hovered ? "#1e2a3f" : "transparent"
1052             border.color: "#1e2a3f"
1053             border.width: 1
1054             Behavior on color { ColorAnimation { duration: 120 } }
1055         }
1056
1057         contentItem: Text {
1058             text: parent.text
1059             color: "#9ca3af"
1060             font: parent.font
1061             horizontalAlignment: Text.AlignHCenter
```

```
1062                 verticalAlignment: Text.AlignVCenter
1063             }
1064         }
1065     }
1066 }
1067 }
1068 }
1069
1070 Popup {
1071     id: csvOnizlemePopup
1072     modal: true
1073     focus: true
1074     anchors.centerIn: parent
1075     width: 640
1076     height: Math.min(680, parent.height - 40)
1077     padding: 0
1078     closePolicy: Popup.CloseOnEscape
1079
1080     property var satirlar: []
1081
1082     background: Rectangle {
1083         color: "#0f1420"
1084         radius: 12
1085         border.color: "#1e2a3f"
1086         border.width: 1
1087     }
1088
1089     Overlay.modal: Rectangle {
1090         color: "#a6000000"
1091     }
```

```
1092
1093     contentItem: Column {
1094         width: csvOnizlemePopup.width
1095         spacing: 0
1096
1097         Rectangle {
1098             width: parent.width
1099             height: 52
1100             color: "#0a0e17"
1101
1102             Text {
1103                 anchors.left: parent.left
1104                 anchors.leftMargin: 20
1105                 anchors.verticalCenter: parent.verticalCenter
1106                 text: "Excel (CSV) Önizleme"
1107                 color: "#dce8f5"
1108                 font.family: "Segoe UI"
1109                 font.pixelSize: 14
1110                 font.bold: true
1111             }
1112         }
1113
1114         Item {
1115             width: parent.width
1116             height: csvBilgiSutunu.implicitHeight + 32
1117
1118             Column {
1119                 id: csvBilgiSutunu
1120                 anchors.left: parent.left
1121                 anchors.right: parent.right
1122                 anchors.top: parent.top
```

```
1123         anchors.margins: 20

1124         spacing: 4

1125

1126         Text { text: "Ölçüm ID: " + olcumId; color: "#9ca3af"; font.family: "Segoe UI";
font.pixelSize: 12 }

1127         Text { text: "Tarih: " + olcumTarihi; color: "#9ca3af"; font.family: "Segoe UI";
font.pixelSize: 12 }

1128         Text { text: "Müşteri: " + muster Adi; color: "#9ca3af"; font.family: "Segoe UI";
font.pixelSize: 12 }

1129         Text { text: "Reçete: " + receteAdi; color: "#9ca3af"; font.family: "Segoe UI";
font.pixelSize: 12 }

1130         Text { text: "Ağırlık (kg): " + agirlikDegeri.toFixed(1); color: "#9ca3af"; font.family:
"Segoe UI"; font.pixelSize: 12 }

1131     }

1132 }

1133

1134     Rectangle {

1135         width: parent.width

1136         height: csvOnizlemePopup.height - 52 - 130 - 64

1137         color: "#0a0e17"

1138         border.color: "#1e2a3f"

1139         border.width: 1

1140

1141         Row {

1142             id: csvTabloBasligi

1143             anchors.top: parent.top

1144             anchors.left: parent.left

1145             anchors.right: parent.right

1146             anchors.margins: 12

1147

1148             Text { width: parent.width * 0.2; text: "STROKE"; color: "#6b7280"; font.pixelSize: 11;
font.bold: true }

1149             Text { width: parent.width * 0.2; text: "BASINÇ (mbar)"; color: "#6b7280";
font.pixelSize: 11; font.bold: true }
```

```

1150                Text { width: parent.width * 0.2; text: "KONUM (mm)"; color: "#6b7280"; font.pixelSize:
11; font.bold: true }

1151                Text { width: parent.width * 0.2; text: "DEBİ (m³/h)"; color: "#6b7280"; font.pixelSize:
11; font.bold: true }

1152                Text { width: parent.width * 0.2; text: "GEÇERLİ"; color: "#6b7280"; font.pixelSize: 11;
font.bold: true }

1153            }

1154
1155            ListView {

1156                anchors.top: csvTabloBasligi.bottom

1157                anchors.left: parent.left

1158                anchors.right: parent.right

1159                anchors.bottom: parent.bottom

1160                anchors.margins: 12

1161                anchors.topMargin: 8

1162                clip: true

1163
1164                model: csvOnizlemePopup.satirlar

1165
1166                delegate: Rectangle {

1167                    width: ListView.view.width

1168                    height: 26

1169                    color: index % 2 === 0 ? "transparent" : "#0f1420"

1170
1171                    Row {

1172                        anchors.fill: parent

1173
1174                        Text { width: parent.width * 0.2; anchors.verticalCenter: parent.verticalCenter;
text: modelData.stroke; color: "#dce8f5"; font.pixelSize: 12 }

1175                        Text { width: parent.width * 0.2; anchors.verticalCenter: parent.verticalCenter;
text: modelData.basinc.toFixed(1); color: "#dce8f5"; font.pixelSize: 12 }

1176                        Text { width: parent.width * 0.2; anchors.verticalCenter: parent.verticalCenter;
text: modelData.konum.toFixed(1); color: "#dce8f5"; font.pixelSize: 12 }

```

```
1177             Text { width: parent.width * 0.2; anchors.verticalCenter: parent.verticalCenter;
text: modelData.debi.toFixed(2); color: "#dce8f5"; font.pixelSize: 12 }

1178             Text { width: parent.width * 0.2; anchors.verticalCenter: parent.verticalCenter;
text: modelData.gecerli ? "Evet" : "Hayır"; color: modelData.gecerli ? "#4ade80" : "#f87171"; font.pixelSize: 12 }

1179         }

1180     }

1181 }

1182 }

1183

1184     Rectangle {

1185         width: parent.width

1186         height: 64

1187         color: "#0a0e17"

1188

1189         Row {

1190             anchors.centerIn: parent

1191             spacing: 12

1192

1193             Button {

1194                 width: 140

1195                 height: 38

1196                 text: "📄 İndir (CSV)"

1197                 font.pixelSize: 13

1198                 font.bold: true

1199

1200                 onClicked: {

1201                     var yol = database.csvDisaAktar(olcumId)

1202                     csvOnizlemePopup.close()

1203                     if (yol.length > 0) {

1204                         bildirimKutusu.goster("CSV kaydedildi: " + yol, false)

1205                     } else {

1206                         bildirimKutusu.goster("CSV oluşturulamadı.", true)
```



```
1207         }
1208     }
1209
1210     background: Rectangle {
1211         radius: 8
1212         color: parent.hovered ? "#22c55e" : "#16a34a"
1213         Behavior on color { ColorAnimation { duration: 120 } }
1214     }
1215
1216     contentItem: Text {
1217         text: parent.text
1218         color: "#ffffff"
1219         font: parent.font
1220         horizontalAlignment: Text.AlignHCenter
1221         verticalAlignment: Text.AlignVCenter
1222     }
1223 }
1224
1225 Button {
1226     width: 100
1227     height: 38
1228     text: "Kapat"
1229     font.pixelSize: 13
1230
1231     onClicked: csvOnizlemePopup.close()
1232
1233     background: Rectangle {
1234         radius: 8
1235         color: parent.hovered ? "#1e2a3f" : "transparent"
1236         border.color: "#1e2a3f"
```

```
1237         border.width: 1
1238         Behavior on color { ColorAnimation { duration: 120 } }
1239     }
1240
1241     contentItem: Text {
1242         text: parent.text
1243         color: "#9ca3af"
1244         font: parent.font
1245         horizontalAlignment: Text.AlignHCenter
1246         verticalAlignment: Text.AlignVCenter
1247     }
1248 }
1249 }
1250 }
1251 }
1252 }
1253 }
```

src/Backend.cpp

```
1 #include "Backend.h"

2 #include <QDebug>

3

4 Backend::Backend(QObject *parent)

5     : QObject(parent)

6 {

7 }

8

9 QString Backend::durum() const

10 {

11     return "Baglanti bekleniyor...";

12 }

13

14 bool Backend::girisYap(const QString &kullaniciAdi, const QString &sifre)

15 {

16     if (kullaniciAdi == "admin" && sifre == "admin123") {

17         qDebug() << "Giris Basarili:" << kullaniciAdi;

18         return true;

19     }

20

21     qDebug() << "Giris Basarisiz:" << kullaniciAdi;

22     return false;

23 }

24

25 bool Backend::sifreDogrula(const QString &sifre)

26 {

27     return sifre == "admin123";

28 }
```

src/Backend.h

```
1 #pragma once

2 #include <QObject>

3

4 class Backend : public QObject

5 {

6     Q_OBJECT

7     Q_PROPERTY(QString durum READ durum CONSTANT)

8

9 public:

10     explicit Backend(QObject *parent = nullptr);

11     QString durum() const;

12     Q_INVOKABLE bool girisYap(const QString &kullaniciAdi, const QString &sifre);

13     Q_INVOKABLE bool sifreDogrula(const QString &sifre);

14 };
```

src/Calculator.cpp

```
1 #include "Calculator.h"

2

3 Calculator::Calculator(QObject *parent)

4     : QObject(parent)

5 {

6 }

7

8 QString Calculator::durum() const

9 {

10     return m_durum;

11 }

12

13 int Calculator::strokeSayisi() const

14 {

15     return m_strokeSayisi;

16 }

17

18 bool Calculator::duraklatildi() const

19 {

20     return m_duraklatildi;

21 }

22

23 void Calculator::konumGuncelle(double konum)

24 {

25     if (m_duraklatildi) {

26         return;

27     }

28

29     QString yeniDurum;

30
```

```
31     if (konum > 450) {
32         yeniDurum = "YUKARIDA";
33     } else if (konum > 20) {
34         yeniDurum = "INIYOR";
35     } else {
36         yeniDurum = "TAMAMLANDI";
37     }
38
39     if (yeniDurum == "YUKARIDA") {
40         m_yukaridaGorulduMu = true;
41     }
42
43     if (yeniDurum == "TAMAMLANDI" && m_durum != "TAMAMLANDI" && m_yukaridaGorulduMu) {
44         m_strokeSayisi++;
45         m_yukaridaGorulduMu = false;
46         emit strokeSayisiChanged();
47     }
48
49     if (yeniDurum != m_durum) {
50         m_durum = yeniDurum;
51         emit durumChanged();
52     }
53 }
54
55 void Calculator::duraklat()
56 {
57     if (m_duraklatildi) {
58         return;
59     }
60
```

```
61     m_duraklatildi = true;
62     emit duraklatildiChanged();
63 }
64
65 void Calculator::devamEt()
66 {
67     if (!m_duraklatildi) {
68         return;
69     }
70
71     m_duraklatildi = false;
72     emit duraklatildiChanged();
73 }
74
75 void Calculator::sifirla()
76 {
77     m_strokeSayisi = 0;
78     m_durum = "BEKLENIYOR";
79     m_duraklatildi = false;
80     m_yukaridaGorulduMu = false;
81     emit strokeSayisiChanged();
82     emit durumChanged();
83     emit duraklatildiChanged();
84 }
```

src/Calculator.h

```
1 #pragma once

2 #include <QObject>

3

4 class Calculator : public QObject

5 {

6     Q_OBJECT

7     Q_PROPERTY(QString durum READ durum NOTIFY durumChanged)

8     Q_PROPERTY(int strokeSayisi READ strokeSayisi NOTIFY strokeSayisiChanged)

9     Q_PROPERTY(bool duraklatildi READ duraklatildi NOTIFY duraklatildiChanged)

10

11 public:

12     explicit Calculator(QObject *parent = nullptr);

13     QString durum() const;

14     int strokeSayisi() const;

15     bool duraklatildi() const;

16

17     Q_INVOKABLE void konumGuncelle(double konum);

18     Q_INVOKABLE void duraklat();

19     Q_INVOKABLE void devamEt();

20     Q_INVOKABLE void sifirla();

21

22 signals:

23     void durumChanged();

24     void strokeSayisiChanged();

25     void duraklatildiChanged();

26

27 private:

28     QString m_durum = "BEKLENIYOR";

29     int m_strokeSayisi = 0;

30     bool m_duraklatildi = false;
```



```
31     bool m_yukaridaGorulduMu = false;
32 };
```

src/Database.cpp

```
1  #include "Database.h"

2  #include <QSqlQuery>

3  #include <QSqlError>

4  #include <QDebug>

5  #include <QDateTime>

6  #include <QVector>

7  #include <QFile>

8  #include <QTextStream>

9  #include <QStandardPaths>

10 #include <QDir>

11

12 Database::Database(QObject *parent)

13     : QObject(parent)

14 {

15     const QString klasor = QStandardPaths::writableLocation(QStandardPaths::AppLocalDataLocation) +

"/database";

16     QDir().mkpath(klasor);

17

18     m_db = QSqlDatabase::addDatabase("SQLITE");

19     m_db.setDatabaseName(klasor + "/sliper.db");

20

21     if (!m_db.open()) {

22         qWarning() << "Veritabani acilamadi:" << m_db.lastError().text();

23     } else {

24         qDebug() << "Veritabani basariyla acildi.";

25         tablolariOlustur();

26     }

27 }

28

29 void Database::tablolariOlustur()
```

```
30 {
31     QSqlQuery sorgu;
32
33     sorgu.exec(
34         "CREATE TABLE IF NOT EXISTS Olcumler ("
35         "id INTEGER PRIMARY KEY AUTOINCREMENT, "
36         "tarih TEXT, "
37         "musteri TEXT, "
38         "recete TEXT, "
39         "agirlik REAL"
40         ")";
41
42
43     sorgu.exec(
44         "CREATE TABLE IF NOT EXISTS Strokelar ("
45         "id INTEGER PRIMARY KEY AUTOINCREMENT, "
46         "olcumId INTEGER, "
47         "basinc REAL, "
48         "konum REAL, "
49         "debi REAL, "
50         "gecerli INTEGER, "
51         "FOREIGN KEY(olcumId) REFERENCES Olcumler(id)"
52         ")";
53
54
55     QSqlQuery kolonKontrol("PRAGMA table_info(Strokelar)");
56     bool debiKolonuVar = false;
57     while (kolonKontrol.next()) {
58         if (kolonKontrol.value("name").toString() == "debi") {
59             debiKolonuVar = true;
60             break;
```

```
61     }
62 }
63
64 if (!debiKolonuVar) {
65     QSqlQuery migrasyon;
66     migrasyon.exec("ALTER TABLE Strokelar ADD COLUMN debi REAL DEFAULT 0");
67 }
68
69 sorgu.exec(
70     "CREATE TABLE IF NOT EXISTS Kalibrasyonlar ("
71     "sensor TEXT PRIMARY KEY, "
72     "deger1 REAL, "
73     "deger2 REAL, "
74     "tarih TEXT"
75     ")";
76 );
77
78 sorgu.exec(
79     "CREATE TABLE IF NOT EXISTS LoadCellNoktalari ("
80     "id INTEGER PRIMARY KEY AUTOINCREMENT, "
81     "hedefKg REAL, "
82     "hamDeger REAL"
83     ")";
84 );
85
86 sorgu.exec(
87     "CREATE TABLE IF NOT EXISTS MesafeNoktalari ("
88     "id INTEGER PRIMARY KEY AUTOINCREMENT, "
89     "hedefMm REAL, "
90     "hamDeger REAL"
```

```

91         ")"
92     );
93
94     sorgu.exec(
95         "CREATE TABLE IF NOT EXISTS EgimKalibrasyon ("
96         "id INTEGER PRIMARY KEY CHECK (id = 1), "
97         "biasX REAL, biasY REAL, biasZ REAL, "
98         "gainX REAL, gainY REAL, gainZ REAL, "
99         "tarih TEXT"
100     ")"
101 );
102
103 sorgu.exec(
104     "CREATE TABLE IF NOT EXISTS KalibrasyonTarihleri ("
105     "sensor TEXT PRIMARY KEY, "
106     "tarih TEXT"
107     ")"
108 );
109
110 qDebug() << "Tablolar hazir.";
111 }
112
113 int Database::olcumBaslat(const QString &musteri, const QString &recete, double agirlik)
114 {
115     QSqlQuery sorgu;
116     sorgu.prepare(
117         "INSERT INTO Olcumler (tarih, muster_i, recete, agirlik) "
118         "VALUES (:tarih, :muster_i, :recete, :agirlik)"
119     );
120
121     sorgu.bindValue(":tarih", QDateTime::currentDateTime().toString("dd.MM.yyyy HH:mm"));

```

```
122     sorgu.bindValue(":musteri", musteri);
123     sorgu.bindValue(":recete", recete);
124     sorgu.bindValue(":agirlik", agirlik);
125
126     if (!sorgu.exec()) {
127         qWarning() << "Olcum baslatilamadi:" << sorgu.lastError().text();
128         return -1;
129     }
130
131     int yeniId = sorgu.lastInsertId().toInt();
132     qDebug() << "Yeni olcum baslatildi, id:" << yeniId;
133     return yeniId;
134 }
135
136 void Database::strokeKaydet(int olcumId, double basinc, double konum, double debi, bool gecerli)
137 {
138     if (olcumId <= 0) {
139         qWarning() << "Gecersiz olcumId, stroke kaydedilemedi.";
140         return;
141     }
142
143     QSqlQuery sorgu;
144     sorgu.prepare(
145         "INSERT INTO Strokelar (olcumId, basinc, konum, debi, gecerli) "
146         "VALUES (:olcumId, :basinc, :konum, :debi, :gecerli)"
147     );
148
149     sorgu.bindValue(":olcumId", olcumId);
150     sorgu.bindValue(":basinc", basinc);
151     sorgu.bindValue(":konum", konum);
```

```
152     sorgu.bindValue(":debi", debi);
153     sorgu.bindValue(":gecerli", gecerli ? 1 : 0);
154
155     if (!sorgu.exec()) {
156         qWarning() << "Stroke kaydedilemedi:" << sorgu.lastError().text();
157     } else {
158         qDebug() << "Stroke kaydedildi, olcumId:" << olcumId;
159     }
160 }
161
162 bool Database::olcumSil(int olcumId)
163 {
164     if (olcumId <= 0) {
165         qWarning() << "Gecersiz olcumId, olcum silinemedi.";
166         return false;
167     }
168
169     if (!m_db.transaction()) {
170         qWarning() << "Olcum silme icin transaction baslatilamadi:" << m_db.lastError().text();
171         return false;
172     }
173
174     QSqlQuery strokeSil;
175     strokeSil.prepare("DELETE FROM Strokelar WHERE olcumId = :olcumId");
176     strokeSil.bindValue(":olcumId", olcumId);
177     if (!strokeSil.exec()) {
178         qWarning() << "Strokelar silinemedi:" << strokeSil.lastError().text();
179         m_db.rollback();
180         return false;
181     }
182
```

```

183     QSqlQuery olcumSil;
184     olcumSil.prepare("DELETE FROM Olcumler WHERE id = :id");
185     olcumSil.bindValue(":id", olcumId);
186     if (!olcumSil.exec()) {
187         qWarning() << "Olcum silinemedi:" << olcumSil.lastError().text();
188         m_db.rollback();
189         return false;
190     }
191
192     if (!m_db.commit()) {
193         qWarning() << "Olcum silme commit edilemedi:" << m_db.lastError().text();
194         return false;
195     }
196
197     qDebug() << "Olcum silindi, id:" << olcumId;
198     return true;
199 }
200
201 QVariantList Database::tumOlcumleriGetir()
202 {
203     QVariantList sonuclar;
204
205     QSqlQuery sorgu(
206         "SELECT o.id, o.tarih, o.musteri, o.recete, o.agirlik, "
207         "COUNT(s.id) AS strokeSayisi "
208         "FROM Olcumler o "
209         "LEFT JOIN Strokelar s ON s.olcumId = o.id "
210         "GROUP BY o.id "
211         "ORDER BY o.id DESC"
212     );

```



```
213
214     while (sorgu.next()) {
215         QVariantMap satir;
216         satir["id"] = sorgu.value("id");
217         satir["tarih"] = sorgu.value("tarih");
218         satir["musteri"] = sorgu.value("musteri");
219         satir["recete"] = sorgu.value("recete");
220         satir["agirlik"] = sorgu.value("agirlik");
221         satir["strokeSayisi"] = sorgu.value("strokeSayisi");
222         sonuclar.append(satir);
223     }
224
225     return sonuclar;
226 }
227
228 QVariantList Database::strokeVerileriGetir(int olcumId)
229 {
230     QVariantList sonuclar;
231
232     QSqlQuery sorgu;
233     sorgu.prepare(
234         "SELECT basinc, konum, debi, gecerli FROM Strokelar "
235         "WHERE olcumId = :olcumId ORDER BY id ASC"
236     );
237     sorgu.bindValue(":olcumId", olcumId);
238
239     if (!sorgu.exec()) {
240         qWarning() << "Stroke verileri alinamadi:" << sorgu.lastError().text();
241         return sonuclar;
242     }
243
```

```

244     int sira = 1;
245     while (sorgu.next()) {
246         QVariantMap satir;
247         satir["stroke"] = sira++;
248         satir["basinc"] = sorgu.value("basinc");
249         satir["konum"] = sorgu.value("konum");
250         satir["debi"] = sorgu.value("debi");
251         satir["gecerli"] = sorgu.value("gecerli").toBool();
252         sonuclar.append(satir);
253     }
254
255     return sonuclar;
256 }
257
258 QVariantMap Database::binghamHesapla(int olcumId)
259 {
260     QVariantMap sonuc;
261
262     QSqlQuery sorgu;
263     sorgu.prepare(
264         "SELECT basinc, debi FROM Strokelar "
265         "WHERE olcumId = :olcumId AND gecerli = 1"
266     );
267     sorgu.bindValue(":olcumId", olcumId);
268
269     QVector<double> pDegerleri;
270     QVector<double> qDegerleri;
271
272     if (sorgu.exec()) {
273         while (sorgu.next()) {

```

```

274         pDegerleri.append(sorgu.value("basinc").toDouble());
275         qDegerleri.append(sorgu.value("debi").toDouble());
276     }
277 }
278
279 const int n = pDegerleri.size();
280
281 if (n < 2) {
282     sonuc["tau0"] = 0.0;
283     sonuc["mu"] = 0.0;
284     sonuc["r2"] = 0.0;
285     sonuc["yeterliVeri"] = false;
286     return sonuc;
287 }
288
289 double qToplam = 0, pToplam = 0, qpToplam = 0, qKareToplam = 0;
290 for (int i = 0; i < n; ++i) {
291     qToplam += qDegerleri[i];
292     pToplam += pDegerleri[i];
293     qpToplam += qDegerleri[i] * pDegerleri[i];
294     qKareToplam += qDegerleri[i] * qDegerleri[i];
295 }
296
297 const double qOrt = qToplam / n;
298 const double pOrt = pToplam / n;
299
300 const double payda = qKareToplam - n * qOrt * qOrt;
301 const double mu = (payda != 0.0) ? (qpToplam - n * qOrt * pOrt) / payda : 0.0;
302 const double tau0 = pOrt - mu * qOrt;
303
304 double ssTot = 0, ssRes = 0;

```

```

305     for (int i = 0; i < n; ++i) {
306         const double tahmin = tau0 + mu * qDegerleri[i];
307         ssRes += (pDegerleri[i] - tahmin) * (pDegerleri[i] - tahmin);
308         ssTot += (pDegerleri[i] - pOrt) * (pDegerleri[i] - pOrt);
309     }
310
311     const double r2 = (ssTot != 0.0) ? (1.0 - ssRes / ssTot) : 0.0;
312
313     sonuc["tau0"] = tau0;
314     sonuc["mu"] = mu;
315     sonuc["r2"] = r2;
316     sonuc["yeterliVeri"] = true;
317     return sonuc;
318 }
319
320 bool Database::kalibrasyonKaydet(const QString &sensor, double deger1, double deger2)
321 {
322     QSqlQuery sorgu;
323     sorgu.prepare(
324         "INSERT INTO Kalibrasyonlar (sensor, deger1, deger2, tarih) "
325         "VALUES (:sensor, :deger1, :deger2, :tarih) "
326         "ON CONFLICT(sensor) DO UPDATE SET "
327         "deger1 = :deger1, deger2 = :deger2, tarih = :tarih"
328     );
329
330     sorgu.bindValue(":sensor", sensor);
331     sorgu.bindValue(":deger1", deger1);
332     sorgu.bindValue(":deger2", deger2);
333     sorgu.bindValue(":tarih", QDateTime::currentDateTime().toString("dd.MM.yyyy HH:mm"));
334

```

```
335     if (!sorgu.exec()) {
336         qWarning() << "Kalibrasyon kaydedilemedi:" << sorgu.lastError().text();
337         return false;
338     }
339
340     qDebug() << "Kalibrasyon kaydedildi:" << sensor;
341     return true;
342 }
343
344 QVariantMap Database::kalibrasyonGetir(const QString &sensor)
345 {
346     QVariantMap sonuc;
347
348     QSqlQuery sorgu;
349     sorgu.prepare("SELECT deger1, deger2, tarih FROM Kalibrasyonlar WHERE sensor = :sensor");
350     sorgu.bindValue(":sensor", sensor);
351
352     if (sorgu.exec() && sorgu.next()) {
353         sonuc["deger1"] = sorgu.value("deger1");
354         sonuc["deger2"] = sorgu.value("deger2");
355         sonuc["tarih"] = sorgu.value("tarih");
356         sonuc["mevcut"] = true;
357     } else {
358         sonuc["mevcut"] = false;
359     }
360
361     return sonuc;
362 }
363
364 void Database::kalibrasyonTarihiKaydet(const QString &sensor)
365 {
```

```
366     QSqlQuery sorgu;
367     sorgu.prepare(
368         "INSERT INTO KalibrasyonTarihleri (sensor, tarih) VALUES (:sensor, :tarih) "
369         "ON CONFLICT(sensor) DO UPDATE SET tarih = :tarih"
370     );
371
372     sorgu.bindValue(":sensor", sensor);
373     sorgu.bindValue(":tarih", QDateTime::currentDateTime().toString("dd.MM.yyyy HH:mm"));
374
375     if (!sorgu.exec()) {
376         qWarning() << "Kalibrasyon tarihi kaydedilemedi:" << sorgu.lastError().text();
377     }
378 }
379
380 QString Database::kalibrasyonTarihiGetir(const QString &sensor)
381 {
382     QSqlQuery sorgu;
383     sorgu.prepare("SELECT tarih FROM KalibrasyonTarihleri WHERE sensor = :sensor");
384     sorgu.bindValue(":sensor", sensor);
385
386     if (sorgu.exec() && sorgu.next()) {
387         return sorgu.value("tarih").toString();
388     }
389
390     return QString();
391 }
392
393 QVariantMap Database::olcumBilgisiGetir(int olcumId)
394 {
395     QVariantMap sonuc;
```

```
396
397     QSqlQuery sorgu;
398     sorgu.prepare("SELECT tarih, musteri, recete, agirlik FROM Olcumler WHERE id = :id");
399     sorgu.bindValue(":id", olcumId);
400
401     if (sorgu.exec() && sorgu.next()) {
402         sonuc["tarih"] = sorgu.value("tarih");
403         sonuc["musteri"] = sorgu.value("musteri");
404         sonuc["recete"] = sorgu.value("recete");
405         sonuc["agirlik"] = sorgu.value("agirlik");
406         sonuc["bulundu"] = true;
407     } else {
408         sonuc["bulundu"] = false;
409     }
410
411     return sonuc;
412 }
413
414 QString Database::csvDisaAktar(int olcumId)
415 {
416     QVariantMap bilgi = olcumBilgisiGetir(olcumId);
417     if (!bilgi["bulundu"].toBool()) {
418         qWarning() << "Olcum bulunamadi, csv olusturulamadi.";
419         return QString();
420     }
421
422     QString klasor = QStandardPaths::writableLocation(QStandardPaths::DocumentsLocation) +
"/SliperRaporlari";
423     if (!QDir().mkpath(klasor)) {
424         qWarning() << "Rapor klasoru olusturulamadi:" << klasor;
425         return QString();
```

```

426     }
427
428     QString dosyaAdi = klasor + QString("/SLIPER_Veri_%1_%2.csv")
429         .arg(olcumId)
430         .arg(QDateTime::currentDateTime().toString("ddMMyyyy_HHmss"));
431
432     QFile dosya(dosyaAdi);
433     if (!dosya.open(QIODevice::WriteOnly | QIODevice::Text)) {
434         qWarning() << "CSV dosyasi acilamadi:" << dosyaAdi;
435         return QString();
436     }
437
438     QTextStream akis(&dosya);
439     akis.setEncoding(QStringConverter::Utf8);
440
441     akis << "Olcum ID;" << olcumId << "\n";
442     akis << "Tarih;" << bilgi["tarih"].toString() << "\n";
443     akis << "Musteri;" << bilgi["musteri"].toString() << "\n";
444     akis << "Recete;" << bilgi["recete"].toString() << "\n";
445     akis << "Agirlik (kg);" << bilgi["agirlik"].toDouble() << "\n";
446     akis << "\n";
447     akis << "Stroke;Basinc (mbar);Konum (mm);Debi (m3/h);Gecerli\n";
448
449     QVariantList strokelar = strokeVerileriGetir(olcumId);
450     for (const QVariant &v : strokelar) {
451         QVariantMap s = v.toMap();
452         akis << s["stroke"].toInt() << ";";
453         << s["basinc"].toDouble() << ";";
454         << s["konum"].toDouble() << ";";
455         << s["debi"].toDouble() << ";";
456         << (s["gecerli"].toBool() ? "Evet" : "Hayir") << "\n";

```



```
457     }
458
459     dosya.close();
460     qDebug() << "CSV disa aktarildi:" << dosyaAdi;
461     return dosyaAdi;
462 }
463
464 bool Database::loadCellNoktalarKaydet(const QVariantList &noktalar)
465 {
466     if (!m_db.transaction()) {
467         qWarning() << "Load cell noktalarini icin transaction baslatilamadi:" << m_db.lastError().text();
468         return false;
469     }
470
471     QSqlQuery temizle;
472     if (!temizle.exec("DELETE FROM LoadCellNoktalar")) {
473         qWarning() << "Load cell noktalarini temizlenemedi:" << temizle.lastError().text();
474         m_db.rollback();
475         return false;
476     }
477
478     QSqlQuery sorgu;
479     sorgu.prepare(
480         "INSERT INTO LoadCellNoktalar (hedefKg, hamDeger) "
481         "VALUES (:hedefKg, :hamDeger)"
482     );
483
484     for (const QVariant &v : noktalar) {
485         QVariantMap nokta = v.toMap();
486         sorgu.bindValue(":hedefKg", nokta["hedefKg"].toDouble());
```

```
487         sorgu.bindValue(":hamDeger", nokta["hamDeger"].toDouble());
488         if (!sorgu.exec()) {
489             qWarning() << "Load cell noktasi kaydedilemedi:" << sorgu.lastError().text();
490             m_db.rollback();
491             return false;
492         }
493     }
494
495     if (!m_db.commit()) {
496         qWarning() << "Load cell noktaları commit edilemedi:" << m_db.lastError().text();
497         return false;
498     }
499
500     qDebug() << "Load cell" << noktalar.size() << "nokta kaydedildi.";
501     kalibrasyonTarihiKaydet("loadcell");
502     return true;
503 }
504
505 QVariantList Database::loadCellNoktalariGetir()
506 {
507     QVariantList sonuclar;
508
509     QSqlQuery sorgu("SELECT hedefKg, hamDeger FROM LoadCellNoktalari ORDER BY hamDeger ASC");
510
511     while (sorgu.next()) {
512         QVariantMap satir;
513         satir["hedefKg"] = sorgu.value("hedefKg");
514         satir["hamDeger"] = sorgu.value("hamDeger");
515         sonuclar.append(satir);
516     }
517 }
```

```
518     return sonuclar;
519 }
520
521 bool Database::mesafeNoktalarKaydet(const QVariantList &noktalar)
522 {
523     if (!m_db.transaction()) {
524         qWarning() << "Mesafe noktalarini icin transaction baslatilamadi:" << m_db.lastError().text();
525         return false;
526     }
527
528     QSqlQuery temizle;
529     if (!temizle.exec("DELETE FROM MesafeNoktalarini")) {
530         qWarning() << "Mesafe noktalarini temizlenemedi:" << temizle.lastError().text();
531         m_db.rollback();
532         return false;
533     }
534
535     QSqlQuery sorgu;
536     sorgu.prepare(
537         "INSERT INTO MesafeNoktalarini (hedefMm, hamDeger) "
538         "VALUES (:hedefMm, :hamDeger)"
539     );
540
541     for (const QVariant &v : noktalar) {
542         QVariantMap nokta = v.toMap();
543         sorgu.bindValue(":hedefMm", nokta["hedefMm"].toDouble());
544         sorgu.bindValue(":hamDeger", nokta["hamDeger"].toDouble());
545         if (!sorgu.exec()) {
546             qWarning() << "Mesafe noktasi kaydedilemedi:" << sorgu.lastError().text();
547             m_db.rollback();
```

```
548         return false;
549     }
550 }
551
552 if (!m_db.commit()) {
553     qWarning() << "Mesafe noktaları commit edilemedi:" << m_db.lastError().text();
554     return false;
555 }
556
557 qDebug() << "Mesafe" << noktalar.size() << "nokta kaydedildi.";
558 kalibrasyonTarihiKaydet("mesafe_olcum");
559 return true;
560 }
561
562 QVariantList Database::mesafeNoktalarıGetir()
563 {
564     QVariantList sonuclar;
565
566     QSqlQuery sorgu("SELECT hedefMm, hamDeger FROM MesafeNoktaları ORDER BY hamDeger ASC");
567
568     while (sorgu.next()) {
569         QVariantMap satir;
570         satir["hedefMm"] = sorgu.value("hedefMm");
571         satir["hamDeger"] = sorgu.value("hamDeger");
572         sonuclar.append(satir);
573     }
574
575     return sonuclar;
576 }
577
578 bool Database::egimKalibrasyonuKaydet(double biasX, double biasY, double biasZ,
```

```

579             double gainX, double gainY, double gainZ)
580 {
581     QSqlQuery sorgu;
582     sorgu.prepare(
583         "INSERT OR REPLACE INTO EgimKalibrasyon (id, biasX, biasY, biasZ, gainX, gainY, gainZ, tarih) "
584         "VALUES (1, :biasX, :biasY, :biasZ, :gainX, :gainY, :gainZ, :tarih)"
585     );
586
587     sorgu.bindValue(":biasX", biasX);
588     sorgu.bindValue(":biasY", biasY);
589     sorgu.bindValue(":biasZ", biasZ);
590     sorgu.bindValue(":gainX", gainX);
591     sorgu.bindValue(":gainY", gainY);
592     sorgu.bindValue(":gainZ", gainZ);
593     sorgu.bindValue(":tarih", QDateTime::currentDateTime().toString("dd.MM.yyyy HH:mm"));
594
595     if (!sorgu.exec()) {
596         qWarning() << "Egim kalibrasyonu kaydedilemedi:" << sorgu.lastError().text();
597         return false;
598     }
599
600     qDebug() << "Egim kalibrasyonu kaydedildi.";
601     return true;
602 }
603
604 QVariantMap Database::egimKalibrasyonuGetir()
605 {
606     QVariantMap sonuc;
607
608     QSqlQuery sorgu("SELECT biasX, biasY, biasZ, gainX, gainY, gainZ, tarih FROM EgimKalibrasyon WHERE id =
1");

```

```
609
610     if (sorgu.exec() && sorgu.next()) {
611         sonuc["biasX"] = sorgu.value("biasX");
612         sonuc["biasY"] = sorgu.value("biasY");
613         sonuc["biasZ"] = sorgu.value("biasZ");
614         sonuc["gainX"] = sorgu.value("gainX");
615         sonuc["gainY"] = sorgu.value("gainY");
616         sonuc["gainZ"] = sorgu.value("gainZ");
617         sonuc["tarih"] = sorgu.value("tarih");
618         sonuc["mevcut"] = true;
619     } else {
620         sonuc["mevcut"] = false;
621     }
622
623     return sonuc;
624 }
```

src/Database.h

```
1 #pragma once

2 #include <QObject>

3 #include <QSqlDatabase>

4 #include <QVariantList>

5 #include <QVariantMap>

6

7 class Database : public QObject

8 {

9     Q_OBJECT

10

11 public:

12     explicit Database(QObject *parent = nullptr);

13

14     Q_INVOKABLE int olcumBaslat(const QString &musteri, const QString &recete, double agirlik);

15     Q_INVOKABLE void strokeKaydet(int olcumId, double basinc, double konum, double debi, bool gecerli);

16     Q_INVOKABLE bool olcumSil(int olcumId);

17     Q_INVOKABLE QVariantList tumOlcumleriGetir();

18     Q_INVOKABLE QVariantList strokeVerileriGetir(int olcumId);

19     Q_INVOKABLE QVariantMap binghamHesapla(int olcumId);

20     Q_INVOKABLE bool kalibrasyonKaydet(const QString &sensor, double deger1, double deger2);

21     Q_INVOKABLE QVariantMap kalibrasyonGetir(const QString &sensor);

22     Q_INVOKABLE QVariantMap olcumBilgisiGetir(int olcumId);

23     Q_INVOKABLE QString csvDisaAktar(int olcumId);

24     Q_INVOKABLE bool loadCellNoktalarKaydet(const QVariantList &noktalar);

25     Q_INVOKABLE QVariantList loadCellNoktalarGetir();

26     Q_INVOKABLE bool mesafeNoktalarKaydet(const QVariantList &noktalar);

27     Q_INVOKABLE QVariantList mesafeNoktalarGetir();

28     Q_INVOKABLE bool egimKalibrasyonuKaydet(double biasX, double biasY, double biasZ,

29                                             double gainX, double gainY, double gainZ);

30     Q_INVOKABLE QVariantMap egimKalibrasyonuGetir();
```

```
31     Q_INVOKABLE void kalibrasyonTarihiKaydet(const QString &sensor);
32     Q_INVOKABLE QString kalibrasyonTarihiGetir(const QString &sensor);
33
34 private:
35     void tablolariOlustur();
36     QSqlDatabase m_db;
37 };
```


src/ReportManager.cpp

```
1 #include "ReportManager.h"

2 #include <QPrinter>

3 #include <QPainter>

4 #include <QTextDocument>

5 #include <QStandardPaths>

6 #include <QDir>

7 #include <QDateTime>

8 #include <QFile>

9 #include <QDebug>

10

11 ReportManager::ReportManager(QObject *parent)

12     : QObject(parent)

13 {

14 }

15

16 // Önizleme ve gerçek PDF çıktısı bu tek fonksiyondan beslenir; ikisi arasında

17 // tasarım/veri tutarsızlığı oluşmasın diye HTML üretimi burada merkezileştirildi.

18 QString ReportManager::raporHtmlOlustur(int olcumId, const QString &musteri, const QString &recete,

19                                         double tau0, double mu, double r2)

20 {

21     const bool pompalanabilir = tau0 < 5 && mu < 10;

22     const QString durumRengi = pompalanabilir ? "#16a34a" : "#dc2626";

23     const QString durumMetni = pompalanabilir ? "POMPALANABİLİR" : "POMPALAMA SORUNU";

24

25     return QString(

26         "<html><body style='font-family:Segoe UI, Arial, sans-serif; color:#111827;'"

27         "<h1 style='color:#3b82f6; margin-bottom:4px;'>SLIPER Analiz Raporu</h1>"

28         "<p style='color:#6b7280; font-size:11px; margin-top:0;'>Liya Laboratuvar Test Cihazları</p>"

29         "<hr style='border:none; border-top:1px solid #d1d5db;'"

30         "<table style='width:100%; border-collapse:collapse; margin-top:10px;'"
```

```

31         "<tr><td style='padding:4px 0; width:140px;'><b>Ölçüm ID</b></td><td>%1</td></tr>"
32         "<tr><td style='padding:4px 0;'><b>Müşteri</b></td><td>%2</td></tr>"
33         "<tr><td style='padding:4px 0;'><b>Reçete</b></td><td>%3</td></tr>"
34         "<tr><td style='padding:4px 0;'><b>Tarih</b></td><td>%4</td></tr>"
35         "</table>"
36         "<h2 style='color:#111827; margin-top:24px; margin-bottom:8px; font-size:16px;'>Bingham Model
Sonuçları</h2>"
37         "<table style='width:100%; border-collapse:collapse;' cellpadding='6'>"
38         "<tr style='background:#f3f4f6;'>"
39         "<th style='text-align:left; border:1px solid #d1d5db;'>Parametre</th>"
40         "<th style='text-align:left; border:1px solid #d1d5db;'>Değer</th></tr>"
41         "<tr><td style='border:1px solid #d1d5db;'>Akma Gerilmesi ( $\tau_0$ )</td><td style='border:1px solid
#d1d5db;'>%5 mbar</td></tr>"
42         "<tr><td style='border:1px solid #d1d5db;'>Plastik Viskozite ( $\mu$ )</td><td style='border:1px solid
#d1d5db;'>%6 mbar·h/m³</td></tr>"
43         "<tr><td style='border:1px solid #d1d5db;'>Uyum Kalitesi ( $R^2$ )</td><td style='border:1px solid
#d1d5db;'>%7</td></tr>"
44         "</table>"
45         "<p style='margin-top:20px;'><b>Durum:</b> "
46         "<span style='color:%8; font-weight:bold;'>%9</span></p>"
47         "<hr style='border:none; border-top:1px solid #d1d5db; margin-top:24px;'>"
48         "<p style='color:#6b7280; font-size:10px;'>Liya Laboratuvar Test Cihazları - SLIPER Analiz
Sistemi</p>"
49         "</body></html>"
50     ).arg(olcumId)
51     .arg(musteri.toHtmlEscaped())
52     .arg(recete.toHtmlEscaped())
53     .arg(QDateTime::currentDateTime().toString("dd.MM.yyyy HH:mm"))
54     .arg(QString::number(tau0, 'f', 2))
55     .arg(QString::number(mu, 'f', 2))
56     .arg(QString::number(r2, 'f', 2))
57     .arg(durumRengi)
58     .arg(durumMetni);

```

```

59 }

60

61 QString ReportManager::pdfOnizlemeHtml(int olcumId, const QString &musteri, const QString &recete,
62
63         double tau0, double mu, double r2)
64 {
65     return raporHtmlOlustur(olcumId, muster, recete, tau0, mu, r2);
66 }
67
68 QString ReportManager::pdfOlustur(int olcumId, const QString &musteri, const QString &recete,
69
70         double tau0, double mu, double r2)
71 {
72     QString klasor = QStandardPaths::writableLocation(QStandardPaths::DocumentsLocation) +
73     "/SliperRaporlari";
74
75     if (!QDir().mkpath(klasor)) {
76
77         qWarning() << "Rapor klasoru olusturulamadi:" << klasor;
78
79         return QString();
80     }
81
82     QString dosyaAdi = klasor + QString("/SLIPER_Rapor_%1_%2.pdf")
83
84         .arg(olcumId)
85
86         .arg(QDateTime::currentDateTime().toString("ddMMyyyy_HHmss"));
87
88     QTextDocument belge;
89
90     belge.setHtml(raporHtmlOlustur(olcumId, muster, recete, tau0, mu, r2));
91
92
93     QPrinter yazici(QPrinter::HighResolution);
94
95     yazici.setOutputFormat(QPrinter::PdfFormat);
96
97     yazici.setOutputFileName(dosyaAdi);
98
99
100     belge.print(&yazici);
101
102

```

```
89     if (!QFile::exists(dosyaAdi)) {
90         qWarning() << "PDF dosyasi yazilamadi:" << dosyaAdi;
91         return QString();
92     }
93
94     qDebug() << "PDF rapor olusturuldu:" << dosyaAdi;
95     return dosyaAdi;
96 }
```

src/ReportManager.h

```
1 #pragma once

2 #include <QObject>

3

4 class ReportManager : public QObject

5 {

6     Q_OBJECT

7

8 public:

9     explicit ReportManager(QObject *parent = nullptr);

10    Q_INVOKABLE QString pdfOnizlemeHtml(int olcumId, const QString &musteri, const QString &recete,

11                                       double tau0, double mu, double r2);

12    Q_INVOKABLE QString pdfOlustur(int olcumId, const QString &musteri, const QString &recete,

13                                   double tau0, double mu, double r2);

14

15 private:

16    QString raporHtmlOlustur(int olcumId, const QString &musteri, const QString &recete,

17                             double tau0, double mu, double r2);

18 };
```

src/SensorManager.cpp

```
1 #include "SensorManager.h"

2

3 SensorManager::SensorManager(QObject *parent)

4     : QObject(parent)

5 {

6 }

7

8 double SensorManager::basinc() const { return m_basinc; }

9 double SensorManager::konum() const { return m_konum; }

10 double SensorManager::hiz() const { return m_hiz; }

11 double SensorManager::debi() const { return m_debi; }

12 double SensorManager::egimX() const { return m_egimX; }

13 double SensorManager::egimY() const { return m_egimY; }

14 double SensorManager::hamAgirlik() const { return m_hamAgirlik; }

15 double SensorManager::hamMesafe() const { return m_hamMesafe; }

16 double SensorManager::hamAccelX() const { return m_hamAccelX; }

17 double SensorManager::hamAccelY() const { return m_hamAccelY; }

18 double SensorManager::hamAccelZ() const { return m_hamAccelZ; }

19 bool SensorManager::veriGecerli() const { return m_veriGecerli; }

20

21 void SensorManager::veriGuncelle(double basinc, double konum, double hiz, double debi, double egimX, double
egimY)

22 {

23     const bool veriGecerliDegisti = !m_veriGecerli;

24     m_veriGecerli = true;

25

26     if (m_basinc != basinc) { m_basinc = basinc; emit basincChanged(); }

27     if (m_konum != konum) { m_konum = konum; emit konumChanged(); }

28     if (m_hiz != hiz) { m_hiz = hiz; emit hizChanged(); }

29     if (m_debi != debi) { m_debi = debi; emit debiChanged(); }
```

```
30     if (m_egimX != egimX) { m_egimX = egimX; emit egimXChanged(); }
31     if (m_egimY != egimY) { m_egimY = egimY; emit egimYChanged(); }
32     if (veriGecerliDegisti) { emit veriGecerliChanged(); }
33
34     emit veriGuncellendi();
35 }
36
37 void SensorManager::hamAgirlikGuncelle(double hamAgirlik)
38 {
39     if (m_hamAgirlik != hamAgirlik) {
40         m_hamAgirlik = hamAgirlik;
41         emit hamAgirlikChanged();
42     }
43 }
44
45 void SensorManager::hamMesafeGuncelle(double hamMesafe)
46 {
47     if (m_hamMesafe != hamMesafe) {
48         m_hamMesafe = hamMesafe;
49         emit hamMesafeChanged();
50     }
51 }
52
53 void SensorManager::hamAccelGuncelle(double x, double y, double z)
54 {
55     if (m_hamAccelX != x || m_hamAccelY != y || m_hamAccelZ != z) {
56         m_hamAccelX = x;
57         m_hamAccelY = y;
58         m_hamAccelZ = z;
59         emit hamAccelDegisti();
60     }
```

```
61 }  
  
62  
63 void SensorManager::veriyiGecersizYap()  
64 {  
65     if (!m_veriGecerli) return;  
66     m_veriGecerli = false;  
67     emit veriGecerliChanged();  
68 }
```


src/SensorManager.h

```
1 #pragma once

2 #include <QObject>

3

4 class SensorManager : public QObject

5 {

6     Q_OBJECT

7     Q_PROPERTY(double basinc READ basinc NOTIFY basincChanged)

8     Q_PROPERTY(double konum READ konum NOTIFY konumChanged)

9     Q_PROPERTY(double hiz READ hiz NOTIFY hizChanged)

10    Q_PROPERTY(double debi READ debi NOTIFY debiChanged)

11    Q_PROPERTY(double egimX READ egimX NOTIFY egimXChanged)

12    Q_PROPERTY(double egimY READ egimY NOTIFY egimYChanged)

13    Q_PROPERTY(double hamAgirlik READ hamAgirlik NOTIFY hamAgirlikChanged)

14    Q_PROPERTY(double hamMesafe READ hamMesafe NOTIFY hamMesafeChanged)

15    Q_PROPERTY(double hamAccelX READ hamAccelX NOTIFY hamAccelDegisti)

16    Q_PROPERTY(double hamAccelY READ hamAccelY NOTIFY hamAccelDegisti)

17    Q_PROPERTY(double hamAccelZ READ hamAccelZ NOTIFY hamAccelDegisti)

18    Q_PROPERTY(bool veriGecerli READ veriGecerli NOTIFY veriGecerliChanged)

19

20 public:

21     explicit SensorManager(QObject *parent = nullptr);

22

23     double basinc() const;

24     double konum() const;

25     double hiz() const;

26     double debi() const;

27     double egimX() const;

28     double egimY() const;

29     double hamAgirlik() const;

30     double hamMesafe() const;
```

```
31     double hamAccelX() const;
32     double hamAccelY() const;
33     double hamAccelZ() const;
34     bool veriGecerli() const;
35
36     Q_INVOKABLE void veriGuncelle(double basinc, double konum, double hiz, double debi, double egimX, double
egimY);
37     Q_INVOKABLE void hamAgirlikGuncelle(double hamAgirlik);
38     Q_INVOKABLE void hamMesafeGuncelle(double hamMesafe);
39     Q_INVOKABLE void hamAccelGuncelle(double x, double y, double z);
40     Q_INVOKABLE void veriyiGecersizYap();
41
42 signals:
43     void basincChanged();
44     void konumChanged();
45     void hizChanged();
46     void debiChanged();
47     void egimXChanged();
48     void egimYChanged();
49     void hamAgirlikChanged();
50     void hamMesafeChanged();
51     void hamAccelDegisti();
52     void veriGecerliChanged();
53     void veriGuncellendi();
54
55 private:
56     double m_basinc = 0.0;
57     double m_konum = 0.0;
58     double m_hiz = 0.0;
59     double m_debi = 0.0;
60     double m_egimX = 0.0;
```

```
61     double m_egimY = 0.0;
62     double m_hamAgirlik = 0.0;
63     double m_hamMesafe = 0.0;
64     double m_hamAccelX = 0.0;
65     double m_hamAccelY = 0.0;
66     double m_hamAccelZ = 0.0;
67     bool m_veriGecerli = false;
68 };
```

src/WifiManager.cpp

```
1 #include "WifiManager.h"

2 #include <QJsonDocument>

3 #include <QJsonObject>

4 #include <QDebug>

5 #include <cmath>

6

7 WifiManager::WifiManager(SensorManager *sensorManager, Database *database, QObject *parent)

8     : QObject(parent)

9     , m_sensorManager(sensorManager)

10    , m_database(database)

11 {

12     m_soket = new QTcpSocket(this);

13

14     connect(m_soket, &QTcpSocket::connected, this, &WifiManager::soketBaglandi);

15     connect(m_soket, &QTcpSocket::disconnected, this, &WifiManager::soketAyrildi);

16     connect(m_soket, &QTcpSocket::errorOccurred, this, &WifiManager::soketHata);

17     connect(m_soket, &QTcpSocket::readyRead, this, &WifiManager::veriHazir);

18 }

19

20 bool WifiManager::baglandi() const { return m_baglandi; }

21 bool WifiManager::baglaniyor() const { return m_baglaniyor; }

22 QString WifiManager::durumMesaji() const { return m_durumMesaji; }

23

24 void WifiManager::baglan()

25 {

26     if (m_baglandi || m_baglaniyor) return;

27

28     m_baglaniyor = true;

29     emit baglaniyorChanged();

30     durumGuncelle("SLIPER-ESP32'ye baglaniliyor...");
```

```
31
32     m_soket->connectToHost(QString(ESP32_IP), ESP32_PORT);
33 }
34
35 void WifiManager::baglantiyiKes()
36 {
37     m_soket->disconnectFromHost();
38 }
39
40 void WifiManager::kalibrasyonYenidenYukle()
41 {
42     kalibrasyonYukle();
43 }
44
45 void WifiManager::kalibrasyonYukle()
46 {
47     if (!m_database) return;
48
49     m_loadCellHamDegerleri.clear();
50     m_loadCellKiloDegerleri.clear();
51     const QVariantList noktalar = m_database->loadCellNoktalariGetir();
52     for (const QVariant &v : noktalar) {
53         QVariantMap n = v.toMap();
54         m_loadCellHamDegerleri.append(n["hamDeger"].toDouble());
55         m_loadCellKiloDegerleri.append(n["hedefKg"].toDouble());
56     }
57     qDebug() << "Load cell" << m_loadCellHamDegerleri.size() << "nokta yuklendi.";
58
59     m_mesafeHamDegerleri.clear();
60     m_mesafeMmDegerleri.clear();
```

```
61     const QVariantList mesafeNoktaları = m_database->mesafeNoktalarıGetir();
62     for (const QVariant &v : mesafeNoktaları) {
63         QVariantMap n = v.toMap();
64         m_mesafeHamDegerleri.append(n["hamDeger"].toDouble());
65         m_mesafeMmDegerleri.append(n["hedefMm"].toDouble());
66     }
67     qDebug() << "Mesafe" << m_mesafeHamDegerleri.size() << "nokta yuklendi.";
68
69     QVariantMap egimKal = m_database->egimKalibrasyonuGetir();
70     if (egimKal.value("mevcut").toBool()) {
71         m_egimBiasX = egimKal.value("biasX").toDouble();
72         m_egimBiasY = egimKal.value("biasY").toDouble();
73         m_egimBiasZ = egimKal.value("biasZ").toDouble();
74         m_egimGainX = egimKal.value("gainX").toDouble();
75         m_egimGainY = egimKal.value("gainY").toDouble();
76         m_egimGainZ = egimKal.value("gainZ").toDouble();
77         qDebug() << "Egim ivme kalibrasyonu yuklendi.";
78     }
79 }
80
81 void WifiManager::soketBaglandi()
82 {
83     m_baglandi = true;
84     m_baglaniyor = false;
85     emit baglandiChanged();
86     emit baglaniyorChanged();
87     durumGuncelle("SLIPER-ESP32 Bağlı");
88
89     m_ilkPaket = true;
90     m_zamanlayici.restart();
91     kalibrasyonYukle();
```

```
92
93     if (m_sensorManager) {
94         m_sensorManager->veriyiGecersizYap();
95     }
96
97     qDebug() << "ESP32'ye baglanildi:" << ESP32_IP << ESP32_PORT;
98 }
99
100 void WifiManager::soketAyrildi()
101 {
102     m_baglandi = false;
103     m_baglaniyor = false;
104     emit baglandiChanged();
105     emit baglaniyorChanged();
106     durumGuncelle("Bağlı Değil");
107
108     if (m_sensorManager) {
109         m_sensorManager->veriyiGecersizYap();
110     }
111
112     qDebug() << "ESP32 baglantisi kesildi.";
113 }
114
115 void WifiManager::soketHata(QAbstractSocket::SocketError hata)
116 {
117     m_baglandi = false;
118     m_baglaniyor = false;
119     emit baglandiChanged();
120     emit baglaniyorChanged();
121     durumGuncelle("Baglanti hatasi: " + m_soket->errorString());
```

```

122
123     qWarning() << "Wifi soket hatasi:" << hata << m_soket->errorString();
124 }
125
126 void WifiManager::veriHazar()
127 {
128     m_tamponVeri.append(m_soket->readAll());
129
130     int satirSonu;
131     while ((satirSonu = m_tamponVeri.indexOf('\n')) != -1) {
132         QByteArray satir = m_tamponVeri.left(satirSonu).trimmed();
133         m_tamponVeri.remove(0, satirSonu + 1);
134
135         if (!satir.isEmpty()) {
136             jsonSatiriIsle(satir);
137         }
138     }
139 }
140
141 void WifiManager::egimHesapla(double accelX, double accelY, double accelZ, double &egimX, double &egimY)
const
142 {
143     egimX = std::atan2(accelY, accelZ) * 180.0 / M_PI;
144     egimY = std::atan2(-accelX, std::sqrt(accelY * accelY + accelZ * accelZ)) * 180.0 / M_PI;
145 }
146
147 void WifiManager::jsonSatiriIsle(const QByteArray &satir)
148 {
149     QJsonParseError hata;
150     QJsonDocument belge = QJsonDocument::fromJson(satir, &hata);
151

```



```
152     if (hata.error != QJsonParseError::NoError || !belge.isObject()) {
153         qWarning() << "JSON parse hatasi:" << hata.errorString() << satir;
154         return;
155     }
156
157     QJsonObject obj = belge.object();
158
159     const double hamAgirlik = obj.value("hamAgirlik").toDouble();
160     const double hamMesafe = obj.contains("hamMesafe")
161         ? obj.value("hamMesafe").toDouble()
162         : obj.value("konum").toDouble();
163     const double accelX = obj.value("accelX").toDouble();
164     const double accelY = obj.value("accelY").toDouble();
165     const double accelZ = obj.value("accelZ").toDouble();
166
167     // Kalibrasyon ekraninda gerekiyor: ham deger her zaman guncellensin
168     if (m_sensorManager) {
169         m_sensorManager->hamAgirlikGuncelle(hamAgirlik);
170         m_sensorManager->hamMesafeGuncelle(hamMesafe);
171         m_sensorManager->hamAccelGuncelle(accelX, accelY, accelZ);
172     }
173
174     const double konum = m_mesafeHamDegerleri.isEmpty()
175         ? hamMesafe
176         : mesafeInterpolasyon(hamMesafe);
177
178     // --- Basinc: kayitli kalibrasyon ile ---
179     const double agirlikKg = loadCellInterpolasyon(hamAgirlik);
180     const double basinc = agirlikKg * 98.1;
181
182     // --- Hiz ve debi: konum farkindan turetiliyor ---
```

```
183     double hiz = 0.0;
184     double debi = 0.0;
185
186     const qint64 simdikiZamanMs = m_zamanlayici.elapsed();
187
188     if (!m_ilkPaket) {
189         double dt = (simdikiZamanMs - m_oncekiZamanMs) / 1000.0;
190         if (dt <= 0.0) dt = 0.001;
191
192         const double boruAlaniM2 = M_PI * (BORU_CAPI_M / 2.0) * (BORU_CAPI_M / 2.0);
193         hiz = std::fabs(konum - m_oncekiKonum) / 1000.0 / dt;
194         debi = hiz * boruAlaniM2 * 3600.0;
195     } else {
196         m_ilkPaket = false;
197     }
198
199     m_oncekiKonum = konum;
200     m_oncekiZamanMs = simdikiZamanMs;
201
202     // --- Egim: once ham ivme bias/gain ile duzeltilir, sonra aciya cevrilir ---
203     const double duzeltilmisX = (accelX - m_egimBiasX) / m_egimGainX;
204     const double duzeltilmisY = (accelY - m_egimBiasY) / m_egimGainY;
205     const double duzeltilmisZ = (accelZ - m_egimBiasZ) / m_egimGainZ;
206
207     double egimX = 0.0, egimY = 0.0;
208     egimHesapla(duzeltilmisX, duzeltilmisY, duzeltilmisZ, egimX, egimY);
209
210     if (m_sensorManager) {
211         m_sensorManager->veriGuncelle(basinc, konum, hiz, debi, egimX, egimY);
212     }
```

```
213 }
214
215 void WifiManager::durumGuncelle(const QString &mesaj)
216 {
217     if (m_durumMesaji != mesaj) {
218         m_durumMesaji = mesaj;
219         emit durumMesajiChanged();
220     }
221 }
222
223 double WifiManager::loadCellInterpolasyon(double hamDeger) const
224 {
225     const int n = m_loadCellHamDegerleri.size();
226     if (n == 0) return 0.0;
227     if (n == 1) return m_loadCellKiloDegerleri[0];
228
229     int i;
230     if (hamDeger <= m_loadCellHamDegerleri[0]) {
231         i = 0;
232     } else if (hamDeger >= m_loadCellHamDegerleri[n - 1]) {
233         i = n - 2;
234     } else {
235         for (i = 0; i < n - 1; ++i) {
236             if (hamDeger <= m_loadCellHamDegerleri[i + 1]) break;
237         }
238     }
239
240     const double ham1 = m_loadCellHamDegerleri[i];
241     const double ham2 = m_loadCellHamDegerleri[i + 1];
242     const double kg1 = m_loadCellKiloDegerleri[i];
243     const double kg2 = m_loadCellKiloDegerleri[i + 1];
```

```
244
245     if (ham2 == ham1) return kg1;
246     return kg1 + (hamDeger - ham1) * (kg2 - kg1) / (ham2 - ham1);
247 }
248
249 double WifiManager::mesafeInterpolasyon(double hamDeger) const
250 {
251     const int n = m_mesafeHamDegerleri.size();
252     if (n == 0) return 0.0;
253     if (n == 1) return m_mesafeMmDegerleri[0];
254
255     int i;
256     if (hamDeger <= m_mesafeHamDegerleri[0]) {
257         i = 0;
258     } else if (hamDeger >= m_mesafeHamDegerleri[n - 1]) {
259         i = n - 2;
260     } else {
261         for (i = 0; i < n - 1; ++i) {
262             if (hamDeger <= m_mesafeHamDegerleri[i + 1]) break;
263         }
264     }
265
266     const double ham1 = m_mesafeHamDegerleri[i];
267     const double ham2 = m_mesafeHamDegerleri[i + 1];
268     const double mm1 = m_mesafeMmDegerleri[i];
269     const double mm2 = m_mesafeMmDegerleri[i + 1];
270
271     if (ham2 == ham1) return mm1;
272     return mm1 + (hamDeger - ham1) * (mm2 - mm1) / (ham2 - ham1);
273 }
```

src/WifiManager.h

```
1 #pragma once

2 #include <QObject>

3 #include <QTcpSocket>

4 #include <QElapsedTimer>

5 #include <QVector>

6 #include "SensorManager.h"

7 #include "Database.h"

8

9 class WifiManager : public QObject
10 {
11     Q_OBJECT
12     Q_PROPERTY(bool baglandi READ baglandi NOTIFY baglandiChanged)
13     Q_PROPERTY(bool baglaniyor READ baglaniyor NOTIFY baglaniyorChanged)
14     Q_PROPERTY(QString durumMesaji READ durumMesaji NOTIFY durumMesajiChanged)
15
16 public:
17     explicit WifiManager(SensorManager *sensorManager, Database *database, QObject *parent = nullptr);
18
19     bool baglandi() const;
20     bool baglaniyor() const;
21     QString durumMesaji() const;
22
23     Q_INVOKABLE void baglan();
24     Q_INVOKABLE void baglantiyiKes();
25     Q_INVOKABLE void kalibrasyonYenidenYukle();
26
27 signals:
28     void baglandiChanged();
29     void baglaniyorChanged();
30     void durumMesajiChanged();
```

```
31
32 private slots:
33     void soketBaglandi();
34     void soketAyrildi();
35     void soketHata(QAbstractSocket::SocketError hata);
36     void veriHazir();
37
38 private:
39     void durumGuncelle(const QString &mesaj);
40     void jsonSatiriIsle(const QByteArray &satir);
41     void egimHesapla(double accelX, double accelY, double accelZ, double &egimX, double &egimY) const;
42     void kalibrasyonYukle();
43     double loadCellInterpolasyon(double hamDeger) const;
44     double mesafeInterpolasyon(double hamDeger) const;
45
46     QTcpSocket *m_soket = nullptr;
47     SensorManager *m_sensorManager = nullptr;
48     Database *m_database = nullptr;
49     QByteArray m_tamponVeri;
50
51     bool m_baglandi = false;
52     bool m_baglaniyor = false;
53     QString m_durumMesaji = "Bağlı Değil";
54
55     bool m_ilkPaket = true;
56     double m_oncekiKonum = 0.0;
57     qint64 m_oncekiZamanMs = 0;
58     QElapsedTimer m_zamanlayici;
59
60     // --- Kayitli kalibrasyon degerleri (Database'den yuklenir) ---
```

```
61
62     QVector<double> m_loadCellHamDegerleri;
63     QVector<double> m_loadCellKiloDegerleri;
64
65     QVector<double> m_mesafeHamDegerleri;
66     QVector<double> m_mesafeMmDegerleri;
67
68     double m_egimBiasX = 0.0, m_egimBiasY = 0.0, m_egimBiasZ = 0.0;
69     double m_egimGainX = 1.0, m_egimGainY = 1.0, m_egimGainZ = 1.0;
70
71     static constexpr const char *ESP32_IP = "192.168.4.1";
72     static constexpr quint16 ESP32_PORT = 8888;
73
74     static constexpr double BORU_CAPI_M = 0.126;
75 };
```